



Crackz Documentation

AI Image Analysis

Version: 0.90.0

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Home

Crackz



The Crackz desktop app — project dashboard with model configuration and detection metrics.



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crackz-gui

crackz

1. crackz-gui
- 2.
- 3.
- 4.
- 5.

```
crackz init --name demo
crackz import-images --project demo --src ./photos --type raw --size 512 512
crackz detect run --project demo
# Results written to demo/artifacts/detections/
```

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Installation

Installation

uv sync

LICENSE

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-

crackz-gui

crackz-gui.exe
crackz-gui
crackz-gui
crackz-gui-cu128
crackz-gui-cu118

nvidia-smi CUDA Version

```
asset=crackz-gui                      # or crackz-gui-cu128 / crackz-gui-cu118
chmod +x "$asset"
sudo install -m 755 "$asset" /usr/local/bin/crackz-gui
crackz-gui
```

crackz-gui.exe PATH

gh auth login uv

gh auth login uv

```
gh auth login
gh release download --repo Anaxiatech-AB/crackz \
  --pattern 'crackz-*.whl' --pattern 'crackz_gui-*.whl'
uv pip install crackz-*.whl crackz_gui-*.whl
crackz-gui
```

```
# Using pip with GitHub authentication
pip install git+https://github.com/OWNER/crackz.git@vVERSION

# Or download the wheel from the private repository's Releases page
# (Requires GitHub authentication - use personal access token)
pip install https://USERNAME:TOKEN@github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
pip install git+https://github.com/OWNER/crackz.git
```

```
# Login with GitHub CLI
gh auth login

# Download release asset
gh release download vVERSION --repo OWNER/crackz --pattern '*.whl'

# Install the downloaded wheel
pip install crackz-VERSION-py3-none-any.whl
```

```
# Create a personal access token with 'repo' scope at:
# https://github.com/settings/tokens

# Set your token (keep it secure!)
export GITHUB_TOKEN="ghp_XXXXXXXXXX" # gitleaks:allow

# Install using token authentication
pip install git+https://${GITHUB_TOKEN}@github.com/OWNER/crackz.git@vVERSION
```

```
https://github.com/OWNER/crackz
crackz-VERSION-py3-none-any.whl
```

```
pip install crackz-VERSION-py3-none-any.whl
```

```
# Download wheel using GitHub token
curl -L \
  -H "Authorization: token ${GITHUB_TOKEN}" \
  -o crackz.whl \
  https://github.com/OWNER/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Install the downloaded wheel
pip install crackz.whl
```

```
# If your organization has configured a private index
pip install --index-url https://pypi.company.com/simple/ crackz
```

```
# Using the provided index URL and credentials
pip install crackz --index-url https://INDEX_URL/simple/

# Example with authentication (if required):
pip install crackz --index-url https://USERNAME:TOKEN@pypi.company.com/simple/
```

~/.config/pip/pip.conf

%APPDATA%\pip\pip.ini

```
[global]
index-url = https://USERNAME:TOKEN@pypi.company.com/simple/
```

```
pip install crackz
pip install --upgrade crackz # For updates
```

```
# Set credentials via environment variables
export PIP_INDEX_URL="https://pypi.company.com/simple/"
export PIP_EXTRA_INDEX_URL="https://pypi.org/simple" # Fallback to PyPI for dependencies
pip install crackz
```

```
https://dl.cloudsmith.io/TOKEN/org/repo/python/simple/
https://artifactory.company.com/artifactory/api/pypi/pypi-local/simple/
https://pypi.fury.io/TOKEN/USERNAME/
https://aws:TOKEN@domain.d.codeartifact.region.amazonaws.com/pypi/repo/simple/
https://pkgs.dev.azure.com/org/_packaging/feed/pypi/simple/ https://devpi.company.com/username/index/+simple/
```

```
chmod 600 ~/.config/pip/pip.conf # Linux/macOS
```

```
pip install --trusted-host pypi.company.com crackz
# Or add CA cert:
pip install --cert /path/to/ca-bundle.crt crackz
```

```
pip install crackz --extra-index-url https://pypi.org/simple/
```

crackz-gui.exe

crackz

crackz-gui

crackz

crackz-gui

crackz.exe

```
REM Download the executables from GitHub Releases
REM Move them to a permanent location
mkdir C:\Program Files\Crackz
move crackz.exe "C:\Program Files\Crackz\"
move crackz-gui.exe "C:\Program Files\Crackz\"

REM Add to PATH (optional, for CLI access from anywhere)
setx PATH "%PATH%;C:\Program Files\Crackz"
```

```
# Download the executables from GitHub Releases
chmod +x crackz crackz-gui

# Move to a system directory (optional)
sudo mv crackz /usr/local/bin/
sudo mv crackz-gui /usr/local/bin/

# Or keep in a local directory and add to PATH
mkdir -p ~/.Applications/crackz
mv crackz crackz-gui ~/.Applications/crackz/
echo 'export PATH="$HOME/Applications/crackz:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

```
# Run CLI directly (no Python needed)
crackz --help
crackz init --name myproject

# Run GUI by double-clicking crackz-gui.exe (Windows)
# Or run from command line:
crackz-gui
```

cuda.exe

crackz-cuda.exe

crackz-gui-

```
crackz --help
python -c "import crackz; import importlib.metadata as m; print(m.version('crackz'))"
```

- crackz-VERSION-py3-none-any.whl
- crackz_cu118-VERSION-py3-none-any.whl
- crackz_cu128-VERSION-py3-none-any.whl

```
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 11.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu118-VERSION-py3-none-any.whl
```

```
# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz CUDA variant
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cu128-VERSION-py3-none-any.whl
```

```
nvidia-smi
```

```
nvcc --version
# Or check driver
nvidia-smi | grep "CUDA Version"
```

crackz_cu118

crackz_cu128

crackz

```
# Login first
gh auth login

# Download and install CPU version (one step)
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz-*.whl'
pip install crackz-VERSION-py3-none-any.whl
```

```
# Login first
gh auth login

# Step 1: Install PyTorch with CUDA 12.8
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Download and install crackz CUDA variant
gh release download vVERSION --repo OWNER/crackz --pattern 'crackz_cul28-*.whl'
pip install crackz_cul28-VERSION-py3-none-any.whl
```

```
# Check CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"

# Check PyTorch version (should include +cu118 or +cu128 suffix)
python -c "import torch; print(f'PyTorch version: {torch.__version__}')"

# Check GPU device name
python -c "import torch; print(f'Device: {torch.cuda.get_device_name(0) if torch.cuda.is_available() else \"CPU\"}')"

```

```
CUDA available: True
PyTorch version: 2.5.0+cu128
Device: NVIDIA GeForce RTX 4090
```

```
# Step 1: Install specific PyTorch version
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu121

# Step 2: Install crackz CUDA variant (cu118 or cu128, either works)
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz_cul18-VERSION-py3-none-any.whl
```

cpu cu128

```
# This installs CPU PyTorch even with the cu128 extra - BROKEN!
pip install "crackz-VERSION-py3-none-any.whl[cu128]"
```

```
# Step 1: Install PyTorch with CUDA
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128

# Step 2: Install crackz
pip install crackz_cul28-VERSION-py3-none-any.whl
```

keyring

```
# With uv
uv add keyring

# With pip
pip install keyring
```

```
from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)

# Check availability
if is_secure_storage_available():
    store_api_key("your-api-key")
    api_key = retrieve_api_key()
```

1.

```
mkdir crackz_wheels
cd crackz_wheels

# Download Crackz wheel
wget https://github.com/Anaxiatech-AB/crackz/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

# Download all dependencies
pip download -r <(pip show -f crackz | grep Requires) -d .
# Or download from the wheel itself:
pip download crackz-VERSION-py3-none-any.whl -d .
```

2.

3.

```
pip install --no-index --find-links crackz_wheels crackz-VERSION-py3-none-any.whl
```

```
pip install .
```

```
pyproject.toml
```

```
# Download the new version from GitHub Releases
pip install --upgrade https://github.com/Anaxiatech-AB/crackz/releases/download/vNEW_VERSION/crackz-NEW_VERSION-py3-none-any.whl

# Or if using a private index:
pip install -U --index-url https://pypi.company.com/simple/ crackz
```

```
pip uninstall crackz
```

```
crackz --version
crackz init --name sanity-test
crackz detect run --project sanity-test # (Will use default config; may download model weights)
```

```
torch.__version__
```

```
No module named crackz
```

```
CUDA not available
```

```
torch.__version__
```

```
ModuleNotFoundError: No module named
'torch'
```

```
CRACKZ_LOG_DIR
```

```
pip install --force-reinstall crackz-VERSION.whl
```

```
pyproject.toml
```

```
# Check if CUDA is available
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'CUDA device: {torch.cuda.get_device_name(0)}')"
```

```
CUDA available: True
CUDA device: NVIDIA GeForce RTX 3080
```

```
# These commands automatically use GPU if available:
crackz detect train --project my_project
crackz detect run --project my_project
crackz-gui # GUI will use GPU automatically
```

```
cpu
```

```
cuda
```

```
mps
```

```
# Force GPU usage (will error if CUDA not available)
crackz detect train --project my_project --device cuda
```

```
# Force CPU usage (useful for debugging)
crackz detect train --project my_project --device cpu
```

```
# Let Crackz auto-detect (default)
crackz detect train --project my_project
```

```
<project>/<project>_project.yaml
```

```
ai:
model:
device: cuda # or cpu, or mps
```

```
crackz detect train --project my_project
```

```
GPU available: True, used: True
TPU available: False, using: 0 TPU cores
IPU available: False, using: 0 IPUs
HPU available: False, using: 0 HPUs
```

```
# NVIDIA GPUs
watch -n 1 nvidia-smi

# Should show Python process using GPU memory and compute
```

```
crackz_cu118-VERSION-py3-none-any.whl
```

```
crackz_cu128-VERSION-py3-none-any.whl
```

```
crackz_cu118-VERSION-py3-none-any.whl
```

```
nvcc --version # If CUDA toolkit installed
nvidia-smi # Shows driver version (different from CUDA toolkit)
```

1.

```
python -c "import torch; print(torch.__version__)"
# Should show: 2.5.0+cu118 or 2.5.0+cu128
# If it shows just 2.5.0, you have CPU-only version!
```

2.

```
pip uninstall torch torchvision
pip install torch torchvision --index-url https://download.pytorch.org/whl/cu118
```

3.

```
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
```

```
# GUI: Project → Settings → Training → Batch Size
# CLI:
crackz detect train --project my_project --batch-size 2
```

- `nvidia-smi`
-
-

```
CRACKZ_LOG_DIR=my_logs crackz detect run --project myproj
```

```
$Env:CRACKZ_LOG_DIR='my_logs'
```

- `pip install --require-hashes -r requirements.txt`
-

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```
crackz migrate --project ./my_project          # applies changes + backup
crackz migrate --project ./my_project --dry-run # preview only
```

CRACKZ_AUTO_MIGRATE=1

Quick Start

Getting Started with Crackz

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```
# Install UV package manager
curl -Lsf https://astral.sh/uv/install.sh | sh # macOS/Linux
# OR
irm https://astral.sh/uv/install.ps1 | iex # Windows PowerShell

# Authenticate with GitHub
gh auth login

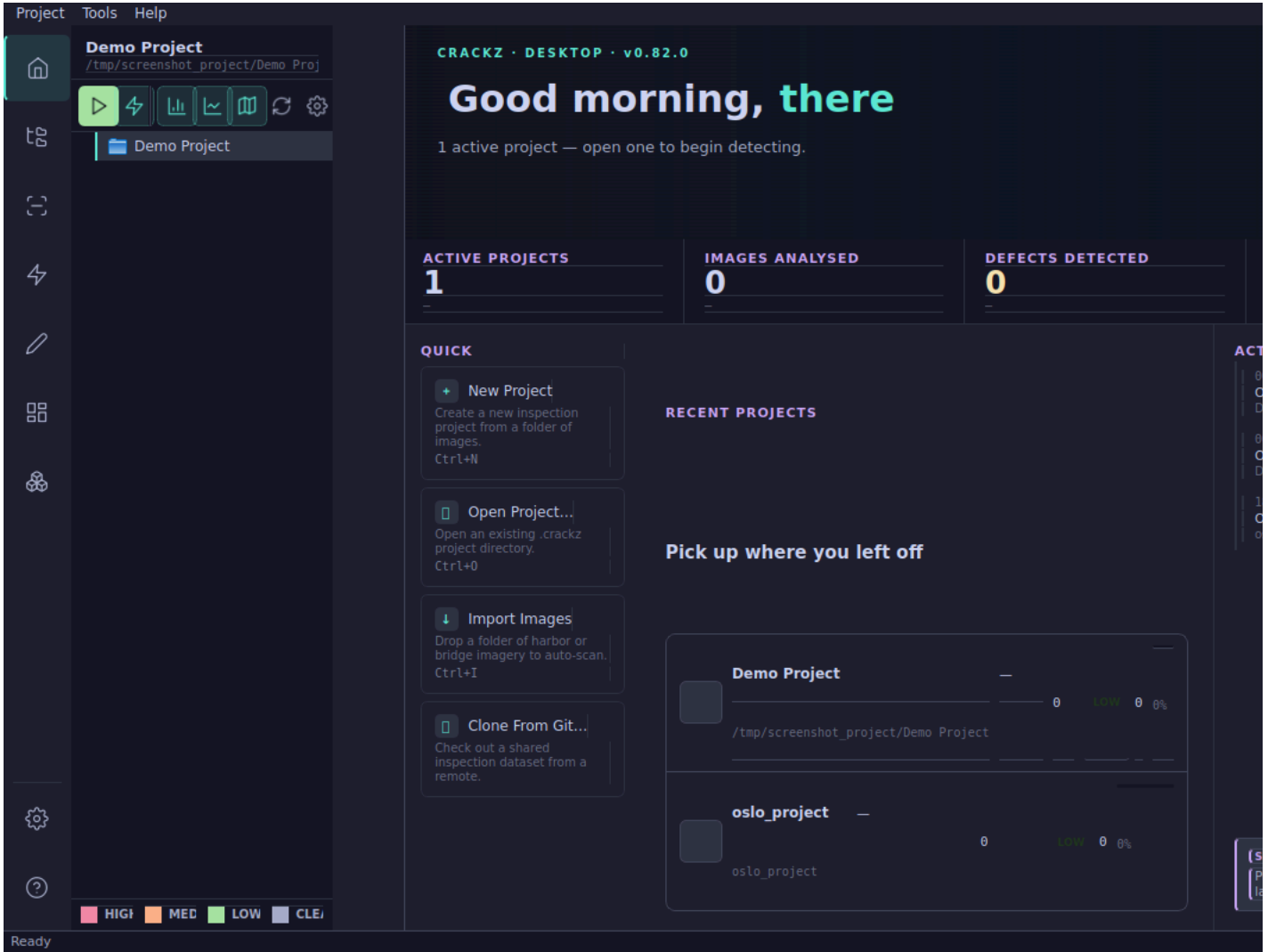
# Download and install Crackz
gh release download --repo anaxiatech/crackz --pattern '*.whl'
uv pip install crackz-*.whl

# Verify installation
crackz --version
```

```
# Pull the Docker image
docker pull ghcr.io/anaxiatech-ab/crackz:latest

# Verify it works
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

```
crackz-gui
```



- 1.
2. my-first-project
- 3.
- 4.
- 5.
- 6.
- 7.

- 1.
- 2.
- 3.
- 4.

IMPORT IMAGES WITH AUTOMATIC SPLIT

Select a source directory with images and optional masks.
They will be split into train / val / test sets based on the ratios below.

SOURCE DIRECTORY

Select a directory containing images/ and optionally masks/ subdirectories.
Masks will be automatically matched to images by filename.

No directory selected

Select Source Directory...

SPLIT RATIOS (AUTO-ADJUSTING)

Train:0.70

Validation:0.15

Test:0.15

Total: 1.00 ✓

RANDOM SEED (OPTIONAL)

Use random seed:

e.g. 42

IMAGE PROCESSING SETTINGS

Image size:512 x 512 pixels

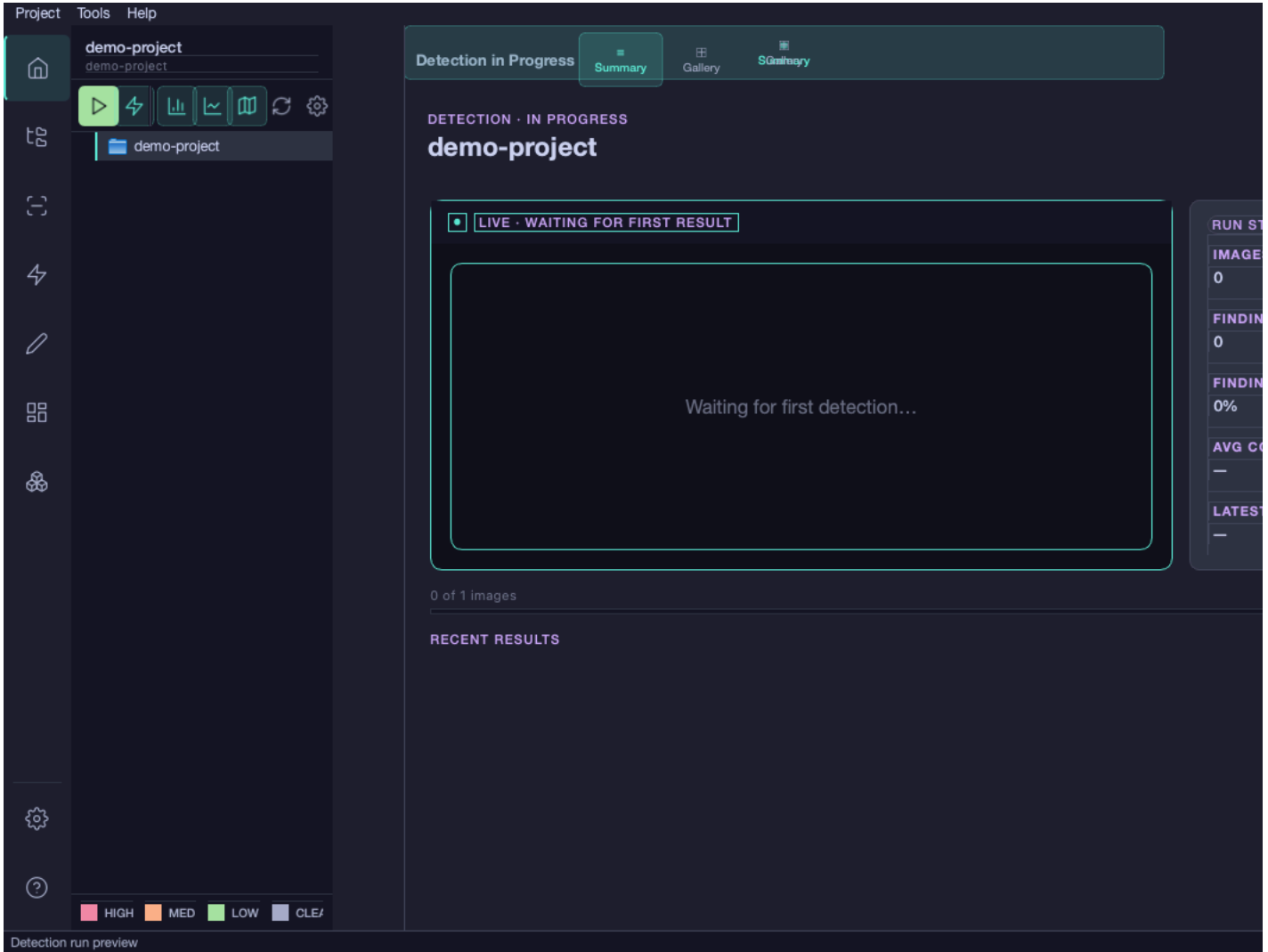
Mask label map:Optional, e.g. 255=1,128=2

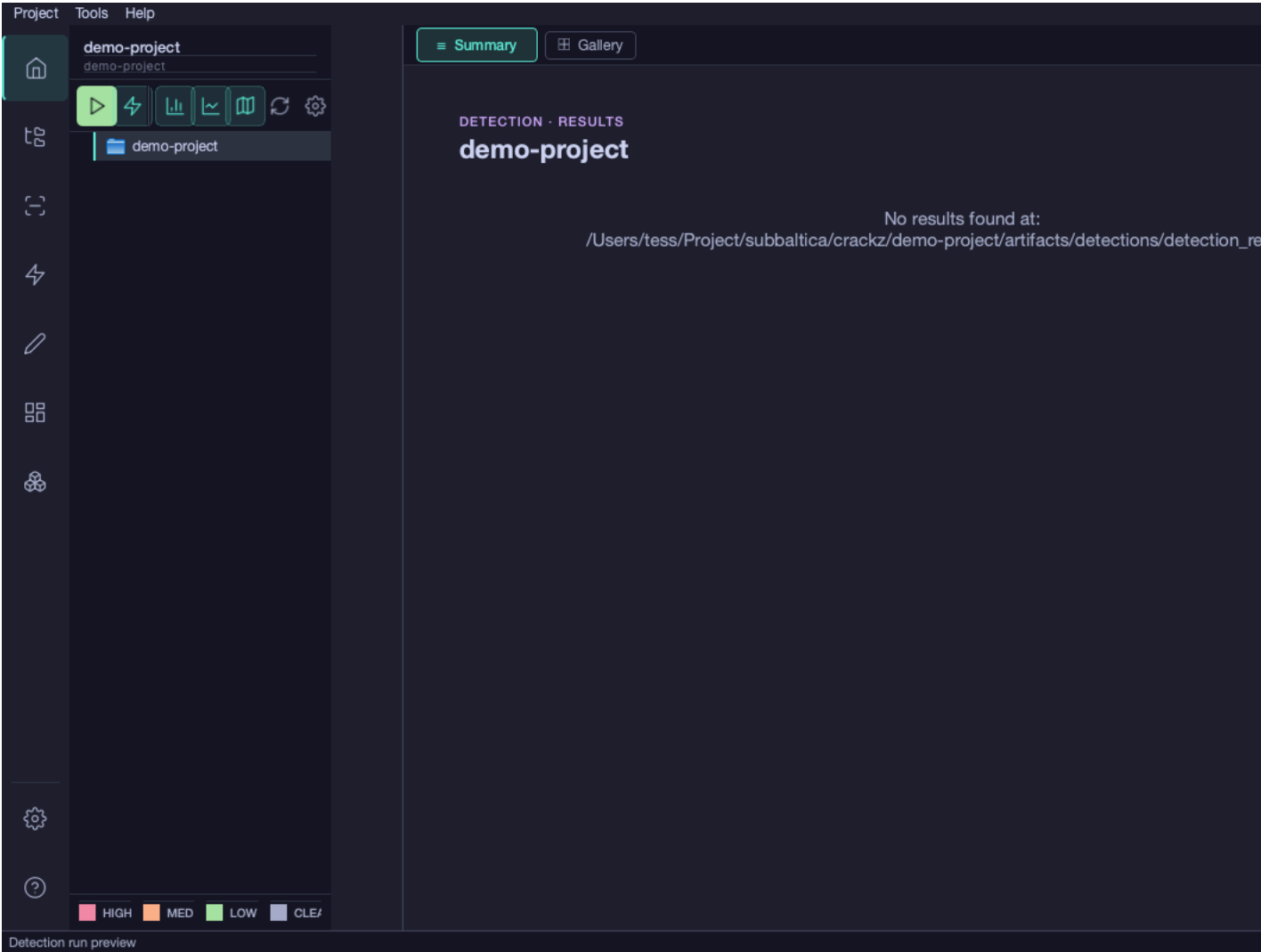
JPEG quality:90

Import

Cancel

- 1.
- 2.
- 3.

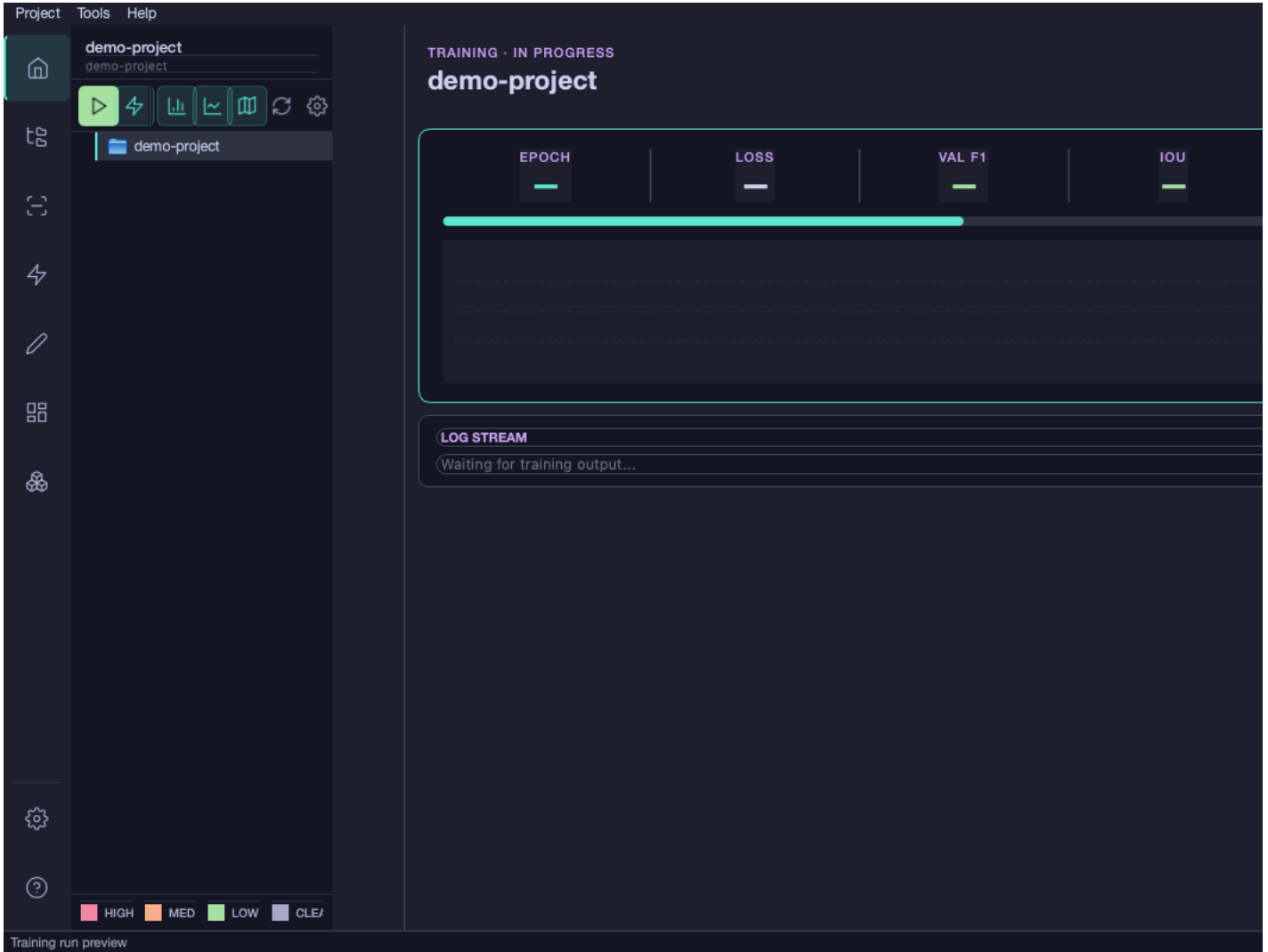




my-first-project/artifacts/detections/visualizations/
project/artifacts/detections/detection_results.json my-first-

- 1. raw
- 2.
- 3.
- 4.

- 1. train val
- 2.
- 3.
- 4.



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```
my-first-project/  
├── images/  
│   ├── raw/      # Original images for detection  
│   ├── train/    # Training images (if training a model)  
│   ├── val/      # Validation images  
│   └── test/     # Test images  
├── masks/  
│   ├── train/    # Ground truth masks for training  
│   ├── val/      # Validation masks  
│   └── test/     # Test masks  
├── artifacts/  
│   ├── models/   # Model checkpoints (best.ckpt, last.ckpt)  
│   └── runtime/  # Detection results (images + JSON), training metrics
```

```
└─ logs/           # Project-specific logs
└─ my-first-project_project.yaml # Configuration file (schema v6)
```

my-first-project/my-first-project_project.yaml

```
ai:
  model:
    threshold: 0.5 # Increase to 0.7 for fewer false positives
                  # Decrease to 0.3 to catch more cracks
```

```
# Process several inspection campaigns
for project in site-A site-B site-C; do
  crackz detect run --project $project
done
```

```
# Run detection in container
docker run --rm \
  -v $(pwd)/my-first-project:/home/app \
  ghcr.io/anaxiatech-ab/crackz:latest \
  detect run --project /home/app
```



```
# Verify CUDA installation
nvidia-smi

# Check PyTorch CUDA support
python -c "import torch; print(torch.cuda.is_available())"

# Reinstall with CUDA support (see Installation Guide)
```

images/raw/ .jpg .jpeg .png .bmp .tif .tiff
project.yaml

```
# Reduce batch size
crackz detect train --project my-project --batch-size 1

# Or reduce image size during import
crackz import-images --project my-project --src ./data --type train --size 256 256
```

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- _____
- _____
- _____



```
crackz init --name my-first-project
```

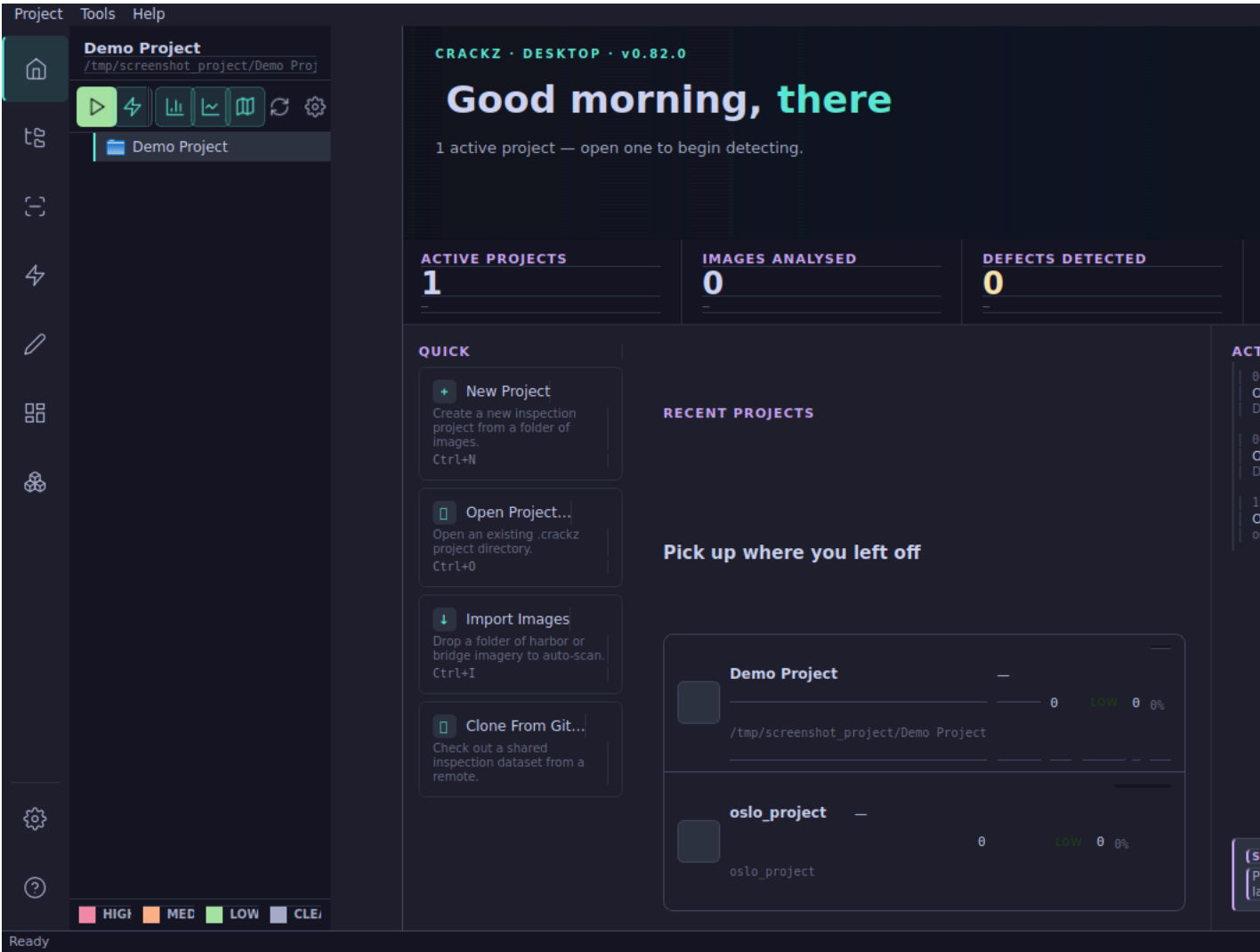
GUI Overview

GUI Guide

```
crackz-gui
```

```
# After installing Crackz
crackz-gui
```

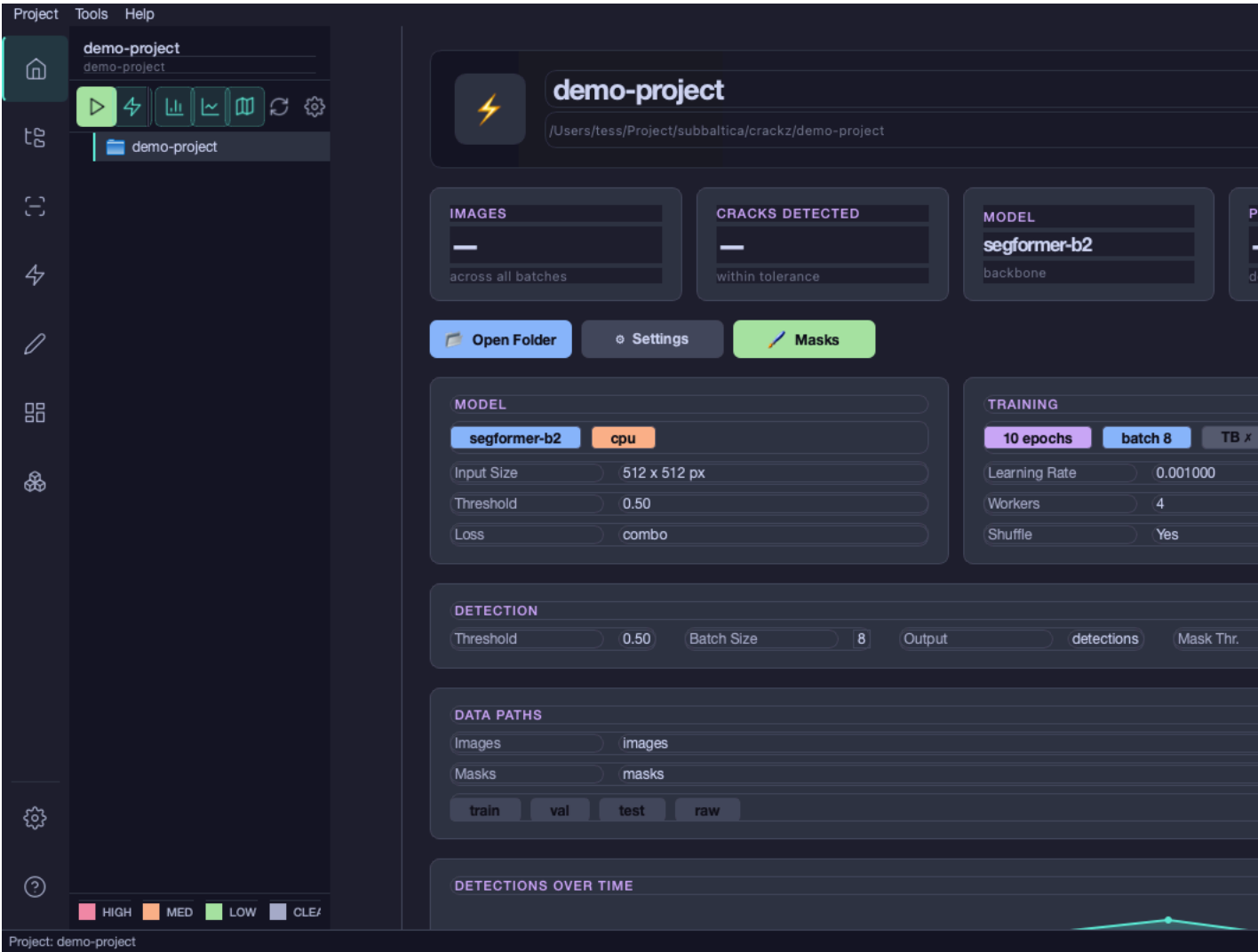
```
gui crackz-gui.exe ./crackz-
```



- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

- 1.
- 2.
- 3.

project.yaml



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-
-

Project

Paths

Logging

Data

Annotation

AI · Model

AI · Training

AI · ▶

Name:

demo-project

Description:

Reset

Save

Cancel

Project

Paths

Logging

Data

Annotation

AI · Model

AI · Training

AI · ▶

Mask Threshold (0-255):

50

Threshold used when importing or converting masks for crack-only projects.
Pixels >= threshold become foreground, others become background.
Multi-category projects preserve indexed label ids instead.

Reset

Save

Cancel

Project

Paths

Logging

Data

Annotation

AI · Model

AI · Training

AI · ▶

ANNOTATION CATEGORIES

Foreground categories are stored per project.
Background is fixed to label id 0, and foreground ids are derived from row order.
Background: id 0 · Foreground categories: 1

#1

crack

Crack

#ef4444

Remove

Add Category

Reset

Save

Cancel

Project Paths Logging Data Annotation **AI · Model** AI · Training AI · ▶

Default Backbone (new training runs): segformer-b2

Used as the default when starting a new training run. Existing trained models keep their own backbone.

Task Mode:

Blank = auto (1 category → binary, more → multiclass).
binary collapses all foreground categories into a single defect-vs-none class.
Switching invalidates checkpoints trained under the previous mode.

Device: cpu **Test**

Input Channels: 3

Num Classes: 1 (task mode: binary)

Checkpoint Path:

Threshold: 0.5

Norm Mean (comma-separated): 0.485,0.456,0.406

Norm Std (comma-separated): 0.229,0.224,0.225

Loss Function: combo

Dice Weight: 0.7

Focal Weight: 0.3

Reset **Save** **Cancel**

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-

Project Paths Logging Data Annotation AI · Model **AI · Training** AI · ▶

Batch Size: 8

Num Workers: 4

Epochs: 10

Learning Rate: 0.001

Shuffle: ☒

Enable TensorBoard: ☐

Loss Override (optional):

Metrics Override (optional):

Random Seed (optional):

Precision (optional):

Devices (optional):

Strategy (optional):

Save Top K Checkpoints: 1

Save Last Checkpoint: ☒

Reset **Save** **Cancel**

Project

Paths

Logging

Data

Annotation

AI · Model

AI · Training

AI · ▶

Project Path:

/Users/tess/Project/subbaltica/crackz/demo-project

Images Path:

/Users/tess/Project/subbaltica/crackz/demo-project/images

Masks Path:

/Users/tess/Project/subbaltica/crackz/demo-project/masks

Artifacts Path:

/Users/tess/Project/subbaltica/crackz/demo-project/artifacts

Checkpoints Path:

/Users/tess/Project/subbaltica/crackz/demo-project/artifacts/models

Reset

Save

Cancel

Project

Paths

Logging

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Annotation

AI · Model

AI · Training

AI · ▶

Log Level:

INFO

Log Dir:

logs

Reset

Save

Cancel

crack spall

#RRGGBB

• binary

- `multiclass` `1 + N`

- 1.
2. `masks/`
- 3.
- 4.
- 5.
- 6.

IMPORT IMAGES WITH AUTOMATIC SPLIT

Select a source directory with images and optional masks.
They will be split into train / val / test sets based on the ratios below.

SOURCE DIRECTORY

Select a directory containing images/ and optionally masks/ subdirectories.
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No directory selected

Select Source Directory...

SPLIT RATIOS (AUTO-ADJUSTING)

Train:0.70

Validation:0.15

Test:0.15

Total: 1.00 ✓

RANDOM SEED (OPTIONAL)

Use random seed:

e.g. 42

IMAGE PROCESSING SETTINGS

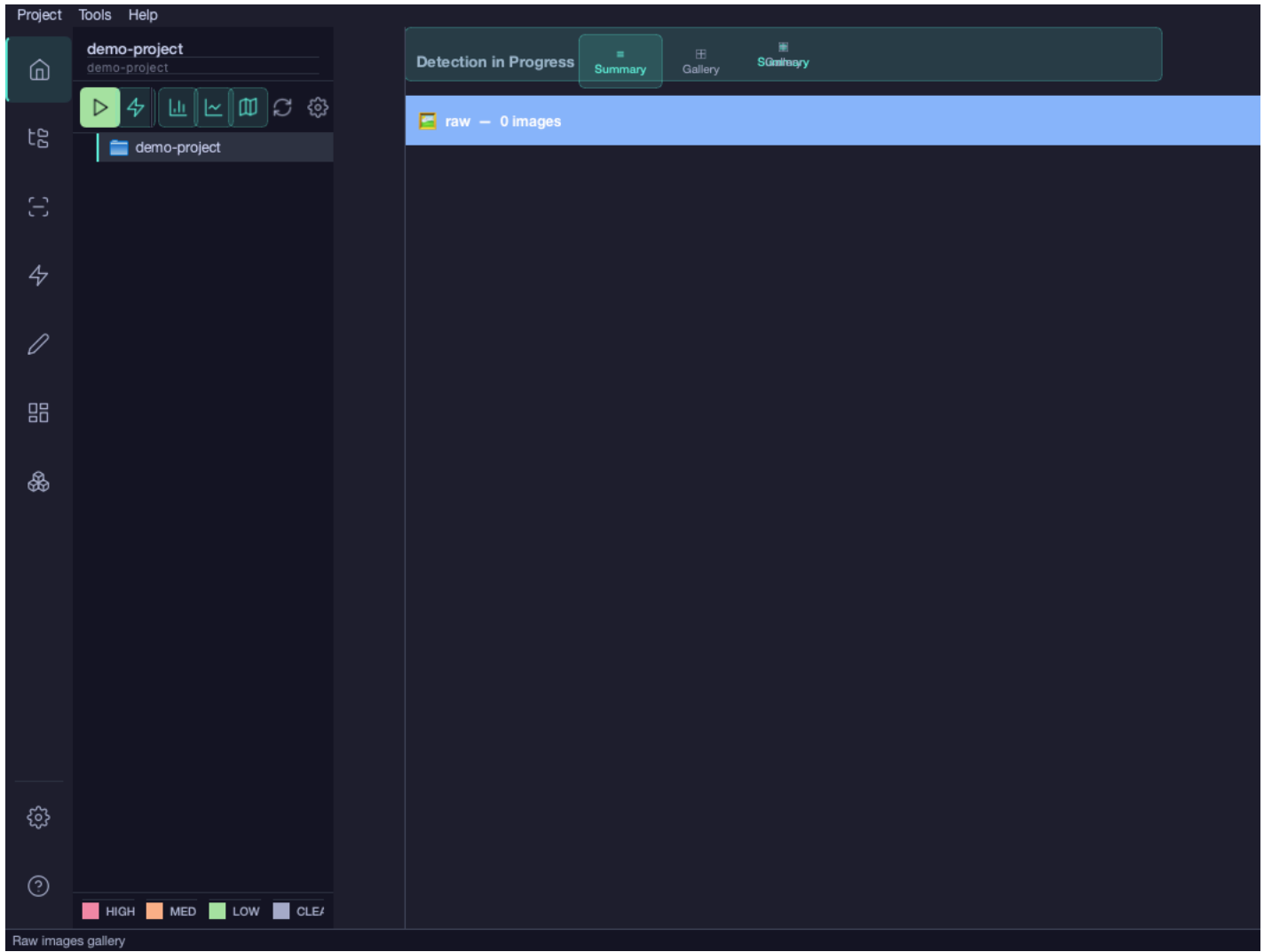
Image size:512 x 512 pixels

Mask label map:Optional, e.g. 255=1,128=2

JPEG quality:90

Import

Cancel



-

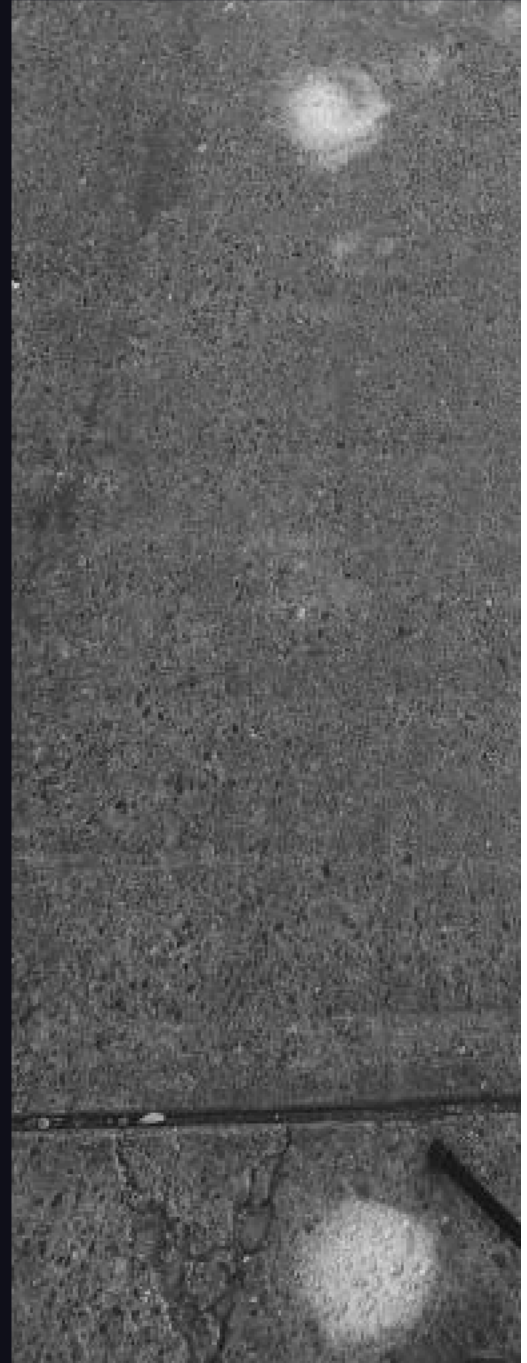
-

images/

images/

images/

MASK EDITOR · GARAYMC_CRACKS_TRAIN_0.JPG



masks/

-
-
-
-
-



B
E
Ctrl+Z Cmd+Z
Ctrl+Y Cmd+Shift+Z

Ctrl+S Cmd+S
Esc

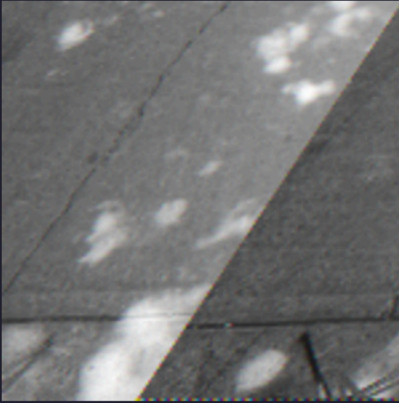
```
project/
├─ images/
│   └─ train/
│       └─ crack_001.jpg      # Original image
├─ masks/
│   └─ train/
│       └─ crack_001.png      # Indexed mask (auto-created)
```

train/ val/ test/ raw/

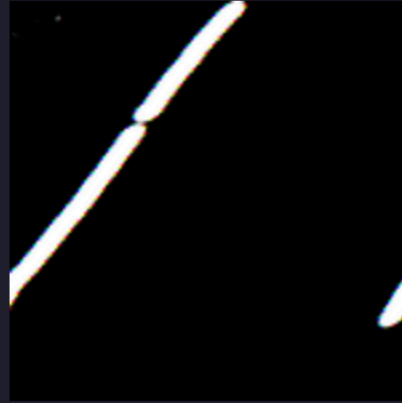
-
- 1.
 - 2.
 - 3.
 - 4.

Mask Propagation — GarayMC_cracks_train_0.jpg

Reference Image



Reference Mask



Similarity Threshold:

Min Matches:

Feature Detector:

SIFT

Find Similar Images

Similar Images (select to propagate)

Select All

Deselect All



- 1.
- 2.
- 3.
- 4.
- 5.

Page 1 / 1 | 2 masks | 0 accepted | 0 rejected

View: overlav ▼

GarayMC_cracks_train_0.jpg | conf 0.85

✓ Accept✗ Reject✎ Edit

GarayMC_cracks_train_2.jpg | c

✓ Accept✗ RejectPrev

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	A	
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	Right Arrow	
	Left Arrow	

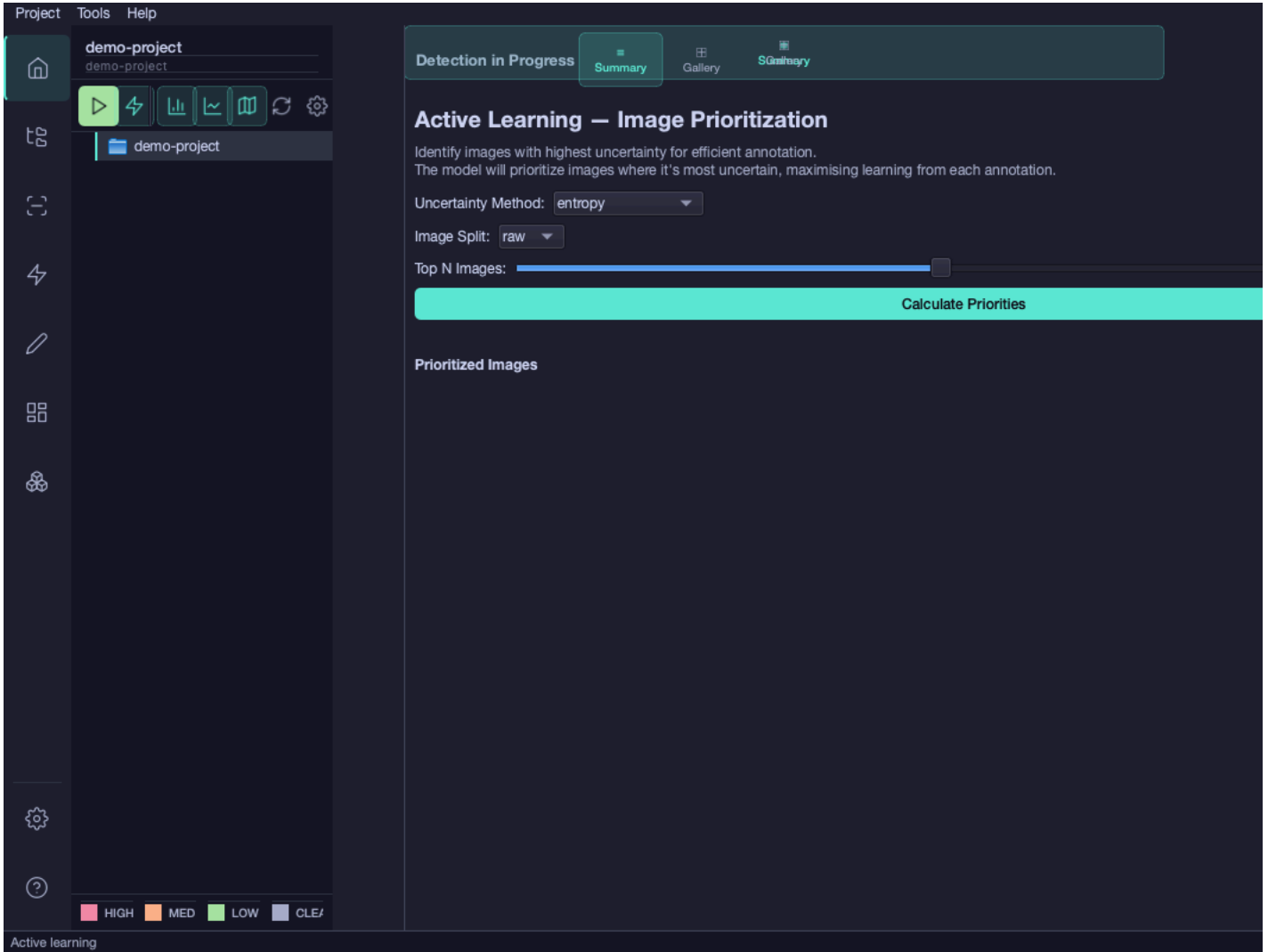
masks/

-
-
-

R

E

A



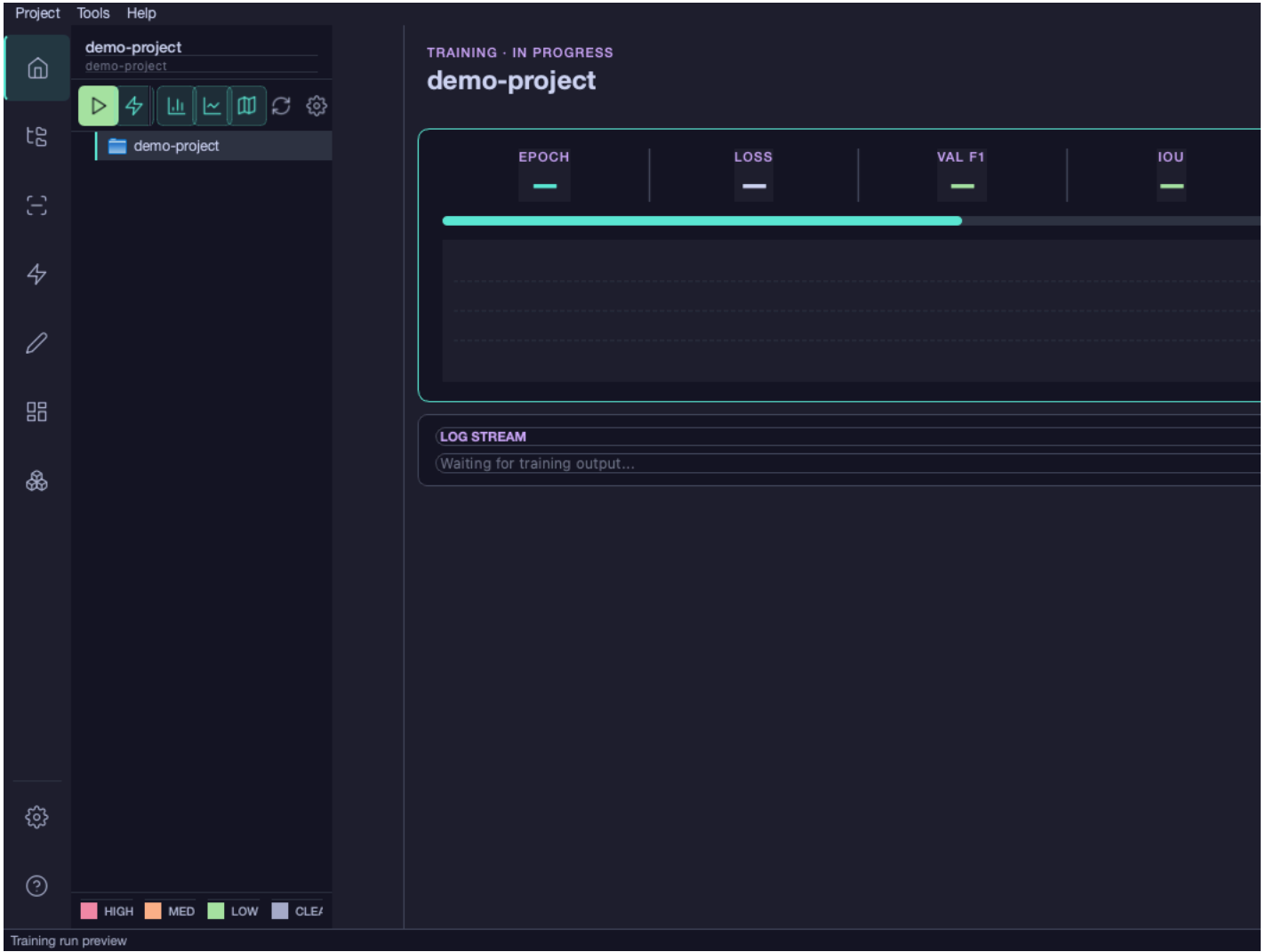
raw/

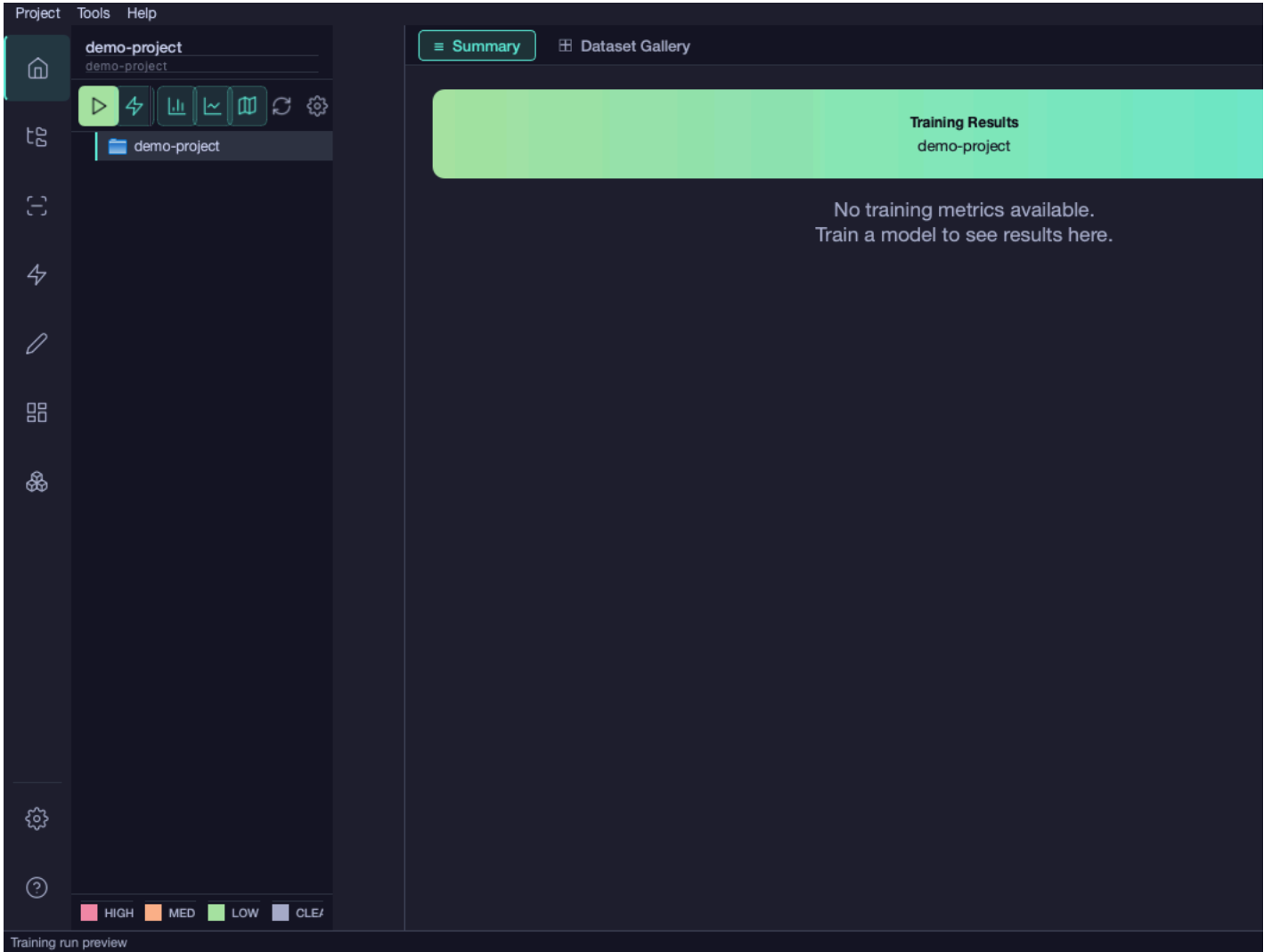
raw/

```
# Move annotated images from raw to train split
crackz move-images --project ./my-project --source raw --target train --pattern "crack_*.jpg"
```

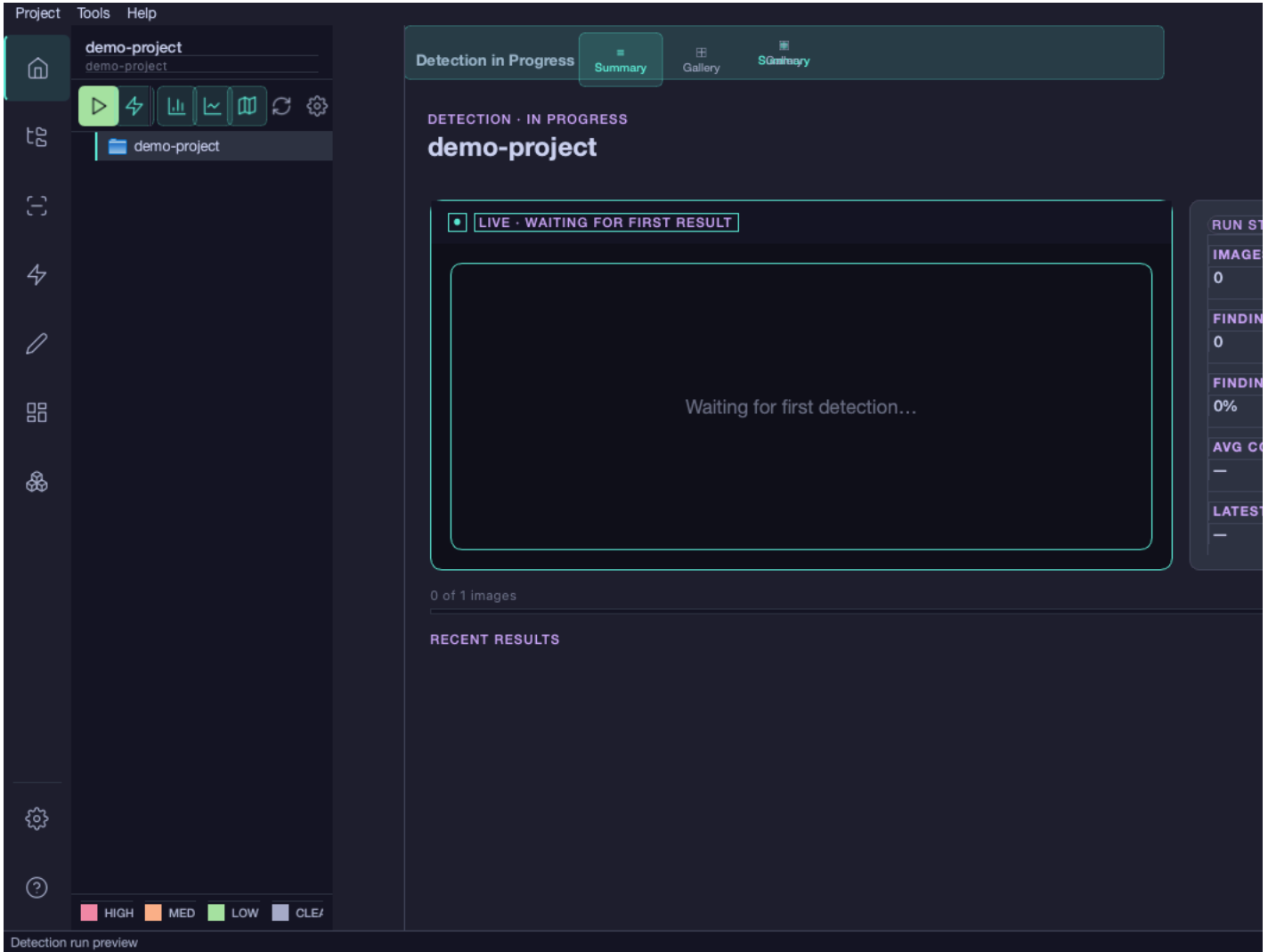
```
images/raw/
  train  val  test
```

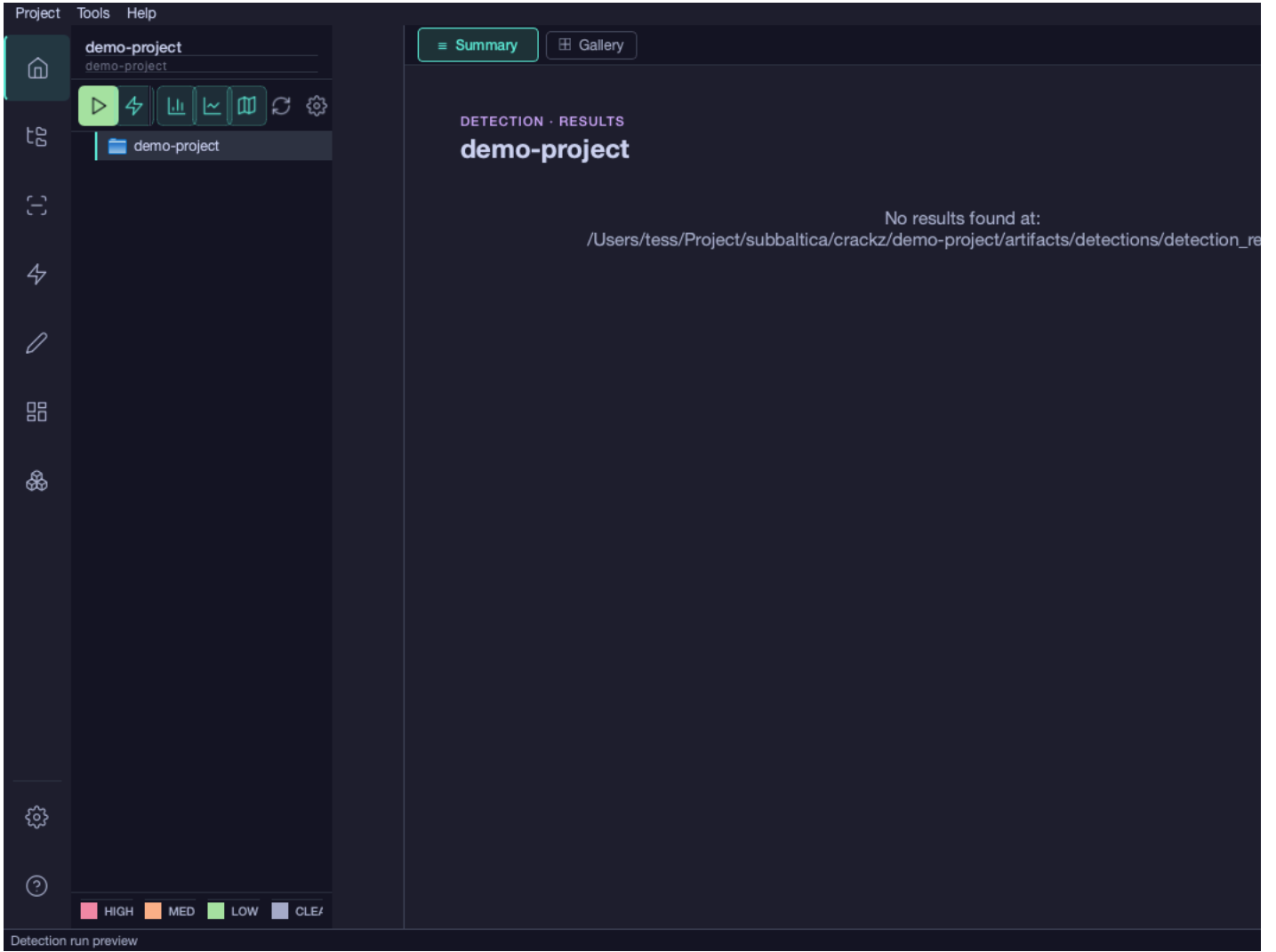
- 1.
- 2.
- 3.





1. raw
- 2.
- 3.
- 4.
- 5.





Project Tools Help

demo-project
demo-project

demo-project

0 of 0 - 0 flagged for review **All**

No detection results

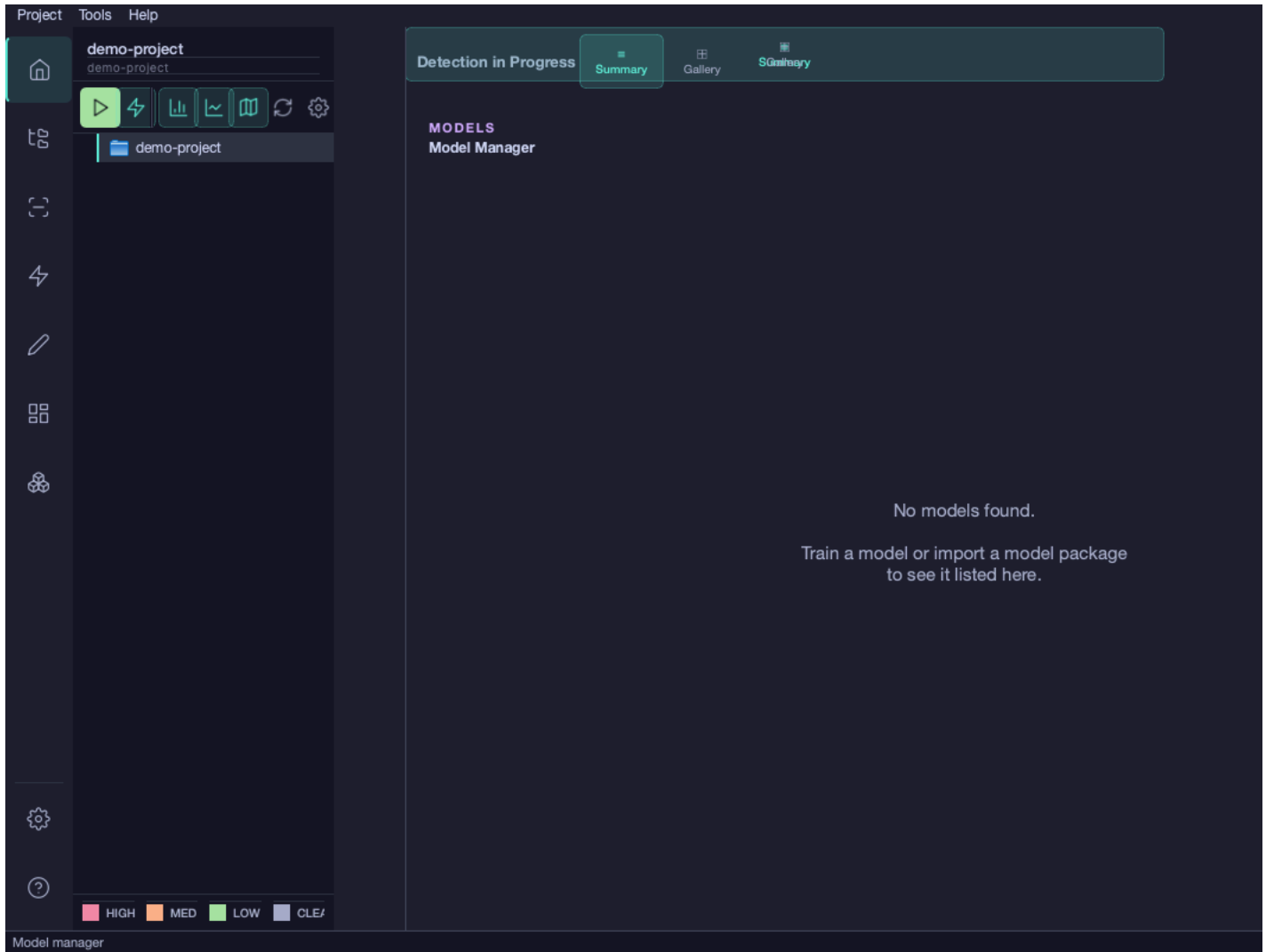
CONFIDENCE	AREA	SEVERITY
-	-	-

0 / 0

Open **✓ Approve** **Flag**

■ HIGH ■ MED ■ LOW ■ CLE/

Detection results — gallery



Binary Segmentation

Multiclass Segmentation

- 1.
- 2.
- 3.
- 4.

Export Model Package

Export a trained model checkpoint as a portable package.

Checkpoint:

File: best_model.ckpt
Size: 45.2 MB
Model: mit_b2
Classes: 1
Input Size: (512, 512)

Output file:

- 1.
2.
- 3.
- 4.

Import Model Package

Import a trained model checkpoint from a package file.

Package file:

Package Information:

Select a model package file to view details...

```
crackz clean --project ./my-project
```

```
project.yaml
```

The screenshot shows the Crackz application interface. On the left, there is a sidebar with a home icon, a folder icon, and a search icon. The main area is titled 'demo-project' and contains a 'Detection in Progress' button, a 'Summary' button, a 'Gallery' button, and a 'Screenshot' button. Below these buttons is a 'Text preview' section showing a log file named 'crackz.log'. The log contains various system messages, including information about project saving, image import, and detection progress. A legend at the bottom left indicates log levels: HIGH (red), MED (orange), LOW (green), and CLE/ (blue).

Project Tools Help

demo-project

demo-project

Detection in Progress Summary Gallery Screenshot

← Back Text preview

crackz.log

```
2026-05-06 10:21:46.797 | INFO | crackz.project.project:save_project:419 - Project 'demo-project' saved successfully to /U
2026-05-06 10:21:48.015 | INFO | crackz.project.project:save_project:419 - Project 'demo-project' saved successfully to /U
2026-05-06 10:21:48.016 | INFO | crackz.cli.image_import_cli:import_images:265 - Set target_size=512 and input_size=
2026-05-06 10:21:48.018 | INFO | crackz.project.image_operations:_import_with_split:363 - Splitting 4 images into train
2026-05-06 10:21:48.018 | INFO | crackz.project.image_operations:_import_with_split:375 - Using random seed 42 for r
2026-05-06 10:21:48.019 | INFO | crackz.project.image_operations:_import_with_split:390 - Split sizes - train: 2, val: 1,
2026-05-06 10:21:48.019 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 2 train images → /U
2026-05-06 10:21:48.049 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 2 train masks → /U
2026-05-06 10:21:48.053 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 1 val images → /U
2026-05-06 10:21:48.064 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 1 val masks → /U
2026-05-06 10:21:48.066 | INFO | crackz.project.image_operations:_import_with_split:412 - Importing 1 test images → /U
2026-05-06 10:21:48.076 | INFO | crackz.project.image_operations:_import_with_split:435 - Importing 1 test masks → /U
2026-05-06 10:21:48.080 | INFO | crackz.project.image_operations:_import_with_split:450 - Completed automatic split
2026-05-06 10:21:49.397 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in /U
2026-05-06 10:21:53.939 | INFO | crackz.cli.detection_cli:callback:248 - Starting detection
2026-05-06 10:21:53.941 | INFO | crackz.cli.detection_cli:callback:248 - No trained model found. Detection skipped. ✖
2026-05-06 10:21:53.942 | WARNING | crackz.ai.detection.detection_runner:detect:538 - Detection skipped: no trained m
2026-05-06 10:21:53.943 | INFO | crackz.cli.detection_cli:_do_detect:261 - ✔ Detection finished
2026-05-06 10:22:18.660 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in /U
2026-05-06 10:22:18.790 | INFO | crackz_gui.qt.util.screenshot_capture:capture_doc_screenshots:398 - Loaded project
2026-05-06 10:22:19.257 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:19.259 | ERROR | crackz.ai.model.checkpoints:select_checkpoint:109 - No checkpoint files found in /U
2026-05-06 10:22:19.633 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:19.980 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:20.323 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:20.683 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:21.065 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:21.426 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:21.787 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:22.934 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
2026-05-06 10:22:23.308 | INFO | crackz_gui.qt.util.screenshot_capture:_save_widget_screenshot:373 - Saved screens
```

Legend: HIGH (red), MED (orange), LOW (green), CLE/ (blue)

Logs viewer

- 1.
- 2.
- 3.

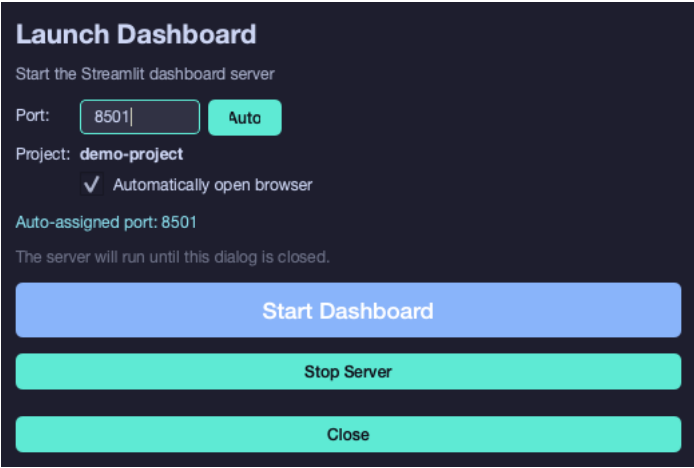
The dialog box has a title bar with 'Data split: raw' and a dropdown menu. Below the title bar are two buttons: 'Refresh' and 'Open in Browser'. The main text area contains the message: 'No GPS data found for split 'raw'. Images must contain EXIF GPS metadata.' At the bottom right is a 'Close' button.

Data split: raw Refresh Open in Browser

No GPS data found for split 'raw'.
Images must contain EXIF GPS metadata.

Close

- 1.
- 2.
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Cmd/Ctrl+,

APPEARANCE

Display name:

GENERAL

Theme:

Log level:

Open last project on start:

Remember last project: ☒

Save

Cancel

dark

light

system

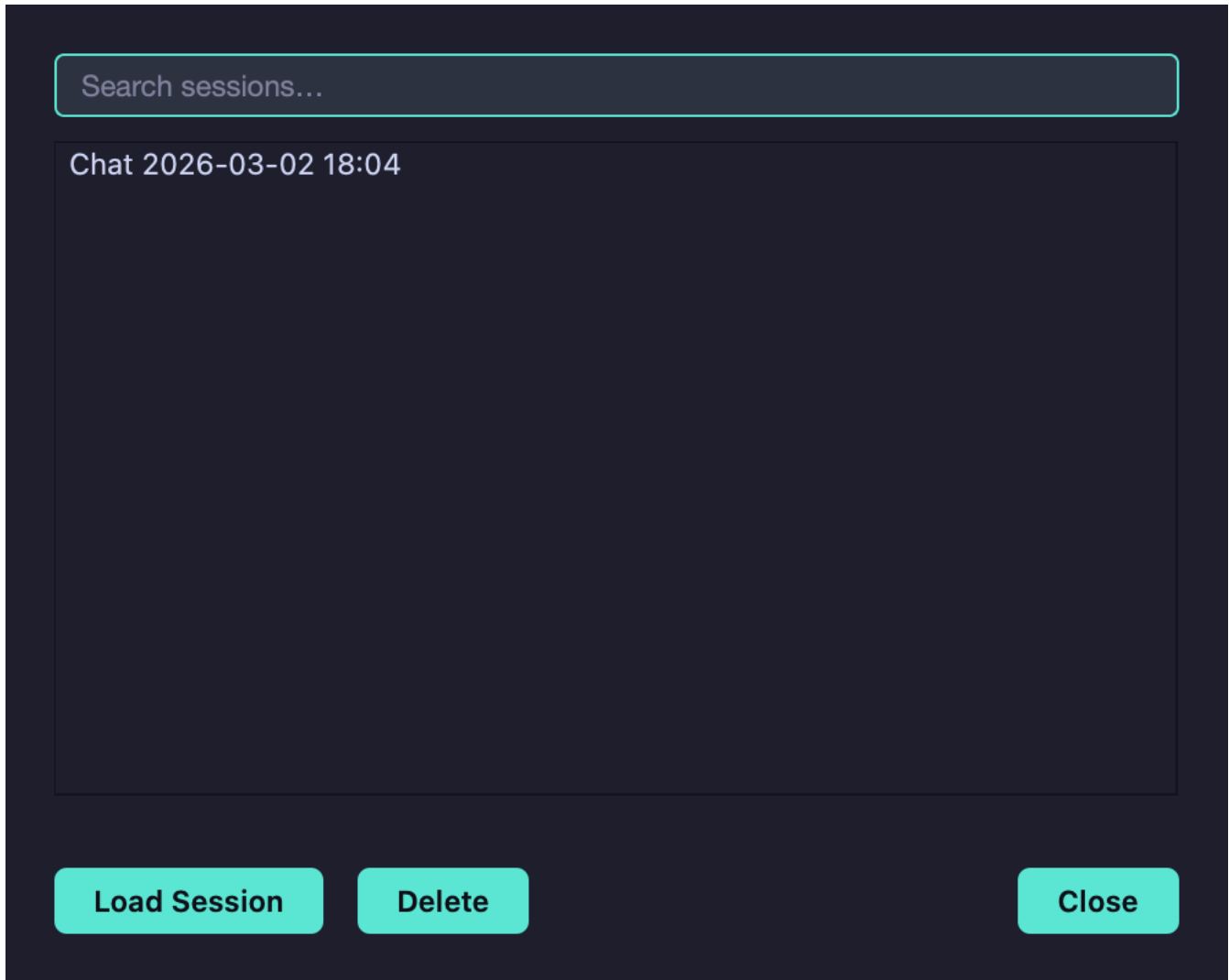
\$

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§ ~ Cmd/Ctrl+/

Category: All ▼

Search...

Analyze Detection Results
Parameter Tuning Help
Project Health Check
Training Recommendations

Preview:

Use Template

New...

Delete

```
~/.crackz/chat_history.db
```

```
# Check installation
python -c "from PySide6.QtWidgets import QApplication; print('PySide6 OK')"
```

```
# Launch with debug output
crackz-gui --debug
```

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CRACKZ_LOG_DIR/gui_app.log

<project>/logs/gui_operations.log

```
# Create and configure in GUI
crackz-gui
```

```
# Batch process via CLI
crackz detect run --project my-project --batch-size 16
```

```
# Review results in GUI
crackz-gui
```

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AI Chat Assistant

AI Chat Assistant

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```
# Install keyring support
uv add keyring

# Or with pip
pip install keyring
```

```
from crackz_gui.state.secure_storage import (
    store_api_key,
    retrieve_api_key,
    is_secure_storage_available,
)
```

```
# Check if keyring is available
if is_secure_storage_available():
    # Store API key securely
    store_api_key("your-api-key-here")

    # Retrieve API key
    api_key = retrieve_api_key()

    # Delete API key
    from crackz_gui.state.secure_storage import delete_api_key
    delete_api_key()
else:
    print("Keyring not available - falling back to plaintext storage")
```

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- 1.
2. \$
3. ~
4. Ctrl+ / Cmd+ /
- 5.
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 - o sk-ant-... sk-or-...
- 10.
- 11.



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- 4.

- \$
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- Ctrl+ / Cmd+ /

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You: My detection is finding too many false positives. How can I reduce them?

AI: Based on your project, I recommend:

1. Increase confidence_threshold from 0.25 to 0.4
2. Adjust iou_threshold to 0.5 for stricter overlap filtering
3. Consider retraining with more negative examples

Would you like me to explain how to adjust these in your project?

You: How many epochs should I train for?

AI: For your dataset with 150 training images:

- Start with 50 epochs as a baseline
- Monitor validation loss - it should decrease steadily
- If validation loss plateaus before 50 epochs, you can stop early
- If still improving at 50, try 75-100 epochs

Your current batch_size of 8 is good for this dataset size.

You: Analyze my project configuration

AI: Here's what I found in your oslo_project:

- Good splits: 70/20/10 train/val/test distribution
- △ Low image count (150 total) - consider data augmentation
- Reasonable batch size (8) for dataset
- Suggestion: Enable data augmentation in training config
- Suggestion: Consider collecting more diverse training data

You: How do I export detection results to CSV?

AI: To export detection results:

1. Run detection: ``crackz detect run <project>``
2. Results are saved in `project/artifacts/detections/`
3. Use export command: ``crackz export --format csv <project>``

Or in GUI:

1. Click "Run Detection" button
2. Results appear in Detection Results panel
3. Click "Export" → "CSV Format"

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5-20251001 claude-opus-4-1-20250805

claude-sonnet-4-5-20250929 claude-haiku-4-

§ ~ Ctrl+ /

- 1.
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- 6.
- 7.

`chat_window.py`

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`ai_chat.py`

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`chat_settings_dialog.py`

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1. `ai_chat.py` `_get_context()`

```
def _get_context(self) -> str:  
    if self.app_state is None or self.app_state.project is None:
```

```
        return "No project loaded"

    project = self.app_state.project

    # Add your new context here
    return f"""
    Current project: {project.name}

    Your new context:
    - Additional info: {project.additional_info}
    """
```

2.

```
uv run task tests
```

- _____
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Dashboard

Streamlit Dashboard

app

 Detection Results

 GPS Map

 Training Metrics

 Parameter Tuning

Project Selection

Project Path

demo-project

✓ Loaded: demo-project

Navigation

Use the sidebar to navigate between:

-  **Detection Results** - View detections and findings
-  **GPS Map** - Geographic visualization
-  **Training Metrics** - Compare training runs
-  **Parameter Tuning** - Adjust detection settings

Crackz - Detection Dashboard

AI-powered analysis for harbor and bridge inspection

Project Overview

Project Name

demo-project

Crackz Version

0.81.0



Use the **Pages** menu in the sidebar to navigate to different views

app

 **Detection Results**

 GPS Map

 Training Metrics

 Parameter Tuning



Detection Results

View detection results with visualizations and statistics



No project loaded. Please select a project from the home page.

app

 Detection Results **GPS Map** Training Metrics Parameter Tuning

GPS Map Visualization

View detections and findings on an interactive map



No project loaded. Please select a project from the home page.

app

 Detection Results

 GPS Map

 **Training Metrics**

 Parameter Tuning



Training Metrics

Monitor training progress and model performance



No project loaded. Please select a project from the home page.

app

 Detection Results

 GPS Map

 Training Metrics

 **Parameter Tuning**



Parameter Tuning

Interactively adjust detection parameters and preview results



No project loaded. Please select a project from the home page.

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```
# Launch with a specific project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8502
```

1. `crackz-gui`
- 2.
- 3.
- 4.
- 5.
- 6.

```
streamlit run core/src/crackz/dashboard/app.py -- --project my-project
```

```
from crackz.dashboard import launch_dashboard
launch_dashboard(project_name="my-project", port=8501)
```

```
crackz geo extract-gps --project <name>
```

```
artifacts/training/training_metrics.json
```

```
crackz train --project <name>
```

```
core/src/crackz/dashboard/
├── init .py          # Launch function
├── app.py            # Main dashboard app
├── pages/           # Dashboard pages
│   ├── 1_0_Detection_Results.py
│   ├── 2_0_GPS_Map.py
│   ├── 3_0_Training_Metrics.py
│   └── 4_0_Parameter_Tuning.py
├── utils/           # Utility modules
│   ├── data_loader.py # Data loading functions
│   ├── image_utils.py # Image processing utilities
│   ├── map_utils.py   # GPS mapping utilities
│   └── viz_utils.py    # Visualization functions
```

```
artifacts/detections/visualizations/
artifacts/detections/detection_results.json
artifacts/training/training_metrics.json
data/{test,val,train,raw}/
```

```
crackz dashboard --project my-project --port 8080
```

```
.streamlit/config.toml
```

```
[server]
port = 8501
headless = true

[theme]
primaryColor = "#FF4B4B"
backgroundColor = "#FFFFFF"
secondaryBackgroundColor = "#F0F2F6"
```

```
textColor = "#262730"  
font = "sans serif"
```

- streamlit>=1.30.0
- streamlit-folium>=0.15.0
- folium>=0.15.0
- plotly>=5.18.0
- pandas>=2.0.0

1.

```
crackz detect --project my-project
```

2.

3.

1.

```
crackz geo extract-gps --project my-project
```

2.

3.

1.

2.

3.

1.

2.

3.

8502

```
uv sync
```

```
pip install streamlit
```

```
--port
```

```
crackz detect --project <name>
```

```
artifacts/detections/visualizations/
```

```
crackz geo extract-gps
```

```
crackz train --project <name>
```

```
artifacts/training/training_metrics.json
```

- 1.
- 2.
- 3.
- 4.

```
# 1. Initialize project
crackz init my-project

# 2. Import images
crackz import-images --source ./photos --project my-project

# 3. Train model
crackz train --project my-project

# 4. Run detection
crackz detect --project my-project

# 5. Launch dashboard to view results
crackz dashboard --project my-project
```

```
from crackz.dashboard import launch_dashboard

# Launch dashboard programmatically
launch_dashboard(
    project_name="my-project", # Optional: project to load
    port=8501                 # Optional: server port
)
```

```
# Show help
crackz dashboard --help

# Launch with project
crackz dashboard --project my-project

# Launch with custom port
crackz dashboard --project my-project --port 8080

# Launch without initial project
crackz dashboard
```

```
# 1. Create and setup project
crackz init harbor-inspection
crackz import-images --source ./drone-photos --project harbor-inspection

# 2. Train model
crackz train --project harbor-inspection --epochs 50

# 3. View training progress
crackz dashboard --project harbor-inspection
# Navigate to "Training Metrics" page to review

# 4. Run detection on test set
crackz detect --project harbor-inspection --split test

# 5. Analyze results
crackz dashboard --project harbor-inspection
# Navigate to "Detection Results" page
# Navigate to "GPS Map" page to see locations

# 6. Tune parameters if needed
# Use "Parameter Tuning" page to optimize thresholds

# 7. Re-run with optimized parameters
crackz detect --project harbor-inspection --threshold 0.6

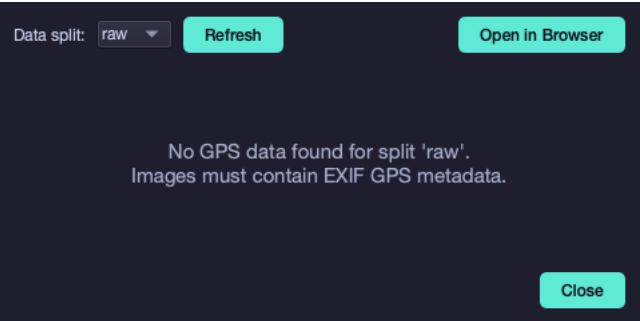
# 8. Generate final report
crackz report generate --project harbor-inspection --format pdf
```

- _____
- _____
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1. _____
2. _____
3. _____ dashboard

Map Viewer

Map Viewer Feature



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- `gui/src/crackz_gui/components/map_viewer.py`
-
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-
- `webbrowser.open()`
-

- 1.
- 2.
- 3.
- 4.
- 5.

```
from pathlib import Path
from crackz_gui.qt.views.map_viewer_view import MapViewerDialog
from crackz.project.project import load_project

# Load a project
project = load_project(Path("path/to/project"))

# Note: MapViewerDialog requires a parent GUI app instance and is typically
# used within the Crackz GUI application. For command-line access to GPS data, use:
from crackz.project.geoposition import get_project_geopositions

# Get GPS data for all images in the project
gps_data = get_project_geopositions(project)
for image_path, geo_pos in gps_data.items():
    print(f"{image_path}: {geo_pos.latitude}, {geo_pos.longitude}")
```

-
-
-

-
-
-

-
-
-

tests/test_data/ oslo_project/

- tests/test_data/images/
- oslo_project/images/

scripts/add_test_gps_data.py

```
python scripts/add_test_gps_data.py
```

- 1.
2. `logs/crackz.log`
- 3.

-
- `crackz.project.geoposition.get_project_geopositions()`
- `python scripts/add_test_gps_data.py`

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- `gui/src/crackz_gui/components/map_viewer.py`
- `gui/src/crackz_gui/components/browser/toolbar.py`
- `core/src/crackz/project/geoposition.py`
- `scripts/add_test_gps_data.py`

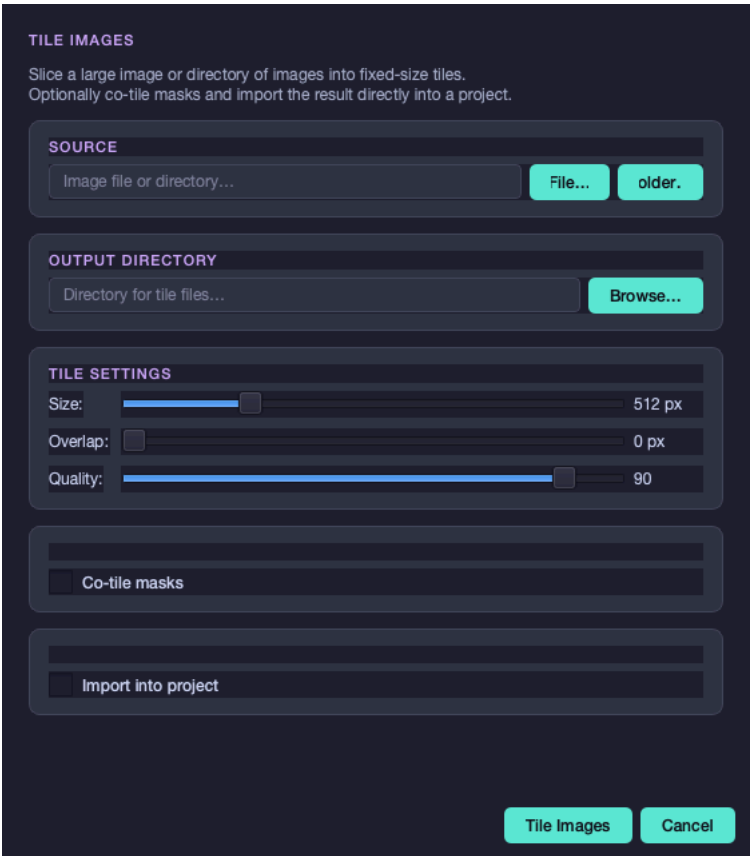
Image Tiling

Image Tiling

crackz tile-image

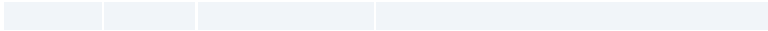
-
-
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-

```
uv run python scripts/capture_qt_screenshots.py \
--scenes tile-images-dialog,tile-images-dialog-masks,tile-images-dialog-import
```



-
-

2048 × 2048 px



≈ 16 tiles · 4 × 4 grid · 512 × 512 px each

size = 512

overlap = 64

TILE IMAGES

Slice a large image or directory of images into fixed-size tiles.
Optionally co-tile masks and import the result directly into a project.

SOURCE
Image file or directory... File... older.

OUTPUT DIRECTORY
Directory for tile files... Browse...

TILE SETTINGS
Size: 512 px
Overlap: 0 px
Quality: 90

☒ Co-tile masks
Masks source: Mask file or directory... File... older.
Masks output: Output directory for mask tiles... Browse...

Tile Images Cancel



TILE IMAGES

Slice a large image or directory of images into fixed-size tiles.
Optionally co-tile masks and import the result directly into a project.

SOURCE
Image file or directory... File... older.

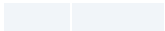
OUTPUT DIRECTORY
Directory for tile files... Browse...

TILE SETTINGS
Size: 512 px
Overlap: ☐ 0 px
Quality: 90

☐ Co-tile masks

☒ Import into project
Image type: Raw
☒ Auto-split into train / val / test

Tile Images Cancel



```
crackz tile-image --src <source> --out <output> [OPTIONS]
```

```
crackz tile-image --src mosaic.jpg --out tiles/
```

```
crackz tile-image --src raw_images/ --out tiles/ --tile-size 256 --overlap 32
```

```
crackz tile-image \
--src raw_images/ \
--out tiles/images/ \
--masks-src annotations/ \
--masks-out tiles/masks/ \
--tile-size 512
```

```
crackz tile-image \
--src mosaic.jpg \
--out tiles/ \
--project ~/projects/harbor_survey \
--type raw
```

```
crackz tile-image \
--src raw_images/ \
--out tiles/ \
--masks-src annotations/ \
--masks-out tiles/masks/ \
--project ~/projects/harbor_survey \
--split \
--train-ratio 0.7 \
--val-ratio 0.15 \
--test-ratio 0.15 \
--random-seed 42
```

<code>--src</code>	<code>-s</code>				
<code>--out</code>	<code>-o</code>				
<code>--masks-src</code>	<code>-ms</code>				
<code>--masks-out</code>	<code>-mo</code>				
<code>--tile-size</code>	<code>-z</code>	512			
<code>--overlap</code>		0			
<code>--quality</code>	<code>-q</code>	90			
<code>--project</code>	<code>-p</code>				
<code>--cfg</code>	<code>-c</code>			<code>--project</code>	
<code>--type</code>	<code>-t</code>	raw		raw	train val test
<code>--split</code>					
<code>--train-ratio</code>		0.7		<code>--split</code>	
<code>--val-ratio</code>		0.15		<code>--split</code>	
<code>--test-ratio</code>		0.15		<code>--split</code>	
<code>--random-seed</code>					

```
{stem}_{row:04d}_{col:04d}.{ext}
```

```
mosaic.jpg
```

```
mosaic_0000_0000.jpg mosaic_0000_0001.jpg
mosaic_0001_0000.jpg mosaic_0001_0001.jpg
```

```
.png
```

```
mosaic_0000_0000.png mosaic_0000_0001.png
mosaic_0001_0000.png mosaic_0001_0001.png
```

```
step = max(tile_size - overlap, 1)
cols = max(ceil((width - overlap) / step), 1)
```

```
rows = max(ceil((height - overlap) / step), 1)
n_tiles = cols × rows
```

```
step = 512 - 64 = 448
cols = ceil((4096 - 64) / 448) = ceil(9.0) = 9
rows = 9
n_tiles = 81
```

.jpg	.jpeg	.png	.tif	.tiff	.bmp	.webp		.jpg
	.png	.jpg	.jpeg	.tif	.tiff			.png

crackz.project.tiling

```
from crackz.project.tiling import tile_image, tile_image_with_mask, tile_image_dir

# Tile a single image
tiles = tile_image(
    source=Path("mosaic.jpg"),
    dest_dir=Path("tiles/"),
    tile_size=(512, 512),
    overlap=0,
    quality=90,
)

# Tile image + mask together
img_tiles, msk_tiles = tile_image_with_mask(
    source=Path("mosaic.jpg"),
    mask=Path("mask.png"),
    dest_dir=Path("tiles/images/"),
    mask_dest_dir=Path("tiles/masks/"),
    tile_size=(512, 512),
    overlap=32,
    quality=90,
)

# Batch-tile a directory
img_tiles, msk_tiles = tile_image_dir(
    source_dir=Path("raw_images/"),
    dest_dir=Path("tiles/images/"),
    tile_size=(512, 512),
    overlap=0,
    quality=90,
    mask_source_dir=Path("annotations/"), # optional
    mask_dest_dir=Path("tiles/masks/"), # optional
)
```

Report Generation

Report Generation

-
-
-
-
-

```
crackz report generate --project demo-project
demo-project_report.docx
```

```
crackz report generate --project demo-project --output inspection_report.docx
```

```
crackz report generate --cfg project_config.yaml --output report.docx
```

-
-
-
-

-
-
-

```
ai:
  detection:
    min_confidence: 0.005    # 0.5% - Possible Crack threshold
    medium_confidence: 0.02  # 2% - Probable Crack threshold
    high_confidence: 0.05    # 5% - Crack Detected threshold
```

detections/

```
crackz report generate --project harbor_inspection_2024 \
  --output "Harbor_Inspection_Report_2024-Q4.docx"
```

```
# Monthly reports
crackz report generate --project monthly_monitoring \
  --output "reports/2024-11_monitoring.docx"
```

```
crackz report generate --project compliance_check \
  --output "compliance/inspection_$(date +%Y%m%d).docx"
```

```
crackz report generate --project qa_validation \
  --output "internal/qa_review.docx"
```

-
-
-
-

- `crackz detect run`
-
-

```
crackz detect run --project demo-project
```

```
# Check for visualization images (new structure)
ls -la demo-project/artifacts/detections/visualizations/

# Or legacy location (older projects)
ls -la demo-project/artifacts/detections/*.png
```

```
crackz detect run --project demo-project
```

```
artifacts/detections/visualizations/
```

```
data:
size: [512, 512] # Use smaller images
```

```
--output
```

```
crackz report generate --project harbor_main \  
--output "Harbor_Main_Inspection_2024-11-05.docx"
```

```
for project in project1 project2 project3; do  
  crackz report generate --project "$project" \  
  --output "reports/${project}_${date +%Y%m%d}.docx"  
done
```

```
crackz report generate --project demo-project # Creates demo-project/demo-project_report.docx
```

```
# Add to project notes  
project:  
  notes: "Report generated 2024-11-05 with default thresholds"
```

- _____
- _____
- _____
- _____

CLI Reference

app

[OPTIONS] COMMAND [ARGS]...

-v, --version	Show version and exit
--debug	Enable debug logging
--quiet	Only warnings and errors
--log-dir DIRECTORY	Central log directory (precedence: env CRACKZ_LOG_DIR > --log-dir > system temp). Created if missing. Project logs are always under <project>/logs.
--telemetry	Opt in to local telemetry: append structured events to logs/events.jsonl. No image data, no PII. Also enabled by CRACKZ_TELEMETRY_ENABLED=1. Off by default. See ADR-021.
--install-completion	Install completion for the current shell.
--show-completion	Show completion for the current shell, to copy it or customize the installation.

dashboard

crackz dashboard --project my-project

http://localhost:8501

crackz dashboard --project my-project --port 8080

crackz dashboard

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.

15.

16.

17.

18.

```
crackz detect --project my-project
crackz dashboard --project my-project
# Navigate to "Detection Results" page
```

```
crackz train --project my-project --epochs 50
crackz dashboard --project my-project
# Navigate to "Training Metrics" page
```

```
crackz dashboard --project my-project
# Navigate to "GPS Map" page
```

```
crackz dashboard --project my-project
# Navigate to "Parameter Tuning" page
# Adjust thresholds and preview results
```

clean

```
crackz clean --project ./my_project
```

```
crackz clean -p ./my_project
```

- *.log
-
- models/ checkpoints/
-
-

```
crackz clean --project ./my_project --dry-run
```

```
Dry run: The following would be cleaned:
• /path/to/project/logs/crackz.log
• /path/to/project/artifacts/training/tensorboard
• /path/to/project/artifacts/models
• /path/to/project/artifacts/detections

Re-run without --dry-run to apply changes.
```

```
crackz clean --project ./my_project --logs-only
```

```
*.log
```

```
models/
```

```
checkpoints/
```

```
$ crackz clean --project ./my_project
```

```
Project cleaned! Removed 5 items (logs and artifacts).
```

```
--logs-only
```

```
$ crackz clean --project ./my_project --logs-only
```

```
Project cleaned! Removed 2 items (logs).
```

```
crackz clean -p ./my_project
```

```
crackz clean -p ./my_project --logs-only
```

```
crackz clean -p ./my_project --dry-run
```

```
migrate
```

```
1      ai.training.tensorboard      false      data.mask_threshold: 1      schema_version:
      crackz_version
```

```
      training/tensorboard/      runtime/tensorboard/      checkpoints/      models/
      runtime/lightning_logs/      detections/*.png      training/lightning_logs/
      detections/visualizations/
```

```
crackz migrate --project ./my_project
```

```
crackz migrate --project ./my_project --migrate-files
```

```
crackz migrate --cfg ./my_project/my_project_project.yaml --migrate-files
```

```
crackz migrate -p ./my_project --migrate-files
```

```
crackz migrate -c ./my_project/my_project_project.yaml --migrate-files
```

```
# Schema changes only
crackz migrate --project ./my_project --dry-run

# Schema + file changes
crackz migrate --project ./my_project --migrate-files --dry-run
```

would

`.v{version}.{timestamp}.bak`

```
crackz migrate --project ./my_project --no-backup
```

```
[migrated] my_project schema 0 -> 1 (current=1) steps: 0->1: added tensorboard default + schema_version=1
```

```
[up-to-date] my_project schema 1 -> 1 (current=1) steps: none
```

```
build.yml
integration-tests.yml    pytest -m integration
release.yml
docker-build.yml
smoke-tests.yml
benchmark-tests.yml     pytest -m benchmark
docs.yml
```

```
pytest -m integration -vv
```

```
pytest -m "not integration"
```

```
pytest -m benchmark --benchmark-only
```

`crackz/cli/*_cli.py`

schema_version

- .bak
- CRACKZ_AUTO_MIGRATE=1

```
crackz migrate --project <project-dir>
```

```
--dry-run --no-backup
```

migration.md

model

```
crackz model export --project ./my_project --out my_model.zip
```

```
crackz model export -p ./my_project -o my_model.zip
```

<project_name>_model.zip

```
crackz model export -p ./my_project -c final -o final_model.zip
```

```
best last final
```

```
crackz model import --project ./my_project --model my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip
```

```
crackz model import -p ./my_project -m my_model.zip --force
```

```
--force
```

```
# Export from training project
crackz model export -p ./training_project -o harbor_crack_v1.zip

# Import into deployment project
crackz model import -p ./deployment_project -m harbor_crack_v1.zip
```

```
# Export pre-trained model
crackz model export -p ./pretrained_project -o base_model.zip

# Import into new project for fine-tuning
crackz model import -p ./finetuning_project -m base_model.zip
```

```
# Export after successful training
crackz model export -p ./my_project -o backups/model_$(date +%Y%m%d).zip
```

<checkpoint_name>.ckpt

metadata.json

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

CLI Cookbook

CLI Cookbook

```
crackz init --name demo-project
```

```
demo-project/
images/
  raw/ train/ val/ test/
masks/
artifacts/
detections/
checkpoints/
project.yaml
```

- `crackz --help` `--help`
- `--project` `--cfg`
- `--debug` `--quiet`
-

```
crackz init --name demo-project
```

```
crackz init --name demo-project --force
```

```
# Preview what will be cleaned (safe)
crackz clean --project demo-project --dry-run

# Clean logs and artifacts
crackz clean --project demo-project

# Clean only logs, keep checkpoints and detections
crackz clean --project demo-project --logs-only
```

```
--logs-only *.log --logs-only
```

```
--type
```

```
crackz import-images \
--project demo-project \
--src ./samples \
--type raw \
--size 512 512 \
--quality 90
```

```
crackz import-images --project demo-project --src ./samples --type train --size 512 512
crackz import-images --project demo-project --src ./samples --type val --size 512 512
crackz import-images --project demo-project --src ./samples --type test --size 512 512
```

```
crackz import-images \
--project demo-project \
--src ./samples \
--split \
--train-ratio 0.7 \
--val-ratio 0.15 \
--test-ratio 0.15 \
--size 512 512
```

```
--split
```

```
--split
```

```
--type
```

```
--random-seed
```

```
crackz import-images --project demo-project --src ./samples --split --random-seed 42
```

```
--src/masks
```

```
crackz import-images --project demo-project --src ./samples --masks_src ./samples/masks --type train
```

```
crackz import-images --project demo-project --src ./samples --type train --overwrite
```

```
train
```

```
images/train
train/
```

```
val test
crackz detect train
```

```
raw/
--split
```

```
--type
```

```
scripts/import_masks.py
```

```
# Basic usage - import from organized mask directories
```

```
python scripts/import_masks.py <project_dir> <masks_source_dir>
```

```
# Example: Import masks for skytteren_projekt from smasks directory
```

```
python scripts/import_masks.py ./skytteren_projekt ./smasks
```

```
# Custom size and quality
```

```
python scripts/import_masks.py ./my_project ./mask_sources --size 1024 1024 --quality 95
```

```
masks_source_dir
```

```
masks_source_dir/
train/
  mask_001.jpg
  mask_002.png
  ...
val/
  mask_101.jpg
  ...
test/
  mask_201.png
  ...
```

```
data.mask_threshold
_mask
```

```
--size WIDTH HEIGHT
```

```
--quality QUALITY
```

```
data.mask_threshold
```

```
val/
```

```
# Basic usage - edit an image (mask path auto-derived)
```

```
crackz masks edit images/train/crack_001.jpg
```

```
# With custom mask path
```

```
crackz masks edit photo.jpg --mask output/my_mask.png
```

```
# With project context (smart path derivation)
```

```
crackz masks edit images/raw/inspection.jpg --project my_harbor_project
```

```
--mask
```

1.

```
--project
```

```
Image: project/images/train/crack_001.jpg
```

```
Mask: project/masks/train/crack_001.png ← Auto-derived
```

2.

```
Image: data/images/train/photo.jpg
```

```
Mask: data/masks/train/photo.png ← Auto-derived
```

3.

```
Image: /tmp/photo.jpg
```

```
Mask: /tmp/masks/photo.png ← Auto-derived
```

pyside6

```
for img in images/train/*.jpg; do
    crackz masks edit "$img" --project my_project
done
```

```
ssh -X user@server
```

```
crackz masks edit /data/harbor/crack.jpg
```

```
# Convert all masks in a project (uses project threshold)
crackz masks convert --project my_harbor_project
```

```
# Convert with custom threshold
```

```
crackz masks convert --project my_harbor_project --threshold 128
```

-
- train val test raw
- raw/

1. train val test image_01.jpg
image_01.png
2. raw/
3. crackz detect train
4. crackz detect run --project <p> raw/
5. test

```
--type train
```

1. train val test labelme
pip install labelme
- 2.
3. image_001.png images/train/image_001.jpg
masks/train/image_001.png
4. masks/<split>/ masks/train/ masks/val/ masks/test/
- 5.
- 6.
7. crackz import-images --masks_src masks/
--split

```
cv2.threshold
```

```
crackz detect run --project demo-project
```

```
--project
```

```
crackz detect run --cfg config/detect.yaml
```

```
--model
```

```
best.ckpt last.ckpt final.ckpt
```

```
# Use specific checkpoint by name
crackz detect run --project demo-project --model best
crackz detect run --project demo-project --model last
crackz detect run --project demo-project --model final

# Use specific epoch checkpoint (e.g., epoch-0.9234.ckpt)
crackz detect run --project demo-project --model epoch-0.9234
```

```
--model
final.ckpt
```

```
best.ckpt
```

```
last.ckpt
```

```
--model last
```

```
--model best
```

```
--model epoch-X
```

```
crackz detect train --project demo-project
train/val/test
```

```
crackz detect train --project demo-project --epochs 1 --batch-size 1
```

```
# Auto-selects best checkpoint, outputs to <project-name>_model.zip
crackz model export --project demo-project

# Specify output path
crackz model export --project demo-project --out models/harbor_v1.zip

# Export specific checkpoint (best, last, or final)
crackz model export --project demo-project --checkpoint best --out best_model.zip
```

```
crackz model export -p demo-project -o my_model.zip
crackz model export -p demo-project -c final -o final_model.zip
```

```
.ckpt
```

```
best.ckpt last.ckpt final.ckpt
```

```
<project_name>_model.zip --out
```

```
# Import model package
crackz model import --project demo-project --model harbor_v1.zip

# Overwrite existing checkpoint with same name
crackz model import --project demo-project --model harbor_v1.zip --force
```

```
crackz model import -p demo-project -m my_model.zip
crackz model import -p demo-project -m my_model.zip --force
```

```
checkpoints/
```

```
--force
```

```
# Team member A: Export trained model
crackz model export -p training-project -o shared/harbor_crack_v1.zip

# Team member B: Import into their project
crackz model import -p deployment-project -m shared/harbor_crack_v1.zip
```

```
# Export base model trained on general cracks
crackz model export -p general-cracks -o pretrained/base_model.zip

# Import into specialized project for fine-tuning
crackz model import -p harbor-specific -m pretrained/base_model.zip

# Fine-tune on harbor-specific data
crackz detect train -p harbor-specific --epochs 10
```

```
# Backup current best model before trying new hyperparameters
crackz model export -p demo-project -c best -o backups/before_experiment.zip

# Run experiment
crackz detect train -p demo-project --epochs 50

# If results are worse, can import the backup
crackz model import -p demo-project -m backups/before_experiment.zip --force
```

```
# Export with timestamp
crackz model export -p demo-project -o models/model_2025-10-11.zip

# Or use git-style versioning
crackz model export -p demo-project -o models/v1.0.0.zip
```

```
# Export from training environment
crackz model export -p training-project -o production/model.zip

# Transfer to production server and import
crackz model import -p production-project -m production/model.zip

# Run detection in production
crackz detect run -p production-project
```

<checkpoint_name>.ckpt

metadata.json

```
# Linux/Mac
unzip -l my_model.zip

# Windows PowerShell
Expand-Archive -Path my_model.zip -DestinationPath temp -Force
cat temp\metadata.json
```

metadata.json

```
{
  "project_name": "harbor_detection",
  "checkpoint_name": "best.ckpt",
  "model_config": {
    "encoder": "resnet34",
    "encoder_weights": "imagenet",
    "in_channels": 3,
    "num_classes": 1,
    "threshold": 0.5,
    "loss_name": "combo",
    "dice_weight": 0.7,
    "focal_weight": 0.3
  }
}
```

```
# Generate map for raw split (default)
crackz geo create-map --project demo-project --output harbor_map.html

# Map for specific split (train/val/test)
crackz geo create-map --project demo-project --output train_map.html --split train
```

```
crackz geo create-map -p demo-project -o map.html -s raw
```

```
# Export all images with GPS data
crackz geo export-geojson --project demo-project --output detections.geojson

# Export specific split
crackz geo export-geojson --project demo-project --output train.geojson --split train
```

```
crackz geo export-geojson -p demo-project -o data.geojson -s raw
```

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [longitude, latitude]
      },
      "properties": {
        "image_name": "IMG_0001.jpg",
        "image_path": "/path/to/image.jpg",
        "has_detection": true,
        "latitude": 59.123456,
        "longitude": 10.123456
      }
    }
  ],
  "metadata": {
    "project": "demo-project",
    "split": "raw",
    "total_images": 42,
    "images_with_detections": 15
  }
}
```

```
# Linux/Mac
exiftool image.jpg | grep GPS

# Python (Pillow)
from PIL import Image
img = Image.open("image.jpg")
exif = img.getexif()
gps_ifd = exif.get_ifd(0x8825) # 0x8825 is GPS IFD tag
print(gps_ifd)
```

```
# 1. Import drone photos (GPS data preserved in EXIF)
crackz import-images --project harbor --src ./drone_photos --type raw --size 512 512

# 2. Run detection
crackz detect run --project harbor

# 3. Generate map to visualize results
```

```
crackz geo create-map --project harbor --output harbor_results.html

# 4. Export to GeoJSON for GIS analysis
crackz geo export-geojson --project harbor --output harbor.geojson

# 5. Open map in browser to review
# (or import GeoJSON into QGIS/ArcGIS)
```

```
crackz --debug detect run --project demo-project
crackz --quiet detect run --project demo-project
crackz --log-dir run_logs detect run --project demo-project
```

```
CRACKZ_LOG_DIR=logs_debug crackz detect run --project demo-project
```

```
--debug
```

```
# Start training
crackz detect train --project demo-project

# Press Ctrl+C to cancel at any time
# The operation will be cancelled gracefully
# Partial results (e.g., checkpoints) are preserved
```

```
crackz detect run --cfg configs/detect.yaml
crackz detect train --cfg configs/train.yaml
```

```
for P in demo-project other-project; do
  crackz detect run --project "$P" || echo "Failed: $P" >&2
done
```

```
/home/app
```

```
# Init project (host ./project mapped to /home/app)
mkdir -p project
docker run --rm -v %cd%\project:/home/app crackz-cli init --name demo-project

# Import images (mount samples to /data)
docker run --rm -v %cd%\project:/home/app -v %cd%\samples:/data \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v %cd%\project:/home/app crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu detect run --project /home/app/demo-project
```

```
--project --cfg
```

```
--src
```

```
images/train
```

```
raw
```

```
torch.cuda.is_available()
```

```
# Generate report with default name (project_name_report.docx)
crackz report generate --project demo-project

# Specify custom output path
crackz report generate --project demo-project --output inspection_report.docx

# Use config file instead of project path
crackz report generate --cfg project.yaml --output report.docx
```

```
crackz detect run
```

```
# Find top 50 most uncertain images
uv run crackz active-learning run \
  --project demo-project \
  --top-n 50 \
  --method entropy \
  --out priorities.json
```

```
entropy
variance
```

```
mean_confidence
```

```
max_probability
```

```
uv run crackz active-learning info
```

```
# Prioritize images in 'raw' split (default)
uv run crackz active-learning run --project demo-project --split raw

# Prioritize images in 'train' split
uv run crackz active-learning run --project demo-project --split train
```

```
{
  "method": "entropy",
  "total_images": 500,
  "prioritized_count": 50,
  "images": [
    {
      "path": "image_042.jpg",
      "uncertainty_score": 0.8234,
      "rank": 1
    }
  ]
}
```

```

]
}

```

```

# Find similar images and propagate mask
uv run crackz propagate-masks run \
  --project demo-project \
  --reference wall_01.jpg \
  --mask wall_01_mask.png \
  --split raw \
  --threshold 0.8 \
  --out ./propagated_masks

```

```

--threshold
sift --orb --min-matches --

```

```

uv run crackz propagate-masks find-similar \
  --project demo-project \
  --reference wall_01.jpg \
  --split raw \
  --out similarities.json

```

```

# More permissive matching (more propagations, less accurate)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.6 \
  --min-matches 5

# Stricter matching (fewer propagations, more reliable)
uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --threshold 0.9 \
  --min-matches 20 \
  --sift

```

```

uv run crackz propagate-masks run \
  --reference ref.jpg \
  --mask ref.png \
  --project demo-project \
  --orb

```

- [api.md](#) [detect\(\)](#) [train\(\)](#)

[crackz export-config](#)

Shell Completion

Shell Completion

crackz

-
-
-
-

```
crackz --show-completion bash
crackz --show-completion zsh
crackz --show-completion fish
crackz --show-completion powershell
```

source Invoke-Expression

```
crackz --install-completion SHELL
```

SHELL bash zsh fish powershell

```
crackz det<TAB>
```

```
crackz detect
```

```
crackz --install-completion bash
# New shell OR
source ~/.bash_completion
```

bash-completion

--show-completion

```
crackz --install-completion zsh
# Then restart shell or:
autoload -U compinit && compinit
```

\$fpath

fpath

```
crackz --install-completion fish
# Typically installs into ~/.config/fish/completions/
# Reload:
exec fish
```

```
pwsh -File scripts/setup_completion.ps1
# Or force re-install:
pwsh -File scripts/setup_completion.ps1 -Force
```

crackz

crackz --install-completion powershell

```
crackz --show-completion powershell | Out-String | Invoke-Expression
```

```
. $PROFILE
```

```
crackz --install-completion powershell
# If the profile wasn't updated, add manually:
Add-Content $PROFILE "crackz --show-completion powershell | Out-String | Invoke-Expression"
. $PROFILE
```

```
crackz --show-completion bash > /tmp/crackz.sh && source /tmp/crackz.sh
source <(crackz --show-completion zsh)
crackz --show-completion fish | source
crackz --show-completion powershell | Out-String | Invoke-Expression
```

```
. $PROFILE
```

```
uv run crackz --install-completion ...
uv run crackz --show-completion ...
```

```
Force
```

```
--
```

```
-Force
```

```
autoload -U compinit && compinit
```

```
_completion_loader
```

```
--show-completion
```

```
~/.config/fish/completions/
```

```
~/.bashrc
```

```
~/.zshrc
```

```
$PROFILE
```

```
crackz --show-completion bash > .crackz_comp.sh
source .crackz_comp.sh
# run tests that exercise completion
```

```
: _____
: _____
```

Project Configuration

Configuration Reference

```
crackz init
```

```
CLI flag > ENV var > Project YAML > Code default
```

```
crackz init --name demo-project
```

```
demo-project/_project.yaml
```

```
<name>_project.yaml
```

```
schema_version: 6
project:
  name: demo-project
  description: Demo project
  created: 2025-01-01T12:00:00+00:00
annotation:
  background:
    id: 0
    name: Background
    color: "#000000"
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
paths:
  project_path: demo-project
  image_path: demo-project/images
  image_mask_path: demo-project/masks
  artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
ai:
  model:
    backbone: segformer-b2
    active_checkpoint: best
    device: cpu
    in_channels: 3
    input_size: [1024, 1024]
    threshold: 0.5
    loss_name: combo
    dice_weight: 0.7
    focal_weight: 0.3
  training:
    batch_size: 8
    num_workers: 4
    epochs: 10
    learning_rate: 0.001
    shuffle: true
    augmentation: true
  detection:
    threshold: 0.5
    batch_size: 8
    output_dir_name: detections
data:
  mask_threshold: 50
  image_quality: 90
  target_size: 1024
```

```
crackz migrate
```

```
--project <path>
```

```
ProjectConfig
```

```
from pathlib import Path
from crackz.config.project_config import ProjectConfig
```

```
pc = ProjectConfig(
    name="demo-project",
    project_path=Path("demo-project"),
    image_path=Path("demo-project/images"),
    image_mask_path=Path("demo-project/masks"),
    artifacts_path=Path("demo-project/artifacts"),
)
pc.save(Path("project_config.yaml"))

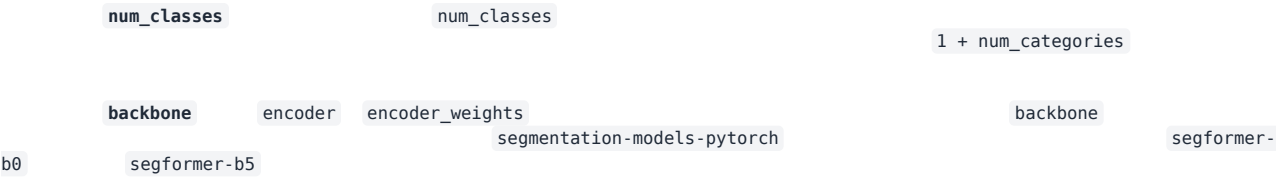
# Later
loaded = ProjectConfig.load(Path("project_config.yaml"))
project = loaded.to_project()
```

Project

```
from crackz.config.project_config import ProjectConfig
pc = ProjectConfig.from_project(project)
pc.save(Path("backup_config.yaml"))
```

- ModelConfig
- TrainingConfig
- DetectionConfig

binary multiclass



- segformer-b0
- segformer-b1
- segformer-b2**
- segformer-b3
- segformer-b4
- segformer-b5

segmentation-models-pytorch

efficientnet-b0
efficientnet-b3
efficientnet-b7
resnet18
resnet34
resnet50
mobilenet_v2

swsl imagenet None null ssl

Binary Segmentation — SegFormer (Default)

```
ai:
  model:
    backbone: segformer-b2 # Default SegFormer model
    input_size: [1024, 1024]
    threshold: 0.5
  annotation:
    categories:
      - id: 1
        key: crack
        name: Crack
        color: "#ef4444"
```

Multi-Class Segmentation — Harbor Defects

```
ai:
  annotation:
    categories:
      - id: 1
        key: crack
        name: Crack
        color: "#ef4444"
      - id: 2
        key: spall
        name: Spall
        color: "#f59e0b"
      - id: 3
        key: corrosion
        name: Corrosion
        color: "#3b82f6"
  model:
    backbone: segformer-b2
    # num_classes is derived automatically: 1 + 3 categories = 4 output channels
```

Binary Segmentation — U-Net with EfficientNet (legacy or fallback)

```
ai:
  model:
    backbone: efficientnet-b3 # U-Net decoder, EfficientNet-B3 encoder
    input_size: [512, 512]
    threshold: 0.5
```

Grayscale Input

```
ai:
  model:
    backbone: segformer-b2
    in_channels: 1 # Grayscale images
    normalization:
      mean: [0.5]
      std: [0.5]
```

Force binary task mode with multiple annotation categories

```
ai:
  model:
    backbone: segformer-b2
    task_mode: binary # All categories collapse into one defect-vs-background class
```

[\[Link\]](#)

--	--	--

TrainingConfig

int int | str

tensorboard
artifacts/runtime/tensorboard/
getattr(project.ai.training, "tensorboard", False) False

```
# Safe access pattern inside runtime code
use_tb = getattr(project.ai.training, "tensorboard", False)
if use_tb:
    ... # create logger
```

getattr(..., default)

False tests/unit/crackz/ai/test_detection.py::test_tensorboard_logger_disabled_by_default
model_fields

TrainingConfig

--	--	--

min_confidence medium_confidence high_confidence

-
-
-

--	--	--

```
ai:
  detection:
    min_confidence: 0.005 # Save detections with ≥0.5% crack area
    medium_confidence: 0.02 # "Probable Crack" at ≥2% crack area
    high_confidence: 0.05 # "Crack Detected" at ≥5% crack area
    min_crack_area_px: 100 # Suppress detections smaller than 100 pixels
```

data

--	--	--

DataConfig

--	--	--

mask_threshold

-
-
-

--	--	--

```
data:
  mask_threshold: 50 # Default - balanced filtering
```

crackz import-images --masks_src <path>
scripts/import_masks.py

project.import_images()

crackz migrate

mask_threshold:

annotation

```

annotation:
  background:
    id: 0
    name: Background
    color: "#000000"
  categories:
    - id: 1
      key: crack
      name: Crack
      color: "#ef4444"
    - id: 2
      key: spill
      name: Spill
      color: "#f59e0b"
    - id: 3
      key: corrosion
      name: Corrosion
      color: "#3b82f6"

```

1..N

crack spall

Crack Spall

#RRGGBB

1, 2, 3, ...N

1

 $1 + N$

- task_mode: binary

crack-only

harbor-defects

survey-general

```
# CLI: initialize a project with a harbor-defects annotation template
crackz init --name harbor-project --annotation-template harbor-defects
```

```
crackz export-config --project demo-project --out tuned_config.yaml
```

```
from crackz.config.project_config import ProjectConfig
pcfg = ProjectConfig.from_project(project)
pcfg.save(Path("tuned_config.yaml"))
```

```
CRACKZ_LOG_DIR=run_logs crackz detect run --project demo-project
```

```
$env:CRACKZ_LOG_DIR='run_logs'; crackz detect run --project demo-project
```

```

                                ai.model.device
resolve_device_and_accelerator
cuda          python -c "import torch; print(torch.cuda.is_available())"
```

```
ai.model.device:
```

```
tensorboard: true
```

```
ai:
  training:
    tensorboard: true
    epochs: 10
    batch_size: 8
```

```
from crackz import Project
project = Project.load(Path("demo-project"))
project.ai.training.tensorboard = True
project.save()
```

```
<artifacts_path>/runtime/tensorboard/
```

```
tensorboard --logdir demo-project/artifacts/runtime/tensorboard
```

```
docker run --rm -v $(pwd)/demo-project:/workspace -p 6006:6006 crackz-cli \
  tensorboard --logdir /workspace/artifacts/runtime/tensorboard
```

```
crackz detect run --project p --cfg cfg.yaml
```

```
    training.epochs    batch_size
```

```
    detection.batch_size
```

```
<artifacts_path>/models/                                best.ckpt                                last.ckpt
epoch={N}-val_iou={score}.ckpt    epoch=9-val_iou=0.8234.ckpt
<artifacts_path>/checkpoints/
```

```
ai:
  training:
    save_top_k: 1      # Keep N best checkpoints (by val_iou), default: 1
    save_last: true    # Always save last.ckpt, default: true
    auto_prune_checkpoints: false # Auto-prune after training, default: false
    checkpoint_keep: 3  # Checkpoints to keep when auto-pruning, default: 3
```

1. save_top_k: 1, save_last: true
2. save_top_k: 3, auto_prune_checkpoints: true, checkpoint_keep: 3
3. save_top_k: 1, save_last: false, auto_prune_checkpoints: true, checkpoint_keep: 1

```
crackz detect prune-checkpoints --project demo-project --keep 3
```

```
docker run --gpus all --rm -v %cd%\project:/workspace crackz-cli:gpu \
  detect prune-checkpoints --project /workspace/demo-project --keep 2
```

```
from crackz.ai.checkpoints import prune_checkpoints
removed = prune_checkpoints(project, keep=2)
```

```
0.8234.ckpt                                best.ckpt                                003-
last.ckpt
```

```
<artifacts_path>/detections/detection_results.json
```

```
from crackz.ai.results import load_detection_results
from crackz.ai.eval import basic_summary
results = load_detection_results(project)
print(basic_summary(results))
```

crack_severity

seed

```
# project_config.yaml (unified example)
name: demo-project
description: Example project
project_path: demo-project
image_path: demo-project/images
image_mask_path: demo-project/masks
artifacts_path: demo-project/artifacts
logging:
  level: INFO
  log_dir: logs
model:
  encoder: resnet34
  device: cpu
  input_size: [512, 512]
training:
  epochs: 5
  batch_size: 4
  learning_rate: 0.0005
  tensorboard: false # Set to true to enable TensorBoard logging
detection:
  threshold: 0.55
  batch_size: 4
```

schema_version

```
crackz migrate --project ./my_project
```

```

false          schema_version          schema_version: 1          tensorboard:
                .bak
```

```
crackz migrate --cfg ./my_project/my_project_project.yaml
```

```
crackz migrate --project ./my_project --dry-run
```

```
crackz migrate --project ./my_project --no-backup
```

up-to-date noop

```
for d in projects/*; do
[ -d "$d" ] && crackz migrate --project "$d" --no-backup;
done
```

ProjectConfig

Training Configuration

TrainingConfig Contract

TrainingConfig	crackz.project.config	ai.training
batch_size		
num_workers		
epochs		
learning_rate		
shuffle		
loss		
metrics		
seed		
precision		
devices		
strategy		
tensorboard	artifacts/training/tensorboard	
tests/unit/crackz/project/test_training_config_schema.py		
TrainingConfig		
1.		
2.		
3.		
4.	CURRENT_SCHEMA_VERSION	
1.		
2.		
3.		
4.		
	--epochs	--batch-size
tensorboard=True	<project>/artifacts/training/tensorboard	
seed		

Data Splits & Import

Data Splits and Import Guide

- 1. _____
 - 2. _____
 - 3. _____
 - 4. _____
 - 5. _____
-



images/raw/

raw/

```
crackz import-images --project my-project --src ./photos --type raw --size 512 512
```

images/train/ masks/train/

img001.jpg img001.png

images/val/ masks/val/

`images/test/` `masks/test/`

`--type raw`

```
crackz import-images \  
--project my-project \  
--src ./labeled-data/images \  
--src-masks ./labeled-data/masks \  

```

```
--split-ratios 0.7 0.15 0.15 \
--size 512 512
```

```
crackz import-images \
--project my-project \
--src ./labeled-data/images \
--src-masks ./labeled-data/masks \
--split-ratios 0.7 0.15 0.15 \
--seed 42 \
--size 512 512
```

```
# Small dataset: larger validation set
crackz import-images \
--project my-project \
--src ./small-dataset/images \
--src-masks ./small-dataset/masks \
--split-ratios 0.6 0.25 0.15 \
--size 512 512

# Large dataset: more for training
crackz import-images \
--project my-project \
--src ./large-dataset/images \
--src-masks ./large-dataset/masks \
--split-ratios 0.8 0.1 0.1 \
--size 512 512
```

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.
- 15.

-
-

- 16.
- 17.
- 18.
- 19.
- 20.
- 21.
- 22.

- 23.
- 24.
- 25.
- 26.

-
-
-
-

```
crackz init --name harbor-inspection
cd harbor-inspection
```

```
crackz import-images \
--project . \
--src ~/datasets/harbor-cracks/images \
--src-masks ~/datasets/harbor-cracks/masks \
--split-ratios 0.7 0.15 0.15 \
--seed 42 \
--size 512 512
```

images/train/
masks/

images/val/

images/test/

```
# Count images
ls images/train/ | wc -l # Should show 140
ls images/val/ | wc -l # Should show 30
ls images/test/ | wc -l # Should show 30

# Count masks (should match images)
ls masks/train/ | wc -l # Should show 140
ls masks/val/ | wc -l # Should show 30
ls masks/test/ | wc -l # Should show 30
```

```
crackz detect train --epochs 50 --batch-size 8
```

images/val/ masks/val/ images/train/ masks/train/ artifacts/metrics.json artifacts/loss_curve.png checkpoints/best.ckpt

```
crackz detect run --split test
```

masks/test/ images/test/ detections/

```
# Import new unlabeled images
crackz import-images \
  --project . \
  --src ~/new-harbor-photos \
  --type raw \
  --size 512 512

# Run detection
crackz detect run --split raw
```

detections/

1.

```
--seed 42 # Same split every time
```

2.

```
cp -r original-data backup/
```

3.

```
# Check if splits are representative
ls images/train/ | wc -l
ls images/val/ | wc -l
```

4.

```
# For 50 images: 0.6, 0.25, 0.15 ensures val gets ~13 images
--split-ratios 0.6 0.25 0.15
```

5.

```
# In project documentation
split_info:
  seed: 42
  ratios: [0.7, 0.15, 0.15]
  date: 2025-10-19
```

1.

2.

3.

4.

5.

6.

- -
 -
 -
 -
 -
 -
-

```
# Wrong: 0.7 + 0.2 + 0.2 = 1.1 x
# Right: 0.7 + 0.15 + 0.15 = 1.0
--split-ratios 0.7 0.15 0.15
```

images/val/

```
# If you have 10 images and use 0.7/0.15/0.15:
# train=7, val=1, test=2 (works)

# If you have 5 images and use 0.8/0.1/0.1:
# train=4, val=0, test=1 (val is empty!)

# Solution: Use larger val ratio or more images
--split-ratios 0.6 0.25 0.15 # Ensures val gets at least 1
```

```
# Image: images/crack_001.jpg
# Mask must be: masks/crack_001.png (same stem)

# Check your filenames:
ls images/ > image_list.txt
ls masks/ > mask_list.txt
# Compare the stems (filename without extension)
```

```
# Add --seed for reproducible splits
--seed 42
```

```
# Increase validation ratio
--split-ratios 0.65 0.25 0.1 # Gives more to validation
```

```
# Import with smaller size
--size 256 256 # Instead of 512 512

# Or reduce batch size during training
crackz detect train --batch-size 2 # Instead of 8
```

```
# Separate images by class first
mkdir temp/crack temp/no-crack

# Move files to class folders
# ... organize manually or with script ...

# Import each class separately with same ratio
crackz import-images --src temp/crack --split-ratios 0.7 0.15 0.15 --seed 42
crackz import-images --src temp/no-crack --split-ratios 0.7 0.15 0.15 --seed 42
```

```
# Import full dataset to train
crackz import-images --src data --type train --size 512 512

# Manually create folds by moving files between train/val
# Train 5 models with different train/val splits
# Average results for robust performance estimate
```

```
# Import new batch with same seed and ratios
crackz import-images \
  --src new-data \
  --split-ratios 0.7 0.15 0.15 \
  --seed 42 \
  --size 512 512

# WARNING: This adds to existing splits
# Consider fresh re-import of all data for clean splits
```

```
images/raw/
images/train/
images/val/
images/test/
```

```
# Import for training (with split)
crackz import-images --src data/images --src-masks data/masks \
  --split-ratios 0.7 0.15 0.15 --seed 42 --size 512 512

# Import for detection (no split)
crackz import-images --src photos --type raw --size 512 512

# Train on train split, validate on val split
crackz detect train --epochs 50 --batch-size 8

# Test on test split
crackz detect run --split test

# Detect on raw split
crackz detect run --split raw
```

-
-
-

- _____
- _____
- _____
- _____

Docker

Docker Usage

```
# Pull from registry
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu

# Run from registry
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --help
```

latest	Dockerfile		
slim	Dockerfile.slim		
gpu	Dockerfile.gpu		

```
# Full
docker build -t crackz-cli .
# Slim
docker build -f Dockerfile.slim -t crackz-cli:slim .
# GPU (CUDA 12.8)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
docker build --build-arg PYTHON_VERSION=3.13 -t crackz-cli .
```

1.

ghcr.io/astral-sh/uv:bookworm-slim

uv build --wheel
2.
3.

python:3.13-slim
4.

nvidia/cuda:12.8.0-runtime-ubuntu22.04
5.
6.

--build-arg TORCH_INDEX_URL=...

- uv build
-
-
-

- nvidia/cuda:12.8.0-runtime-ubuntu22.04
-
- git libglib2.0-0 libjpeg-turbo8 libgomp1 ca-certificates
-
-

```
# Default: CUDA 12.8 (recommended, matches PyTorch cu128)
docker build -f Dockerfile.gpu -t crackz-cli:gpu .

# Override for CUDA 12.1
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu121 \
  -t crackz-cli:gpu-cu121 .

# Override for CUDA 12.4 (if needed for compatibility)
docker build -f Dockerfile.gpu \
  --build-arg TORCH_INDEX_URL=https://download.pytorch.org/whl/cu124 \
  -t crackz-cli:gpu-cu124 .
```

TORCH_INDEX_URL

```
#18 [runtime 7/9] RUN /app/.venv/bin/python3 -c 'import torch; print("Torch:", torch.__version__, "CUDA available:", torch.cuda.is_available())'
#18 0.699 Torch: 2.8.0+cpu CUDA available: False
```

CUDA available: False

linux/amd64

/home/app

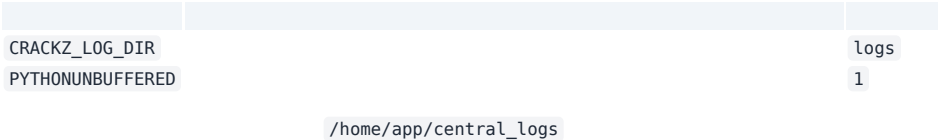
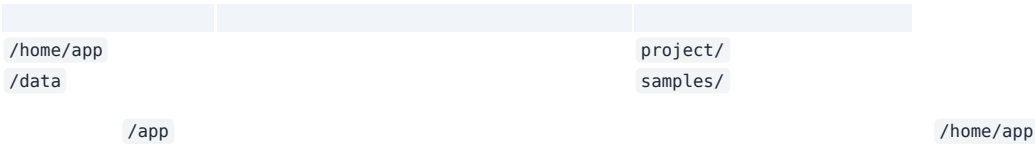
```
# Help
docker run --rm crackz-cli --help

# Init project (creates /home/app/demo-project inside container)
mkdir -p project
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli init --name demo-project

# Import images (mount samples to /data)
mkdir -p samples
# (populate samples/ with images and optional masks/ first)
docker run --rm \
  -v "$(pwd)/project:/home/app" \
  -v "$(pwd)/samples:/data" \
  crackz-cli import-images --project /home/app/demo-project --src /data --type raw --size 512 512

# Detection
docker run --rm -v "$(pwd)/project:/home/app" crackz-cli detect run --project /home/app/demo-project
```

```
docker run --gpus all --rm -v "$(pwd)/project:/home/app" crackz-cli:gpu detect run --project /home/app/demo-project
```



/home/app/central_logs

```
docker run --rm -e CRACKZ_LOG_DIR=central_logs -v "$(pwd)/project:/home/app" \
  crackz-cli detect run --project /home/app/demo-project
```

/home/app

```
make build      # full image
make build-slim # slim image
make build-gpu  # gpu image (CUDA 12.8)
make init NAME=demo-project # creates project under /home/app
make import PROJECT=demo-project DATA_DIR=./samples
make detect PROJECT=demo-project
make detect-gpu PROJECT=demo-project # GPU detection
make train-gpu PROJECT=demo-project  # GPU training
```

act

```
# Install act (macOS)
brew install act

# Test GPU build with make targets
make act-gpu-dryrun # Dry-run to validate workflow
make act-gpu        # Actually run the GPU build locally
make act-docker-build # Test all Docker builds (full, slim, gpu)
```

	latest
	slim
	gpu
gpu	TORCH_INDEX_URL

--	--	--

CRACKZ_LOG_DIR

--gpus all

TORCH_INDEX_URL

- _____
- _____
- _____

Azure VM (GPU)

Azure Windows VM with CUDA for Crackz

```
azure_vm_cuda.sh
```

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-

```
# Deploy default GPU VM
./scripts/azure/azure_vm_cuda.sh

# Deploy with custom configuration
./scripts/azure/azure_vm_cuda.sh \
  --name ml-workstation \
  --size Standard_NC12s_v3 \
  --location westus2

# Connect via RDP and verify
# RDP to the public IP, then run:
nvidia-smi
cd C:\Crackz && .venv\Scripts\Activate.ps1
python -m crackz --version
```

```
Standard_NC6s_v3
Standard_NC12s_v3
Standard_NC24s_v3
```

```
Standard_NC6s_v2
Standard_NV6
```

```
# Default deployment (V100 GPU, East US)
./scripts/azure/azure_vm_cuda.sh

# Custom name and location
./scripts/azure/azure_vm_cuda.sh --name gpu-crackz --location westeurope

# Specific GPU configuration
./scripts/azure/azure_vm_cuda.sh --size Standard_NC12s_v3 --location northeurope
```

Option	Value
--name NAME	crackz-gpu-vm
--location REGION	eastus
--size SIZE	Standard_NC6s_v3
--user USERNAME	crackzuser
--password PASSWORD	CrackzGpu2024!
--mode MODE	standard
--data-disk-size SIZE	1024
--storage-type TYPE	Premium_LRS
--no-data-disk	
--no-cuda	
--no-crackz	
--enable-nested-virt	
--destroy	
--help	

```
# Production inference server
./scripts/azure/azure_vm_cuda.sh \
--name crackz-prod \
--size Standard_NC12s_v3 \
--location westus2 \
--mode production

# Development environment with large storage
./scripts/azure/azure_vm_cuda.sh \
--name crackz-dev \
--size Standard_NV6 \
--mode development \
--data-disk-size 2048 \
--storage-type Premium_LRS

# Budget setup without data disk
./scripts/azure/azure_vm_cuda.sh \
--name crackz-simple \
--size Standard_NV6 \
--no-data-disk

# Manual setup (no auto-install)
./scripts/azure/azure_vm_cuda.sh \
--name crackz-manual \
--size Standard_NC6s_v3 \
--no-cuda \
--no-crackz

# Docker development (with nested virtualization)
./scripts/azure/azure_vm_cuda.sh \
--name crackz-docker \
--size Standard_NC6s_v3 \
--enable-nested-virt

# Clean up resources
./scripts/azure/azure_vm_cuda.sh --name crackz-gpu-vm --destroy
```

Option	Value
--data-disk-size SIZE	
--storage-type TYPE	
--no-data-disk	

```
C:\ (OS Disk - 127 GB)
├─ Windows/
├─ Program Files/
├─ Users/crackzuser/Desktop/Crackz/ # Crackz installation
└─ ...

D:\ (Data Disk - configurable size)
├─ Projects/ # ML project workspace
│   ├── datasets/ # Training datasets
│   ├── models/ # Model checkpoints
│   └─ outputs/ # Detection results
```

	└─ exports/	# Exported models
	└─ temp/	# Temporary processing

--	--	--	--

```
# Production: High-performance storage
./scripts/azure/azure_vm_cuda.sh \
--name ml-prod \
--data-disk-size 2048 \
--storage-type Premium_LRS

# Development: Cost-effective storage
./scripts/azure/azure_vm_cuda.sh \
--name ml-dev \
--data-disk-size 512 \
--storage-type Standard_LRS

# Simple setup: No separate disk
./scripts/azure/azure_vm_cuda.sh \
--name ml-simple \
--no-data-disk
```

- 1.
 - 2.
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 - 4.
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 - 6.
 - 7.
 8. C:\Crackz\.venv
 - 9.
 - 10.
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 - 13.
 - 14.
 - 15.
 - 16.
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```
# 1. Connect via RDP to the VM
# 2. Double-click "File Upload Server" desktop shortcut
# 3. Open browser and go to: http://localhost:8080
# 4. Or access remotely: http://<VM-PUBLIC-IP>:8080
```

```
# From your local computer:
# Windows: \\<VM-PUBLIC-IP>\CrackzUploads
# Mac/Linux: smb://<VM-PUBLIC-IP>/CrackzUploads

# Credentials: crackzuser / CrackzGpu2024!
```

```
# Using any SFTP client (WinSCP, FileZilla, etc.)
sftp crackzuser@<VM-PUBLIC-IP>
# Upload to: /d/Projects/uploads/images/
```

```
D:\Projects\uploads\
├─ images/          # Individual images for analysis
└─ batch/           # Batch processing folders
```

```
# Use Windows built-in Remote Desktop
mstsc /v:<PUBLIC_IP>

# Or download RDP file
az vm show -d -g crackz-gpu-rg -n crackz-gpu-vm --query publicIps -o tsv
```

```
# Check GPU status
nvidia-smi

# Test CUDA availability
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
python -c "import torch; print(f'GPU count: {torch.cuda.device_count()}')"
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Check Crackz version
python -m crackz --version

# Test GPU detection
python -m crackz gui # Should detect CUDA device
```

```
# Activate virtual environment
cd C:\Crackz
.\.venv\Scripts\Activate.ps1

# Option 1: Direct analysis on uploaded images
python -m crackz detect run --images "D:\Projects\uploads\images" --device cuda

# Option 2: Create project and import uploaded images
python -m crackz init harbor-inspection
python -m crackz import-images "D:\Projects\uploads\images" harbor-inspection/images/raw
python -m crackz detect run harbor-inspection --device cuda

# Option 3: Batch processing multiple folders
python -m crackz detect run --images "D:\Projects\uploads\batch\*" --device cuda

# View results
explorer "harbor-inspection\artifacts\detection_results"
```

```
# Create test project
python -m crackz init test-project

# Import sample images
python -m crackz import-images C:\path\to\images test-project/images/raw

# Run GPU-accelerated detection
python -m crackz detect run test-project --device cuda
```

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```
az vm deallocate
```

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```
CrackzGpu2024!
```

```
# Manually install CUDA
# Download from: https://developer.nvidia.com/cuda-11-8-0-download-archive
# Choose Windows x86_64 local installer
```

```
# Manual installation
cd C:\Crackz
.\.venv\Scripts\Activate.ps1
uv pip install --find-links https://download.pytorch.org/whl/cu118 torch torchvision
# Download wheel from GitHub releases and install
```

```
# Check drivers
nvidia-smi

# Verify CUDA installation
nvcc --version

# Check PyTorch CUDA
python -c "import torch; print(torch.version.cuda)"
```

- ```
• nvidia-smi -l 1
```

```
Windows Event Logs
eventvwr.msc

CUDA installation logs
Get-Content C:\ProgramData\NVIDIA Corporation\CUDA Installer\log.txt

Crackz logs
Get-Content $env:TEMP\crackz-*.log
```

```
az vm list-sizes --location eastus --query "[?contains(name, 'NC')]"
```

```
Use spot instances (60-90% discount)
az vm create --priority Spot --max-price -1 ...

Auto-shutdown schedule
az vm auto-shutdown --name crackz-gpu-vm --resource-group crackz-gpu-rg --time 1800

Deallocate when not in use
az vm deallocate --name crackz-gpu-vm --resource-group crackz-gpu-rg
```

- 

```
%TEMP%\crackz-*.log
```

---

# Azure Batch

---

# Azure Batch Deployment

---

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- 1.
- 2.
- 3.
4. `az login`

```
./scripts/azure/azure_batch.sh \
--batch-account mycrackzaccount \
--storage-account mycrackzstorage
```

mycrackzaccount

mycrackzstorage

```
Get storage account key
STORAGE_KEY=$(az storage account keys list \
--account-name mycrackzstorage \
--resource-group subbaltica-crackz-rg \
--query "[0].value" -o tsv)
```

```
Upload project directory
az storage blob upload-batch \
--account-name mycrackzstorage \
--account-key "$STORAGE_KEY" \
--destination crackz-data \
--source ./my-project
```

```
Create job
az batch job create \
--id crackz-detection-job \
--pool-id crackz-pool

Add tasks to the job
az batch task create \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--command-line "docker run crackz-cli:latest detect run --project /mnt/batch/tasks/fsmounts/crackz-data/my-project"
```

```

--batch-account
--resource-group subbaltica-crackz-rg
--location swedencentral
--pool-id crackz-pool
--vm-size Standard_D2s_v3
--node-count 2
--image crackz-cli:latest
--storage-account
--container crackz-data

```

```

Standard_D2s_v3
Standard_D4s_v3
Standard_D8s_v3
Standard_NC6s_v3
Standard_NC4as_T4_v3

```

```

./scripts/azure/azure_batch.sh \
--batch-account crackz-batch \
--pool-id high-performance-pool \
--vm-size Standard_D8s_v3 \
--node-count 10 \
--storage-account crackzstore

```

```

./scripts/azure/azure_batch.sh \
--batch-account crackz-batch-gpu \
--pool-id gpu-pool \
--vm-size Standard_NC4as_T4_v3 \
--node-count 2 \
--image crackz-cli:gpu

```

```

Enable auto-scaling
az batch pool autoscale enable \
--pool-id crackz-pool \
--account-name mycrackzaccount \
--auto-scale-formula '
 startingNumberOfVMs = 2;
 maxNumberOfVMs = 10;
 pendingTaskSamplePercent = $PendingTasks.GetSamplePercent(180 * TimeInterval_Second);
 pendingTaskSamples = pendingTaskSamplePercent < 70 ? startingNumberOfVMs : avg($PendingTasks.GetSample(180 * TimeInterval_Second));
 $TargetDedicatedNodes = min(maxNumberOfVMs, pendingTaskSamples);
'

```

```
#!/bin/bash
Create job
az batch job create --id multi-project-job --pool-id crackz-pool

Submit tasks for each project
for project in project1 project2 project3; do
 az batch task create \
 --job-id multi-project-job \
 --task-id "detect-$project" \
 --command-line "docker run crackz-cli:latest detect run --project /data/$project"
done
```

```
Training task
az batch task create \
 --job-id pipeline-job \
 --task-id train \
 --command-line "docker run crackz-cli:latest detect train --project /data/my-project --epochs 50"

Detection task (depends on training)
az batch task create \
 --job-id pipeline-job \
 --task-id detect \
 --depends-on train \
 --command-line "docker run crackz-cli:latest detect run --project /data/my-project"
```

```
az batch task create \
 --job-id output-job \
 --task-id detect-with-output \
 --command-line "docker run -v /mnt/output:/output crackz-cli:latest detect run --project /data/project" \
 --resource-files '["{"url":"https://mystorage.blob.core.windows.net/input/data.zip","filePath":"data.zip"}"]' \
 --output-files '["{"filePattern":"/mnt/output/**/*","destination":{"container":{"containerUrl":"https://mystorage.blob.core.windows.net/results"},"uploadOptions":{"uploadC
```

```
az batch pool show \
 --pool-id crackz-pool \
 --account-name mycrackzaccount \
 --query "{State:allocationState, DedicatedNodes:currentDedicatedNodes}"
```

```
az batch job list \
 --account-name mycrackzaccount \
 -o table
```

```
az batch task list \
 --job-id crackz-detection-job \
 --account-name mycrackzaccount \
 -o table
```

```
az batch task file download \
 --job-id crackz-detection-job \
 --task-id detect-task-1 \
 --file-path stdout.txt \
 --destination ./task-stdout.txt \
 --account-name mycrackzaccount
```

```
After pool creation, modify to use low-priority nodes
az batch pool resize \
```

```
--pool-id crackz-pool \
--account-name mycrackzaccount \
--target-dedicated-nodes 0 \
--target-low-priority-nodes 4
```

```
./scripts/azure/azure batch.sh \
--batch-account mycrackzaccount \
--pool-id crackz-pool \
--destroy
```

```
az batch location quotas show --location <region>
az batch task file download
```

```
az batch account login
```

```
Get task details
az batch task show \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--account-name mycrackzaccount

Download stderr
az batch task file download \
--job-id crackz-detection-job \
--task-id detect-task-1 \
--file-path stderr.txt \
--destination ./task-stderr.txt \
--account-name mycrackzaccount
```

```
Check current quotas
az batch location quotas show --location swedencentral

Submit quota increase request through Azure Portal
Support > New support request > Issue type: Service and subscription limits
```

```
name: Azure Batch Processing
on:
 push:
 branches: [main]
jobs:
 batch-process:
 runs-on: ubuntu-latest
 steps:
 - uses: actions/checkout@v5

 - name: Azure Login
 uses: azure/login@v1
 with:
 creds: ${ secrets.AZURE_CREDENTIALS }

 - name: Create Batch Pool
 run: |
 ./scripts/azure/azure_batch.sh \
 --batch-account ${ secrets.BATCH_ACCOUNT } \
 --storage-account ${ secrets.STORAGE_ACCOUNT }

 - name: Submit Batch Job
 run: |
```

```
az batch job create --id ci-job-{{ github.run_number }} --pool-id crackz-pool
az batch task create \
 --job-id ci-job-{{ github.run_number }} \
 --task-id detect \
 --command-line "docker run crackz-cli:latest detect run --project /data/project"
```

- 1.
- 2.
- 3.
- 4.
- 5.

taskSlotsPerNode

```
Modify pool to allow multiple tasks per node
az batch pool set \
 --pool-id crackz-pool \
 --account-name mycrackzaccount \
 --json-file pool-config.json
```

pool-config.json

```
{
 "taskSlotsPerNode": 4
}
```

```
ai:
 detection:
 batch_size: 16 # Adjust based on VM size
```

- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

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## VM vs Batch

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# Azure Deployment Comparison

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|  |  |  |
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|--|--|--|

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```
Create Windows 11 VM
./scripts/azure/azure_vm.sh \
 --name crackz-dev \
 --size Standard_D4s_v3

Connect via RDP and use GUI
IP address displayed after creation
```

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```
Create batch pool with 10 nodes
./scripts/azure/azure_batch.sh \
--batch-account mybatchaccount \
--storage-account mystorage \
--node-count 10 \
--vm-size Standard_D4s_v3

Submit 100 detection jobs
for i in {1..100}; do
 az batch task create \
 --job-id detection-job \
 --task-id "task-$i" \
 --command-line "docker run crackz-cli:latest detect run --project /data/project-$i"
done
```

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- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

```
Phase 1: Development on VM
./scripts/azure/azure_vm.sh --name crackz-dev --size Standard_D4s_v3
RDP in, use GUI to develop and test
Export config: crackz export-config --project demo --out optimized.yaml

Phase 2: Production on Batch
./scripts/azure/azure_batch.sh \
--batch-account prod-batch \
--pool-id production-pool \
```

```
--node-count 20 \
--vm-size Standard_D8s_v3

Upload optimized config and data to storage
Submit batch jobs with optimized parameters
```

|  |             |                |
|--|-------------|----------------|
|  |             |                |
|  | azure_vm.sh | azure_batch.sh |

- 1.
- 2.
- 3.
- 4.

- 1.
- 2.
- 3.
- 4.
- 5.

- 1.
- 2.
- 3.
- 4.

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```
1. Develop on VM
./scripts/azure/azure_vm.sh --name dev-vm

2. Test configuration
crackz detect train --project my-project --epochs 50
crackz detect run --project my-project

3. Export configuration
crackz export-config --project my-project --out batch-config.yaml

4. Upload to Azure Storage
az storage blob upload-batch \
 --account-name mystorage \
 --destination crackz-data \
 --source my-project

5. Create Batch pool
./scripts/azure/azure_batch.sh \
 --batch-account mybatch \
 --storage-account mystorage \
 --node-count 10

6. Submit batch jobs
az batch job create --id production-job --pool-id crackz-pool
... add tasks
```

- `scripts/azure/azure_vm.sh --help`
- `_____`
- `_____`
- `_____`

`--size Standard_NC*`

`az batch task file download`

- `_____`
- `_____`
- `_____`
- `_____`

---

## Model Evaluation

---

# Model Evaluation

---

```
Run evaluation on test dataset
crackz evaluate run --project my_harbor

Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,iou

Export results
crackz evaluate run --project my_harbor --output results.json
crackz evaluate run --project my_harbor --output results.csv
```

```
from pathlib import Path
from crackz.ai.evaluation import evaluate_model, export_json

Run evaluation
results = evaluate_model(
 project_path=Path("my_harbor"),
 metrics=["accuracy", "precision", "recall", "iou"]
)

Display results
print(results)

Export to JSON
export_json(results, Path("evaluation_results.json"))
```

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$
$$\text{Precision} = TP / (TP + FP)$$
$$\text{Recall} = TP / (TP + FN)$$
$$F1 = 2 * (Precision * Recall) / (Precision + Recall)$$
$$\text{IoU} = \text{Area of Overlap} / \text{Area of Union}$$

```
Default threshold (0.5)
crackz evaluate run --project my_harbor
```

```
Custom threshold
crackz evaluate run --project my_harbor --threshold 0.7
```

```
All metrics (default)
crackz evaluate run --project my_harbor
```

```
Specific metrics
crackz evaluate run --project my_harbor --metrics accuracy,precision,recall
crackz evaluate run --project my_harbor --metrics iou # Only IoU
```

```
accuracy precision recall f1 iou
```

#### Evaluation Results (500 images)

```
=====
Accuracy: 92.3%
Precision: 89.1%
Recall: 94.8%
F1 Score: 91.9%
Mean IoU: 0.847
Threshold: 0.5
```

```
{
 "num_images": 500,
 "threshold": 0.5,
 "metrics": {
 "accuracy": 0.923,
 "precision": 0.891,
 "recall": 0.948,
 "f1_score": 0.919,
 "iou": 0.847
 }
}
```

```
Metric,Value
Num Images,500
Threshold,0.5
Accuracy,0.9230
Precision,0.8910
Recall,0.9480
F1 Score,0.9190
IoU,0.8470
```

```
project/
├─ images/
│ └─ test/ # Test images
│ ├── img1.png
│ ├── img2.png
│ └─ ...
└─ masks/
 └─ test/ # Ground truth masks
 └─ img1.png
```

```
└─ img2.png
└─ ...
```

```
def evaluate_model(
 project_path: Path,
 test_images: list[Path] | None = None,
 test_masks: list[Path] | None = None,
 metrics: list[str] | None = None,
 threshold: float = 0.5,
) -> EvaluationResults:
 """Evaluate crack detection model on test dataset."""
```

```

 project_path test_images
test_masks threshold metrics
 EvaluationResults
```

```
@dataclass
class EvaluationResults:
 accuracy: float
 precision: float
 recall: float
 f1_score: float
 iou: float
 num_images: int
 threshold: float
```

```
Export to JSON
export_json(results, Path("results.json"))

Export to CSV
export_csv(results, Path("results.csv"))
```

```
ValueError: No test images found
```

```
<project>/images/test/
```

```
ls <project>/images/test/
```

```
ValueError: Image/mask count mismatch: 100 vs 98
```

```
Check counts
ls <project>/images/test/ | wc -l
ls <project>/masks/test/ | wc -l

Find missing masks
diff <(ls <project>/images/test/) <(ls <project>/masks/test/)
```

```
FileNotFoundError: No trained model checkpoint found
```

```
crackz detect train --project <project>
```

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# Advanced Performance

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# Advanced Performance & Reproducibility

|           | ai.training | ai.detection | ProjectConfig |
|-----------|-------------|--------------|---------------|
|           |             |              |               |
| precision | ai.training |              |               |
| devices   | ai.training |              |               |
| strategy  | ai.training |              | ddp           |
| seed      | ai.training | ai.detection |               |

321616-mixedbf16-mixed

```
ai:
 training:
 precision: 16-mixed
 batch_size: 16
```

```
crackz detect train --project demo-project
```

devicesstrategyddp

```
ai:
 training:
 devices: 2
 strategy: ddp
 batch_size: 8
 precision: bf16-mixed
```

```
crackz detect train --project demo-project
```

```
docker run --gpus all --rm -v %cd%\project:/home/app crackz-cli:gpu \
detect train --project /home/app/demo-project
```

```
ddp
ddp_spawn
auto
```

seed

```
ai:
 training:
```

```
seed: 1234
detection:
seed: 1234
```

PYTHONHASHSEED

random

```
import torch
torch.use_deterministic_algorithms(True)
```

- 
- 

```
project:
 name: demo-project
 description: Perf tuned
paths:
 project_path: demo-project
 image_path: demo-project/images
 image_mask_path: demo-project/masks
 artifacts_path: demo-project/artifacts
logging:
 level: INFO
 log_dir: logs
ai:
 model:
 encoder: resnet34
 device: cuda
 input_size: [512, 512]
 threshold: 0.5
 training:
 batch_size: 16
 epochs: 20
 learning_rate: 0.0005
 precision: 16-mixed
 devices: 2
 strategy: ddp
 seed: 4242
 detection:
 batch_size: 8
 seed: 4242
```

crackz export-config

```
crackz export-config --project demo-project --out tuned_config.yaml
```

tuned\_config.yaml

ai.detection.seed

ddp\_spawn

```
torch.use_deterministic_algorithms(True)
```

⋮

---

# Migration Guide

---

# Project Schema Migration

ai.training.tensorboard schema\_version

CURRENT\_SCHEMA\_VERSION crackz.project.migration schema\_version schema\_version

|                   |                                |  |
|-------------------|--------------------------------|--|
|                   |                                |  |
| schema_version    | ai.training.tensorboard        |  |
| schema_version: 1 | ai.training.tensorboard: false |  |
| data              | mask_threshold: 1              |  |
|                   | project.crackz_version         |  |

```
schema_version: 3
project:
 name: my project
 description: My crack detection project
 created: '2025-01-15T10:30:00Z'
 crackz_version: '0.50.0' # NEW: Tracks Crackz app version
```

project.crackz\_version

checkpoints/ models/ runtime/

detections/visualizations/

checkpoints\_path checkpoints/

models/ runtime/ crackz\_version

data

```
data:
 mask_threshold: 1
 schema_version: 2
```

data.mask\_threshold

detect train detect run

- 1.
- 2. Project schema v0 < v1. Migrate now? [y/N]:
- 3. .bak
- 4.
- 5. CRACKZ\_AUTO\_MIGRATE=1
- 6.

```
crackz migrate --project <path-to-project-dir>
or
crackz migrate --cfg <path-to-project-yaml>
```

```
-migrate-files --dry-run --no-backup <file>.bak -
```

```
Using project directory
crackz migrate --project my_project --dry-run
crackz migrate --project my_project

Using direct YAML file path
crackz migrate --cfg my_project/my_project_project.yaml
```

```
Preview what files would be moved
crackz migrate --project my_project --migrate-files --dry-run

Migrate both schema AND files
crackz migrate --project my_project --migrate-files
```

```
runtime/lightning_logs/ checkpoints/ models/ training/tensorboard/ runtime/tensorboard/ training/lightning_logs/
detections/*.png detections/visualizations/
```

```
{filename}.v{version}.{timestamp}.bak
```

```
myproject_project.v2.20250115_143022.bak
```

```
v2
```

```
20250115_143022
```

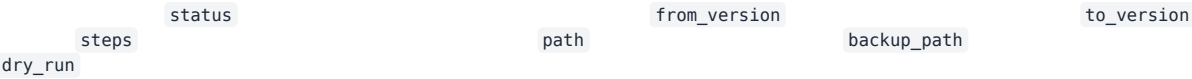
```
myproject_project.v2.20250115_143022.bak
.v2.20250115_143022.bak
```

```
CRACKZ_AUTO_MIGRATE=1
```

```
from pathlib import Path
from crackz.project.migration import migrate_project_file

result = migrate_project_file(Path("my_project")) # or path/to_yaml
print(result)
Example result:
{
'status': 'migrated',
'from_version': 2,
'to_version': 3,
'steps': ['2->3: updated to artifacts structure v3 + added crackz_version tracking'],
'path': 'my_project/my_project_project.yaml',
'backup_path': 'my_project/my_project_project.v2.20250115_143022.bak',
```

```
'dry_run': False
}
```



- 1.
- 2.



# Architecture

# System Architecture

```
crackz/
├── pyproject.toml # Workspace root
├── core/ # Workspace config, dev deps, tool settings (no [project])
│ ├── pyproject.toml # crackz-core package
│ ├── src/crackz/ # Package metadata and dependencies
│ ├── tests/ # Core library (AI, CLI, config, logging, project)
│ └── tests/ # Core unit + integration tests
├── gui/ # crackz-gui package (depends on crackz-core)
│ ├── pyproject.toml # Package metadata, dependency on crackz-core
│ ├── src/crackz gui/ # PySide6 Qt GUI application
│ └── tests/ # GUI unit + integration tests
└── tests/ # Root-level tests (cross-package, legacy)
```

```
core/
crackz_gui.qt import ...

uv sync --dev

gui/
from crackz.ai import ...

core/
from
```



```
graph TD
 subgraph L4["Layer 4 – Presentation"]
 CLI["crackz.cli"]
 GUI["crackz_gui"]
 DASH["crackz.dashboard"]
 end
 subgraph L3["Layer 3 – Application Façade"]
 APP["crackz.app"]
 end
 subgraph L2["Layer 2 – Domain"]
 AI["crackz.ai"]
 PROJECT["crackz.project"]
 REPORT["crackz.report"]
 end
 subgraph L1["Layer 1 – Infrastructure"]
 CONFIG["crackz.config"]
 LOG["crackz.log"]
 TRACING["crackz.tracing"]
 UTIL["crackz.util"]
 end

 CLI --> APP
 GUI --> APP
 DASH --> APP
 APP --> AI
 APP --> PROJECT
 APP --> REPORT
 AI --> CONFIG
 AI --> LOG
 AI --> UTIL
 PROJECT --> CONFIG
 PROJECT --> LOG
 PROJECT --> UTIL
 REPORT --> CONFIG
 REPORT --> LOG
 CONFIG --> LOG
 CONFIG --> UTIL
```

|                     |                  |                |                  |                  |                               |               |    |
|---------------------|------------------|----------------|------------------|------------------|-------------------------------|---------------|----|
|                     |                  |                |                  |                  |                               |               |    |
|                     | crackz.cli       |                | crackz_gui       | crackz.dashboard |                               |               |    |
|                     | crackz_gui       |                | crackz.dashboard |                  | crackz.cli                    |               |    |
|                     | crackz.dashboard |                |                  |                  | crackz.cli                    | crackz_gui    |    |
|                     | crackz.app       |                |                  |                  |                               |               |    |
|                     | crackz.ai        |                |                  |                  |                               |               |    |
|                     | crackz.project   |                |                  |                  |                               |               |    |
|                     | crackz.report    |                |                  |                  |                               |               |    |
|                     | crackz.config    | crackz.project | crackz.log       | crackz.util      | crackz.ai                     | crackz.report |    |
|                     | crackz.log       |                |                  |                  |                               |               |    |
|                     | crackz.tracing   | crackz.log     |                  |                  |                               |               |    |
|                     | crackz.util      | crackz.log     |                  |                  |                               |               |    |
|                     |                  |                |                  |                  |                               |               |    |
|                     |                  |                |                  | .importlinter    | scripts/check_architecture.py |               | uv |
| run task pre_commit |                  |                |                  |                  |                               |               |    |

|         |  |  |                        |                   |                    |  |
|---------|--|--|------------------------|-------------------|--------------------|--|
|         |  |  |                        |                   | pydantic-settings  |  |
|         |  |  | raw/ train/ val/ test/ |                   | CrackzDataset      |  |
| Dataset |  |  |                        |                   |                    |  |
| loguru  |  |  |                        | artifacts/models/ | artifacts/runtime/ |  |

|                                |                                |                               |                             |  |  |           |
|--------------------------------|--------------------------------|-------------------------------|-----------------------------|--|--|-----------|
|                                |                                |                               |                             |  |  |           |
|                                | crackz.db                      |                               | crackz_gui.data.projects_db |  |  |           |
|                                | <project>/artifacts/project.db |                               | crackz.ai.model.registry    |  |  |           |
|                                |                                |                               |                             |  |  |           |
|                                |                                | artifacts/project.db          |                             |  |  | crackz.db |
|                                |                                |                               |                             |  |  |           |
| <project>/artifacts/project.db |                                | <project>/artifacts/crackz.db |                             |  |  |           |
|                                |                                | crackz.db                     |                             |  |  |           |

|            |  |  |  |  |  |
|------------|--|--|--|--|--|
|            |  |  |  |  |  |
| binary     |  |  |  |  |  |
| multiclass |  |  |  |  |  |

num\_classes

- 0 1 255
- {0, 1, 255}

ai.model.task\_mode binary  
task\_mode binary multiclass

- 1.
- 2.
- 3.
4. crackz detect train
- 5.
- 6.
- 7.
- 8.
9. crackz detect run
- 10.

artifacts/models/

segmentation-models-pytorch

project.yaml

```
ai:
 model:
 backbone: segformer-b2 # or segformer-b0/b4, efficientnet-b3, resnet34, etc.
 annotation:
 categories:
 - id: 1
 key: crack
 name: Crack
 color: "#ef4444"
```

crack crackz init --name harbor --

annotation-template crack-only

corrosion joint\_damage crack spall  
harbor-defects crackz init --name harbor --annotation-template

## annotation.categories

```

C4Container
title Crackz – Container View (Implemented)
Person(user, "User")
System_Boundary(crackz, "Crackz Local System") {
 Container(cli, "CLI", "Typer", "Command parsing & dispatch")
 Container(gui, "GUI", "PySide6 (Qt 6)", "Desktop inspection & invocation")
 Container(core, "Core Library", "Python Packages", "project, ai, config, logging")
 Container(fs, "Local Filesystem", "images/ + artifacts/", "Images, masks, checkpoints, metrics")
 ContainerDb(udb, "User DB", "SQLite", "Known projects + welcome aggregates")
 ContainerDb(pdb, "Project DB", "SQLite", "Per-project models, detections, training runs")
}
Rel(user, cli, "Runs commands")
Rel(user, gui, "Interacts (point & click)")
Rel(cli, core, "Invokes APIs")
Rel(gui, core, "Invokes APIs")
Rel(core, fs, "read/write images, masks, checkpoints, metrics")
Rel(gui, udb, "read/write project index")
Rel(core, pdb, "read/write project metadata")

```

```

C4Context
title Crackz – System Context (Local)
Person(user, "User", "Engineer / Operator")
System(local, "Crackz Application", "CLI + GUI + Core Library")
System_Ext(osfs, "Host Filesystem", "Images + Artifacts")
Rel(user, local, "Invoke CLI / Launch GUI")
Rel(local, osfs, "read / write")

```

```

C4Component
title Core Library Components (Implemented)
Container(core, "core", "Python")
Component(project, "Project & Paths", "pydantic", "Project metadata & path helpers")
Component(config, "Configuration", "pydantic-settings", "Merge defaults / env / YAML")
Component(importer, "Image Importer", "Pillow", "Resize, quality, organize splits & masks")
Component(dataset, "Dataset Layer", "Torch Dataset", "CrackzDataset + DataLoader factory")
Component(train, "Training Orchestrator", "PyTorch Lightning", "Fit/validate/test + callbacks")
Component(detect, "Detection Pipeline", "PyTorch", "Batch inference over dataset splits (raw/test)")
Component(metrics, "Metrics & Checkpoints", "Lightning + JSON", "Save val/test metrics & best.ckpt/last.ckpt")
Component(perf, "Performance Utilities", "Seeding", "Seed + trainer overrides")
Component(logging, "Logging", "loguru", "Structured log sink & setup")
Component(evaluation, "Evaluation Framework", "Metrics", "IoU, precision, recall, F1, accuracy")
Component(reporter, "Report Generator", "python-docx", "Branded Word reports")
Component(viz, "Visualization", "Matplotlib", "Detection overlays (3-panel)")
Component(pipeline_orch, "Pipeline Orchestrator", "Workflow", "Train-detect + cancellation")
Component(device_mgmt, "Device Management", "Abstraction", "CPU/GPU/MP5 selection")
Component(severity, "Severity Classifier", "Categorization", "3-tier crack severity")
Component(suggester, "Mask Suggester", "Semi-automation", "Model-assisted annotation")
Component(perf_est, "Performance Estimator", "RuntimeTracker", "Time estimates")
Component(async_cb, "Async Callbacks", "Janus", "Sync/async bridging")
Component(tracing_mod, "Tracing", "OpenTelemetry", "AI API observability")
Component(ai_chat_svc, "AI Chat", "Anthropic", "User assistance")
Rel(importer, project, "Uses path layout")
Rel(dataset, project, "Resolves split directories")
Rel(train, dataset, "Consumes (train/val/test) loaders")
Rel(detect, dataset, "Consumes detection loader (raw or test)")
Rel(train, metrics, "Saves checkpoints + metrics")
Rel(detect, metrics, "Writes detection results JSON/PNGs")
Rel(project, config, "Initializes from effective settings")
Rel(train, perf, "Reads overrides / seed")
Rel(detect, perf, "Sets seed (optional)")
Rel(dataset, logging, "Reports warnings (missing masks / size mismatch)")
Rel(detect, evaluation, "Evaluates performance")
Rel(evaluation, metrics, "Calculates metrics")
Rel(detect, viz, "Visualizes results")
Rel(detect, reporter, "Generates reports")
Rel(train, pipeline_orch, "Orchestrated by")
Rel(detect, pipeline_orch, "Orchestrated by")
Rel(train, device_mgmt, "Selects device")
Rel(detect, device_mgmt, "Selects device")
Rel(detect, severity, "Categorizes cracks")
Rel(suggester, detect, "Uses model")
Rel(train, perf_est, "Tracks runtime")
Rel(detect, perf_est, "Tracks runtime")
Rel(train, async_cb, "Reports progress")
Rel(detect, async_cb, "Reports progress")
Rel(ai_chat_svc, tracing_mod, "Optionally traced")

```

```

flowchart LR
 A["Test Dataset"] --> B["Evaluator"]
 B --> C["Metrics Calculator"]
 C --> D["Evaluation Results"]
 D --> E["JSON/CSV Reporter"]
 E --> F["Report Generator"]
 F --> G["Branded Word Document"]
 I["Detection Results"] --> J["Visualization"]
 J --> K["Detection Overlays"]
 I --> F

```

```

flowchart TB
 A["Training Thread (sync)"] --> B["CallbackDispatcher.sync_q"]
 B --> C["Janus Queue"]
 C --> D["CallbackDispatcher.async_q"]
 D --> E["GUI Event Loop (async)"]
 E --> F["UI Update"]

```

```

flowchart LR
 A["Import Command / GUI Action"] --> B["Image Importer"]
 B --> C["Split Directories (images/*, masks/*)"]
 C --> D["Train Command"]
 D --> E["Training Orchestrator
(Trainer.fit)"]
 E --> F["Checkpoints & Metrics
(best.ckpt / last.ckpt / JSON)"]
 F --> G["Detect Command"]
 G --> H["Inference Loop
(sigmoid + threshold)"]
 H --> I["Annotated Output
(PNGs + detection_results.json)"]

```

```

sequenceDiagram
 participant User
 participant CLI as CLI / GUI
 participant Project
 participant Import as ImageImporter
 participant FS as Filesystem
 User->>CLI: import-images --src ./samples --type train
 CLI->>Project: load_project(path|yaml)
 CLI->>Import: iterate source images
 Import->>FS: resize + normalize + write split file
 Import->>FS: write mask (if provided)
 Import->>CLI: progress callback
 CLI->>User: summary (count, skipped)

```

```

sequenceDiagram
 participant User
 participant CLI as CLI / GUI
 participant Project
 participant DS as CrackzDataset
 participant Trainer as Lightning Trainer
 participant Model
 participant FS as Filesystem
 User->>CLI: detect train --project P
 CLI->>Project: load_project()
 CLI->>DS: build train/val/test datasets
 DS->>CLI: DataLoaders
 CLI->>Trainer: fit(model, loaders)
 Trainer->>FS: save best.ckpt / last.ckpt
 Trainer->>CLI: metrics (val/test)
 CLI->>User: training summary
 User->>CLI: detect run --project P
 CLI->>Project: load_project()
 CLI->>Model: load checkpoint (best or last)
 CLI->>DS: build raw dataset (or test)
 DS->>CLI: DataLoader
 CLI->>Model: forward (batched)
 Model->>CLI: probabilities

```

```
CLI-->CLI: threshold -> mask
CLI-->FS: write annotated PNG + JSON metrics
CLI-->User: detection summary
```

```
flowchart TD
 Defaults[Built-in Defaults] --> Merge
 YAML[Project YAML] --> Merge
 Env[Environment] --> Merge
 Merge --> Effective[Effective Settings]
```

*project defaults*

*catalog of trained models*

```
ai.model.backbone project.yaml
ai.model.active_checkpoint
artifacts/project.db
```

*Start Training*  
*Model Selector*

```
active_checkpoint ai.model.backbone

Start Training Save as new project default
project.yaml
```

```
flowchart LR
 User([User]) --> CLI["CLI (Typer)"]
 User --> GUI["GUI (PySide6)"]
 CLI --> CORE["Core Library"]
 GUI --> CORE
 CORE --> FS["Filesystem
images / artifacts/models / artifacts/runtime"]
 CORE -.optional.-> GPU["GPU"]

 subgraph Optional_Docker [Optional Docker]
 IMG["Container Image
(Python 3.13 + deps)"]
 end

 %% Show that container can host CLI/GUI
 IMG --> CLI
 IMG --> GUI
```

```
crackz.project.project
pydantic-settings
project.import_images
crackz.ai.data
crackz.ai.detection.train
crackz.ai.detection.detect
crackz.ai.model
crackz.ai.checkpoints
crackz.ai.device
```

crackz.util  
crackz.ai.performance  
crackz.log.crackz\_logger  
crackz\_gui.\*  
crackz.ai.evaluation  
crackz.report.generator  
crackz.ai.visualization  
crackz.ai.pipeline  
crackz.ai.pipeline.cancellation  
crackz.ai.detection.severity  
crackz.ai.mask\_suggester  
crackz.ai.performance.runtime\_estimation  
crackz.ai.core.async\_callbacks  
crackz.tracing  
crackz\_gui.services.ai\_chat

test/

raw/

train/

val/

- 
- - 
  - 
  - 
  - 
  - 
  - 
  - 
  - 
  - 
  - 
  -
- 

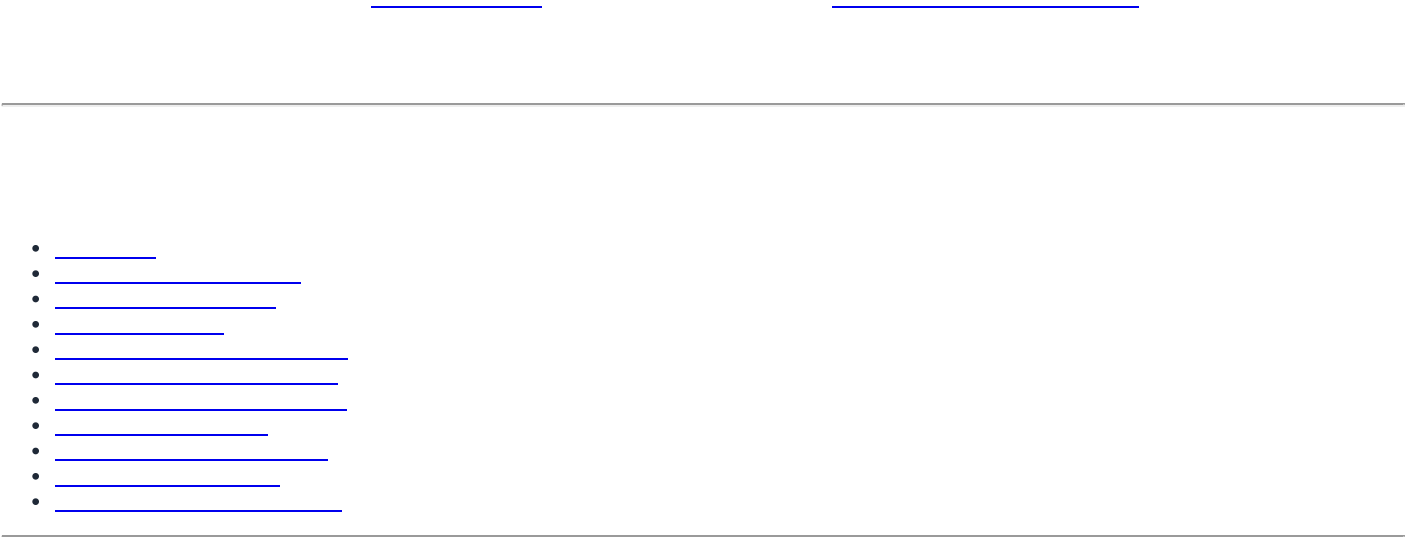
core/

gui/

*Last updated: synchronized with repository implementation on commit timeline of current editing session.*

# Architecture Guidelines

# Architecture Guidelines



|                                       |                      |
|---------------------------------------|----------------------|
| User Interfaces (CLI, GUI)            | ↳ Entry points       |
| Business Logic (Commands, Operations) | ↳ Use cases          |
| Core Domain (Project, AI, Config)     | ↳ Domain models      |
| Infrastructure (Logging, Storage)     | ↳ Technical concerns |

```
[] GOOD: Clear separation
core/src/crackz/cli/detection_cli.py - CLI commands only
core/src/crackz/ai/detection.py - Detection logic only
core/src/crackz/project/structure.py - Project layout only

x BAD: Mixed concerns
core/src/crackz/everything.py - CLI, detection, project management all in one file
```

```
[] GOOD: Depend on protocol
from typing import Protocol

class ModelProtocol(Protocol):
 def predict(self, image: torch.Tensor) -> torch.Tensor: ...

def run_detection(model: ModelProtocol, image: torch.Tensor) -> dict:
 """Works with any model implementing the protocol."""
 return {"mask": model.predict(image)}

x BAD: Depend on concrete class
def run_detection(model: SegmentationModel, image: torch.Tensor) -> dict:
 """Tightly coupled to SegmentationModel."""
 return {"mask": model.predict(image)}
```

```
[] GOOD: Single responsibility
class ProjectLoader:
 """Loads project configuration from disk."""
 def load(self, path: Path) -> ProjectConfig: ...

class ProjectValidator:
 """Validates project structure and configuration."""
 def validate(self, project: ProjectConfig) -> list[str]: ...

x BAD: Multiple responsibilities
class ProjectManager:
 """Loads, validates, trains models, exports results."""
 def load(self, path: Path) -> ProjectConfig: ...
 def validate(self, project: ProjectConfig) -> list[str]: ...
 def train_model(self, project: ProjectConfig) -> None: ...
 def export_results(self, project: ProjectConfig) -> None: ...
```

```
[] GOOD: Extensible via configuration
def create_model(config: ModelConfig) -> nn.Module:
 """Factory method - add new encoders via config."""
 if config.encoder in SUPPORTED_ENCODERS:
 return create_segmentation_model(config)
 raise ValueError(f"Unsupported encoder: {config.encoder}")

x BAD: Requires code changes for new models
def create_model(encoder_name: str) -> nn.Module:
 if encoder_name == "resnet34":
 return ResNet34Model()
 elif encoder_name == "efficientnet":
 return EfficientNetModel()
 # Must modify this function for each new encoder!
```

```
[] GOOD: Explicit parameters
def train_model(
 project_path: Path,
 epochs: int,
 batch_size: int,
 learning_rate: float,
) -> dict[str, float]:
 """All parameters explicit."""
 pass
```

```
x BAD: Implicit global state
GLOBAL_CONFIG = {} # Modified elsewhere

def train_model():
 """What parameters? Where do they come from?"""
 epochs = GLOBAL_CONFIG["epochs"] # Implicit dependency
```

```
core/src/crackz/ # Core library (crackz-core package)
├─ ai/ # AI/ML pipeline (domain logic)
│ ├── data.py # Dataset, data loading
│ ├── models.py # Model definitions
│ ├── training.py # Training pipeline
│ └── detection.py # Inference pipeline
├─ cli/ # CLI interface (entry points)
│ ├── options.py # Shared CLI options
│ ├── exceptions.py # CLI error handling
│ ├── project_cli.py
│ └── detection_cli.py
├─ config/ # Configuration (domain models)
│ ├── project.py # ProjectConfig
│ ├── ai.py # AIConfig
│ └── logging.py # LogConfig
├─ log/ # Logging (infrastructure)
│ ├── setup.py # Loguru configuration
├─ project/ # Project management (domain logic)
│ └── structure.py # Directory layout

gui/src/crackz_gui/ # GUI package (crackz-gui, depends on crackz-core)
├─ qt/ # PySide6 Qt application
│ ├── app_state.py # QtAppState (reactive state)
│ └── async_bridge.py # TaskSignals, run_in_background
├─ services/ # External service integrations
├─ state/ # State management
└─ util/ # GUI utilities
```

```
CLI (core) → Business Logic → Domain Models → Infrastructure
GUI (gui) → (ai/, project/) (config/) (log/)
```

Examples:

```
- core: cli/detection_cli.py → ai/detection.py → config/ai.py → log/setup.py □
- gui: crackz_gui/qt/panels/train.py → crackz.ai.training → crackz.config □
- gui: crackz-gui depends on crackz-core (declared in gui/pyproject.toml) □
```

```
Domain →X- GUI/CLI # Domain shouldn't know about UI
Infrastructure →X- Domain # Infrastructure shouldn't depend on business logic
GUI →X- CLI # GUI shouldn't import CLI modules
```

Examples:

```
- ai/detection.py → cli/options.py x # Domain shouldn't import from CLI
- log/setup.py → config/project.py x # Infrastructure shouldn't import domain
- crackz_gui → crackz.cli x # GUI should use core APIs, not CLI
```

```
x BAD: Domain imports from CLI
from crackz.cli.options import PROJECT_PATH_OPTION
def train_model(project_path: Path = PROJECT_PATH_OPTION): ...

□ GOOD: CLI calls domain with explicit parameters
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.ai.training import train_model

def train_cli(project_path: Path = PROJECT_PATH_OPTION):
 train_model(project_path) # Domain function has no CLI dependency
```

core/src/crackz/ai/models.py

```
def create_segmentation_model(config: ModelConfig) -> nn.Module:
 """Factory for creating segmentation models."""
 import segmentation_models_pytorch as smp

 return smp.Unet(
 encoder_name=config.encoder,
 encoder_weights=config.encoder_weights,
 in_channels=config.in_channels,
 classes=config.num_classes,
)
```

core/src/crackz/ai/detection.py

```
class DetectionStrategy(Protocol):
 def detect(self, images: list[Path]) -> dict[Path, np.ndarray]: ...

class SingleImageDetection:
 def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
 """Process one at a time."""
 return {img: self._detect_one(img) for img in images}

class BatchDetection:
 def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
 """Process in batches for efficiency."""
 results = {}
 for batch in chunk_list(images, self.batch_size):
 results.update(self._detect_batch(batch))
 return results
```

core/src/crackz/config/project.py

```
class ProjectConfig(BaseModel):
 """Pydantic acts as builder with validation."""
 name: str
 project_path: Path
 training: TrainingConfig = Field(default_factory=TrainingConfig)
 logging: LogConfig = Field(default_factory=LogConfig)

 @classmethod
 def load(cls, path: Path) -> ProjectConfig:
 """Build from YAML with defaults."""
 with open(path) as f:
 data = yaml.safe_load(f)
 return cls(**data) # Validation + defaults applied
```

core/src/crackz/cli/\*.py

```
@app.command(name="train")
def train_command(
 project_path: Path = PROJECT_PATH_OPTION,
 epochs: int | None = EPOCHS_OPTION,
) -> None:
 """Each CLI command is a self-contained operation."""
 try:
 # Load configuration
 config = ProjectConfig.load(project_path / f"{project_path.name}_project.yaml")

 # Override with CLI args
 if epochs:
 config.training.epochs = epochs

 # Execute
 train_model(config)
```

```
typer.secho(" Training completed", fg=typer.colors.GREEN)
except Exception as e:
 cli_failure(e, "Training")
```

core/src/crackz/project/structure.py

```
class ProjectRepository:
 """Abstracts project storage."""

 def load(self, path: Path) -> ProjectConfig:
 """Load project from filesystem."""
 return ProjectConfig.load(path / f"{path.name}_project.yaml")

 def save(self, project: ProjectConfig) -> None:
 """Save project to filesystem."""
 project.save(project.project_path / f"{project.name}_project.yaml")

 def exists(self, path: Path) -> bool:
 """Check if project exists."""
 return (path / f"{path.name}_project.yaml").exists()
```

gui/src/crackz\_gui/data/projects\_db.py

core/src/crackz/ai/model/registry.py

<project>/artifacts/project.db

<project>/artifacts/project.db  
crackz.db

crackz.db

crackz.db

artifacts/project.db

```
User-level DB: known projects + cached aggregate stats.
projects_db = ProjectsDB()
projects_db.upsert_project(str(project.root), name=project.name)
projects_db.update_project_stats(
 str(project.root),
 image_count=summary.image_count,
 defect_count=summary.defect_count,
 model_count=summary.model_count,
 best_val_iou=summary.best_val_iou,
)

Project-level DB: authoritative project operation history.
registry = ModelRegistry(project.registry_db_path, project=project)
registry.record_detection_run(detection_run)
registry.upsert_training_epoch(training_epoch)
```

ProjectsDB(db\_path=...)

registry\_db\_path

artifacts/project.db

crackz.db

project.db

core/src/crackz/ai/core/async\_callbacks.py

RuntimeError: no running event loop

sync\_q

async\_q

```
from crackz.ai.core.async_callbacks import CallbackDispatcher

In training setup (main thread)
dispatcher = CallbackDispatcher(maxsize=100)

In PyTorch Lightning callback (training thread)
sync_reporter = dispatcher.create_sync_progress_reporter()
sync_reporter(0.5, "Training epoch 5/10") # Thread-safe put

In GUI (async event loop)
```

```

async for progress, message in dispatcher.iter_progress():
 update_progress_bar(progress, message) # Async get

```

AsyncProgressReporter AsyncMetricsReporter

core/src/crackz/tracing/

--extra tracing

```

In core/src/crackz/tracing/setup.py
from crackz.tracing import setup_tracing, is_tracing_enabled

Automatic instrumentation on setup
if is_tracing_enabled():
 setup_tracing() # Instruments Anthropic client automatically

The instrumentation happens automatically via AnthropicInstrumentor
No manual client instrumentation needed

```

```

Install tracing dependencies
uv sync --extra tracing

Enable at runtime
export CRACKZ_TRACING_ENABLED=1

```

core/src/crackz/ai/evaluation/reporter.py core/src/crackz/report/generator.py

```

from crackz.ai.evaluation.reporter import export_json, export_csv
from crackz.report.generator import ReportGenerator

Aggregation (format-agnostic)
results = {
 "accuracy": 0.95,
 "precision": 0.92,
 "recall": 0.89,
 "f1_score": 0.90,
 "iou": 0.85
}

Export to JSON (lightweight)
export_json(
 results,
 output_path=project_path / "evaluation_results.json"
)

Export to CSV (analysis-friendly)
export_csv(
 results,
 output_path=project_path / "evaluation_results.csv"
)

Export to DOCX (rich formatting)
generator = ReportGenerator(
 project_path=project_path,
 template_path=template_path
)
generator.generate_report(
 detection_results=results,

```

```

 output_path=project_path / "report.docx"
)

```

```
core/src/crackz/ai/device/device.py
```

```
torch.device("cuda")
```

```

from crackz.ai.device.device import AIDevice

Auto-select best available device
device = AIDevice.default() # Returns CUDA if available, else MPS, else CPU

Check availability
available = AIDevice.available_devices() # [AIDevice.CPU, AIDevice.CUDA]

Test device works
success, message, details = AIDevice.test_device("cuda")
if success:
 print(f"CUDA available: {details['device_name']}")

Use in model
model = SegmentationModel()
model.to(device.value) # device.value is "cuda", "mps", "cpu", etc.

```

```
core/src/crackz/ai/detection/severity.py
```

```

from crackz.ai.detection.severity import categorize_crack_severity

Default thresholds: 5% (high), 2% (medium)
severity = categorize_crack_severity(confidence=0.08) # "Crack Detected"
severity = categorize_crack_severity(confidence=0.03) # "Suspected Crack"
severity = categorize_crack_severity(confidence=0.01) # "No Crack"

Custom thresholds via config
severity = categorize_crack_severity(
 confidence=0.04,
 high_threshold=0.06, # Raise "Crack Detected" threshold
 medium_threshold=0.03 # Raise "Suspected" threshold
) # "Suspected Crack" (was "Crack Detected" with defaults)

Override with has_crack flag (confidence + area thresholds met)
severity = categorize_crack_severity(confidence=0.01, has_crack=True)
"Crack Detected" (flag indicates combined criteria met)

```

```

project.yaml
ai:
 detection:
 high_confidence_threshold: 0.05 # 5% crack area
 medium_confidence_threshold: 0.02 # 2% crack area
 min_confidence: 0.005 # 0.5% minimum to save

```

```
core/src/crackz/cli/options.py
PROJECT_PATH_OPTION = typer.Option(
 None,
 "--project",
 help="Path to project directory",
)

FORCE_OPTION = typer.Option(
 False,
 "--force",
 help="Force operation without confirmation",
)

core/src/crackz/cli/some command.py
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION

def my_command(
 project_path: Path | None = PROJECT_PATH_OPTION,
 force: bool = FORCE_OPTION,
) -> None:
 pass
```

```
print() logger typer.echo()
```

```
from crackz.log import logger
import typer

User-facing success/error messages
typer.secho("[] Detection completed", fg=typer.colors.GREEN)
typer.secho("x Training failed", fg=typer.colors.RED)

Technical/operational logs
logger.info("Loading {} images from {}", len(images), path)
logger.debug("Model config: {}", config.model_dump())
logger.error("Failed to load checkpoint: {}", error, exc_info=True)
```

```
typer.secho()
```

```
print() typer.echo()
```

```
logger.*
```

```
from crackz.cli.exceptions import cli_failure

def train_command():
 try:
 # Business logic
 result = train_model(project_path)
 typer.secho("[] Training completed", fg=typer.colors.GREEN)
 except FileNotFoundError as e:
 cli_failure(e, "File access")
 except RuntimeError as e:
 cli_failure(e, "Training execution")
 except Exception as e:
 cli_failure(e, "Unexpected error")
```

```
cli_failure
```

```
logger.error()
```

```
typer.secho()
```

```

class AIConfig(BaseModel):
 model: ModelConfig = Field(default_factory=ModelConfig)
 training: TrainingConfig = Field(default_factory=TrainingConfig)

Usage
config = AIConfig() # Loads from defaults only
config_from_yaml = AIConfig(**yaml_data) # Loads from YAML + defaults

CLI override (doesn't persist to YAML)
if cli_epochs:
 config.training.epochs = cli_epochs # Runtime only

```

```

logger.info("message")
↓
┌───────────┐
│ Loguru Router │
└───────────┘
├───> Central Log: /tmp/crackz_logs/app.log (all operations)
├───> Project Log: <project>/logs/operations.log (project-specific)

```

```

from loguru import logger

Configure at startup
logger.add(
 central_log_path,
 format="{time} {level} {message}",
 level="INFO",
)

Add project-specific handler when project opens
logger.add(
 project_log_path,
 format="{time} {level} {message}",
 level="DEBUG",
 filter=lambda record: record["extra"].get("project") == project_name,
)

Usage
logger.bind(project=project.name).info("Training started") # Goes to both logs

```

frozen=False

schema\_version

Field(description=...)

```

class ProjectConfig(BaseModel):
 """Project configuration with validation and versioning."""

 schema_version: int = Field(
 default=2,
 ge=1,
 description="Config schema version for migration support",
)

 name: str = Field(
 ..., # Required
 min_length=1,
 max_length=100,
 pattern=r"^[a-zA-Z0-9_-]+$",
 description="Project name (alphanumeric, underscore, hyphen)",
)

```

```

root: Path = Field(
 ..., # Required
 description="Project root directory",
)

ai: AIConfig = Field(
 default_factory=AIConfig,
 description="AI/ML configuration",
)

@field_validator("root")
@classmethod
def root_must_exist(cls, v: Path) -> Path:
 """Validate project root exists."""
 if not v.exists():
 raise ValueError(f"Project root does not exist: {v}")
 return v

@model_validator(mode="after")
def name_matches_root(self) -> Self:
 """Validate name matches root directory."""
 if self.root.name != self.name:
 raise ValueError(f"Name '{self.name}' doesn't match directory '{self.root.name}'")
 return self

```

```
schema_version: int
```

```
crackz migrate <project>
```

```

def migrate_v1_to_v2(config_data: dict) -> dict:
 """Migrate project config from v1 to v2."""
 if config_data.get("schema_version", 1) == 1:
 # v2 added 'ai.detection.batch_size'
 if "ai" not in config_data:
 config_data["ai"] = {}
 if "detection" not in config_data["ai"]:
 config_data["ai"]["detection"] = {}
 if "batch_size" not in config_data["ai"]["detection"]:
 config_data["ai"]["detection"]["batch_size"] = 8

 config_data["schema_version"] = 2

 return config_data

```

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

```

class CrackzError(Exception):
 """Base exception for all Crackz errors."""
 pass

class ConfigurationError(CrackzError):
 """Invalid configuration."""
 pass

class ProjectError(CrackzError):

```

```

"""Project-related error."""
pass

class ModelError(CrackzError):
 """Model loading/execution error."""
 pass

```

```

def command():
 try:
 # Business logic
 result = operation()
 typer.secho("✅ Success", fg=typer.colors.GREEN)

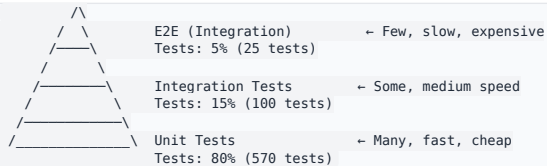
 except FileNotFoundError as e:
 # User error: file doesn't exist
 logger.error("File not found: {}", e)
 typer.secho(f"❌ File not found: {e}", fg=typer.colors.RED)
 raise typer.Exit(1)

 except ConfigurationError as e:
 # User error: invalid config
 logger.error("Configuration error: {}", e)
 typer.secho(f"❌ Configuration error: {e}", fg=typer.colors.RED)
 raise typer.Exit(1)

 except MemoryError:
 # System error: out of memory
 logger.error("Out of memory during operation")
 typer.secho("❌ Out of memory. Try reducing batch size.", fg=typer.colors.RED)
 raise typer.Exit(1)

 except Exception as e:
 # Programmer error: unexpected
 logger.exception("Unexpected error: {}", e)
 typer.secho("❌ Unexpected error. Check logs for details.", fg=typer.colors.RED)
 raise typer.Exit(1)

```



```

core/tests/
├── unit/ # Fast, isolated, no I/O
│ ├── crackz/
│ │ ├── ai/
│ │ │ ├── test_models.py # Model creation, logic
│ │ │ ├── test_training.py # Training loop logic
│ │ │ └── test_detection.py # Detection logic
│ │ ├── cli/
│ │ │ ├── test_options.py # CLI option validation
│ │ │ └── test_commands.py # Command parsing
│ │ └── config/
│ │ └── test_project.py # Config validation
│ └── integration/ # Slower, system-level, I/O allowed
│ ├── crackz/
│ │ ├── test_cli_workflows.py # End-to-end CLI
│ │ └── test_training_pipeline.py # Full training
│ └── docs/
│ └── test_docs_build.py # Documentation build
└── gui/tests/
 ├── unit/ # GUI unit tests
 └── integration/ # GUI integration tests

```

```

def test_model_creation():
 """Test model factory with different configs."""
 # Arrange
 config = ModelConfig(encoder="resnet34", num_classes=1)

 # Act
 model = create_segmentation_model(config)

```

```
Assert
assert isinstance(model, nn.Module)
assert hasattr(model, "encoder")
assert hasattr(model, "decoder")
```

```
@pytest.mark.integration
def test_full_training_workflow(tmp_path):
 """Test complete training workflow."""
 # Arrange
 project = create_test_project(tmp_path, num_images=10)

 # Act
 result = train_model(project, epochs=1)

 # Assert
 assert result["final_loss"] < result["initial_loss"]
 assert (project.project_path / "artifacts" / "models" / "checkpoint.ckpt").exists()
```

```
@pytest.mark.benchmark
def test_detection_performance(benchmark, sample_images):
 """Ensure detection meets performance targets."""
 model = load_model(TEST_CHECKPOINT)

 result = benchmark(detect_cracks, model, sample_images)

 # Verify performance target: <5s per image on CPU
 assert result.mean < 5.0
```

#### core/tests/conftest.py

```
@pytest.fixture
def tiny_model():
 """Fast model for unit tests."""
 return create_segmentation_model(ModelConfig(
 encoder="resnet18",
 encoder_weights=None, # No pre-trained weights
 num_classes=1,
))

@pytest.fixture
def sample_project(tmp_path):
 """Sample project with test data."""
 project_dir = tmp_path / "test_project"
 initialize_project(project_dir, name="test_project")

 # Add minimal test data
 (project_dir / "images" / "raw").mkdir(parents=True)
 create_test_image(project_dir / "images" / "raw" / "test.png")

 return project_dir

@pytest.fixture
def mock_model(mocker):
 """Mock model for testing without actual inference."""
 mock = mocker.Mock(spec=nn.Module)
 mock.predict.return_value = torch.zeros(1, 1, 512, 512)
 return mock
```

```
x BAD: Load everything into memory
all_images = [Image.open(p) for p in image_paths] # OOM for large datasets
for img in all_images:
 process(img)

[] GOOD: Process in batches
for batch_paths in chunk_list(image_paths, batch_size=8):
 batch = [Image.open(p) for p in batch_paths]
 process_batch(batch)
 # batch garbage collected after each iteration
```

```
def train_with_cleanup():
 try:
 # Training code
 model.train()
 for batch in dataloader:
 loss = model.training_step(batch)
 loss.backward()
 optimizer.step()
 finally:
 # Always clean up GPU memory
 if torch.cuda.is_available():
 torch.cuda.empty_cache()
```

```
class Project:
 def __init__(self, root: Path):
 self.root = root
 self._config: ProjectConfig | None = None # Not loaded yet
 self._images: list[Path] | None = None # Not loaded yet

 @property
 def config(self) -> ProjectConfig:
 """Lazy load configuration."""
 if self._config is None:
 self._config = ProjectConfig.load(
 self.root / f"{self.root.name}_project.yaml"
)
 return self._config

 @property
 def images(self) -> list[Path]:
 """Lazy load image list."""
 if self._images is None:
 self._images = list((self.root / "images").rglob("*.png"))
 return self._images
```

- 1.
- 2.
- 3.
- 4.
- 5.

```
from pydantic import BaseModel, Field, validator

class ProjectConfig(BaseModel):
 name: str = Field(
 ...,
 min_length=1,
 max_length=100,
 pattern=r"^[a-zA-Z0-9_-]+$", # Prevent path traversal
)

 @validator("name")
 def no_path_traversal(cls, v: str) -> str:
 """Prevent directory traversal attacks."""
 if "." in v or "/" in v or "\\" in v:
 raise ValueError("Invalid project name: contains path separators")
 return v
```

```
def safe_path_join(base: Path, *parts: str) -> Path:
 """Join paths safely, preventing traversal attacks."""
 result = base
 for part in parts:
```

```

 if ".." in part or part.startswith("/") or part.startswith("\\"):
 raise ValueError(f"Unsafe path component: {part}")
 result = result / part

Verify result is still under base
if not result.resolve().is_relative_to(base.resolve()):
 raise ValueError("Path traversal detected")

return result

```

1. `core/src/crackz/cli/my_command.py`

2.

```

from __future__ import annotations

import typer
from crackz.cli.options import PROJECT_PATH_OPTION
from crackz.cli.exceptions import cli_failure
from crackz.log import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="my-command")
def my_command(
 project_path: Path = PROJECT_PATH_OPTION,
) -> None:
 """Description shown in --help."""
 try:
 # Business logic
 logger.info("Starting operation")
 result = do_operation(project_path)
 typer.secho("[] Operation completed", fg=typer.colors.GREEN)
 except Exception as e:
 cli_failure(e, "Operation name")

```

3. `core/src/crackz/cli/main.py`

4. `core/tests/unit/crackz/cli/test_my_command.py`

5.

1. `core/src/crackz/config/my_feature.py`

```

from pydantic import BaseModel, Field

class MyFeatureConfig(BaseModel):
 """Configuration for my feature."""

 enabled: bool = Field(
 default=False,
 description="Enable my feature",
)

 option: str = Field(
 default="default_value",
 description="Feature option",
)

```

2. `core/src/crackz/config/project.py`

```

class ProjectConfig(BaseModel):
 # ... existing fields ...
 my_feature: MyFeatureConfig = Field(default_factory=MyFeatureConfig)

```

3. `core/src/crackz/project/migrations.py`

4.

5.

1. `core/src/crackz/ai/models/my_model.py`

2.

```

class MyCustomModel(nn.Module):
 def __init__(self, config: ModelConfig):
 super().__init__()
 # Model setup

 def forward(self, x: torch.Tensor) -> torch.Tensor:
 # Forward pass
 return x

 def predict(self, x: torch.Tensor) -> torch.Tensor:
 """Inference interface."""

```

```
self.eval()
with torch.no_grad():
 return self(x)
```

3. `core/src/crackz/ai/models/__init__.py`

```
def create_model(config: ModelConfig) -> nn.Module:
 if config.type == "segmentation":
 return create_segmentation_model(config)
 elif config.type == "my_custom":
 return MyCustomModel(config)
 else:
 raise ValueError(f"Unknown model type: {config.type}")
```

4. `core/src/crackz/config/ai.py`

5.

1. `core/src/crackz/ai/detection/my_strategy.py`

2.

```
class MyDetectionStrategy:
 def __init__(self, model: nn.Module, config: DetectionConfig):
 self.model = model
 self.config = config

 def detect(self, images: list[Path]) -> dict[Path, np.ndarray]:
 """Run detection with my strategy."""
 results = {}
 for image_path in images:
 mask = self._detect_single(image_path)
 results[image_path] = mask
 return results
```

3. `core/src/crackz/ai/detection/__init__.py`

4. `core/src/crackz/config/ai.py`

5.

- \_\_\_\_\_
- \_\_\_\_\_
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- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

---

## API Reference

---

# API Reference

---

```
from pathlib import Path
from crackz.project.project import load_project_from_path
from crackz.ai.detection import detect

project = load_project_from_path(Path("demo-project"))
Optional: custom progress reporter
progress = lambda p, msg=None: print(f"{p:.0%}", msg or "")

detect(project, progress_reporter=progress)
```

```
from crackz.project.project import load_project_from_path
from crackz.ai.training import train

project = load_project_from_path("demo-project")
train(project) # writes checkpoints & metrics under project.artifacts/checkpoints
```

```
from pathlib import Path
from crackz.project.project import (
 ProjectMetadata, ProjectPaths, create_project, save_project
)

project = create_project(
 metadata=ProjectMetadata(name="demo-project", description="Example"),
 paths=ProjectPaths(
 project_path=Path("demo-project"),
 image_path=Path("demo-project/images"),
 image_mask_path=Path("demo-project/masks"),
 artifacts_path=Path("demo-project/artifacts"),
),
 force=False,
)
save_project(project)
```

```
from crackz.project.project import load_project_from_path
project = load_project_from_path("demo-project")
Uses same logic as CLI import-images (see project.import_images signature)
project.import_images(
 source_path="./samples", # directory with raw images
 masks_source_path="./samples/masks", # optional
 image_type=project.ImagesType.RAW if hasattr(project, 'ImagesType') else None, # or enum from crackz.ImagesType
 size=(512, 512),
 quality=90,
 overwrite=False,
)
```

CrackzSettings

CRACKZ\_

CRACKZ\_LOG\_LEVEL=DEBUG

BaseSettings

► core/src/crackz/config/crackz\_settings.py

classmethod

from\_conf(conf\_file)

► core/src/crackz/config/crackz\_settings.py

CrackzError

► core/src/crackz/config/exceptions.py

[ConfigError](#)

► core/src/crackz/config/exceptions.py

[ConfigError](#)

► core/src/crackz/config/exceptions.py

project.project

Project  
Project

BaseModel

► `core/src/crackz/config/project_config.py`

`classmethod`

`load(path)`

► `core/src/crackz/config/project_config.py`

`save(path)`

► `core/src/crackz/config/project_config.py`

`to_project()`

`Project`

`ProjectConfig`

► `core/src/crackz/config/project_config.py`

`classmethod`

`from_project(project)`

`ProjectConfig`

`Project`

► `core/src/crackz/config/project_config.py`

---

`Project`

`BaseModel`

► `core/src/crackz/project/project.py`

`property`

`name`

`property`

root

property

config\_file

property

detections\_path

Path Path

property

models\_path

Path Path

property

checkpoints\_path

Path Path

property

runtime\_path

Path Path

property

training\_path

property

detections\_visualizations\_path

Path

Path

property

detection\_results\_path

property

training\_metrics\_path

property

tensorboard\_logs\_path

property

registry\_db\_path

project.db

crackz.db

property

lightning\_logs\_path

property

val\_path

Path

Path

property

test\_path

Path

Path

property

train\_path

Path Path

property

train\_mask\_path

Path Path

property

val\_mask\_path

Path Path

property

test\_mask\_path

Path Path

property

raw\_path

Path Path

property

raw\_mask\_path



Path Path

property

project\_file\_path

report\_logo\_path(default\_logo\_path)

▶ core/src/crackz/project/project.py

resolve\_task\_mode()

ai.model.task\_mode "binary"

▶ core/src/crackz/project/project.py

sync\_derived\_fields()

▶ core/src/crackz/project/project.py

map\_type\_to\_dir(image\_type)

test val raw train

Path

None

▶ core/src/crackz/project/project.py

map\_type\_to\_masks\_dir(image\_type)

test val raw train

Path

None

▶ core/src/crackz/project/project.py

save\_project()

model\_dump

model\_dump

► core/src/crackz/project/project.py

```
import_image_files(
 files,
 dest_dir,
 size=None,
 quality=90,
 *,
 overwrite=False,
 progress_cb=None,
)
```

- 
- 
- 

► core/src/crackz/project/project.py

```
import_images(
 source_path,
 masks_source_path=None,
 image_type=ImagesType.RAW,
 size=None,
 quality=90,
 *,
 overwrite=False,
 progress_cb=None,
 split=False,
 train_ratio=0.7,
 val_ratio=0.15,
 test_ratio=0.15,
 random_seed=None,
 mask_label_map=None,
)
```

- 
- 
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- 
- 

► core/src/crackz/project/project.py

```
create_project_dir(*, force)
```

force

project\_path

artifacts\_path

► core/src/crackz/project/project.py

```
clean(*, logs_only=False, keep_best_model=True)
```

► core/src/crackz/project/project.py

```
create_project_and_save(
 metadata,
 paths,
 ai=default_ai_config,
 logging=default_logging_config,
 *,
 force=False,
)
```

► `core/src/crackz/project/project.py`

```
create_project(
 metadata, paths, ai=None, logging=None, *, force=False
)
```

► `core/src/crackz/project/project.py`

```
save_project(project)
```

`Project.save_project`

► `core/src/crackz/project/project.py`

```
load_project(project_file_path)
```

`Project`

|                                |                   |  |                 |
|--------------------------------|-------------------|--|-----------------|
|                                |                   |  |                 |
| <code>project_file_path</code> | <code>Path</code> |  | <i>required</i> |

|                         |                      |
|-------------------------|----------------------|
|                         |                      |
| <a href="#">Project</a> | <code>Project</code> |

|                         |                   |
|-------------------------|-------------------|
| <a href="#">Project</a> | <code>None</code> |
|-------------------------|-------------------|

► `core/src/crackz/project/project.py`

```
load_project_from_path(project_path)
```

► `core/src/crackz/project/project.py`

```
load_from_project_path_or_cfg(cfg, project_path)
```

► `core/src/crackz/project/project.py`

`StrEnum`

`RAW`

`TRAIN`

`TEST`

`VAL`

► `core/src/crackz/project/types.py`

`CrackzError`

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

[ProjectError](#)

► `core/src/crackz/project/exceptions.py`

---

[BaseModel](#)

► `core/src/crackz/ai/detection/results.py`

```
def categorize_crack_severity(
 confidence,
 *,
 high_threshold=DEFAULT_HIGH_CONFIDENCE,
 medium_threshold=DEFAULT_MEDIUM_CONFIDENCE,
 has_crack=False,
):
```

|                  |       |                           |
|------------------|-------|---------------------------|
| confidence       | float | required                  |
| high_threshold   | float | DEFAULT_HIGH_CONFIDENCE   |
| medium_threshold | float | DEFAULT_MEDIUM_CONFIDENCE |
| has_crack        | bool  | False                     |

str

```
>>> categorize_crack_severity(0.08) # 8% crack area
'Crack Detected'
>>> categorize_crack_severity(0.03) # 3% crack area
'Suspected Crack'
>>> categorize_crack_severity(0.01) # 1% crack area
'No Crack'
>>> categorize_crack_severity(0.01, has_crack=True)
'Crack Detected' # has_crack flag indicates both confidence and area thresholds met
```

► core/src/crackz/ai/detection/severity.py

```
def create_tensorboard_logger(project):
```

|         |                         |          |
|---------|-------------------------|----------|
| project | <a href="#">Project</a> | required |
|---------|-------------------------|----------|

TensorBoardLogger | None

► core/src/crackz/ai/training/tensorboard.py

```
train(
 project,
 progress_reporter=None,
 cancellation_token=None,
 metrics_callback=None,
 ckpt_path=None,
 backbone_override=None,
)
```

| project            | <a href="#">Project</a>                  |           | required                  |
|--------------------|------------------------------------------|-----------|---------------------------|
| progress_reporter  | ProgressReporter   None                  |           | None                      |
| cancellation_token | <a href="#">CancellationToken</a>   None |           | None                      |
| metrics_callback   | MetricsReporter   None                   |           | None                      |
| ckpt_path          | Path   None                              |           | None                      |
| backbone_override  | str   None                               | ckpt_path | project.ai.model.backbone |
|                    |                                          |           | None                      |

CancellationError

► core/src/crackz/ai/training/training\_runner.py

```
detect(
 project,
 progress_reporter=None,
 cancellation_token=None,
 checkpoint_name=None,
 result_callback=None,
 *,
 on_incompatible="original",
)
```

| project            | <a href="#">Project</a>                  |  | required |
|--------------------|------------------------------------------|--|----------|
| progress_reporter  | ProgressReporter   None                  |  | None     |
| cancellation_token | <a href="#">CancellationToken</a>   None |  | None     |

|                 |                                         |                        |                    |
|-----------------|-----------------------------------------|------------------------|--------------------|
|                 |                                         |                        |                    |
| checkpoint_name | str   None                              |                        | None               |
| result_callback | Callable[[dict[str, Any]], None]   None |                        | None               |
| on_incompatible | IncompatibleMode                        | "untrained" "original" | "raise" 'original' |

|      |  |
|------|--|
|      |  |
| None |  |
| None |  |

|                             |                                                  |
|-----------------------------|--------------------------------------------------|
|                             |                                                  |
| CancellationError           |                                                  |
| CheckpointIncompatibleError | on_incompatible "raise"                          |
| ▶                           | core/src/crackz/ai/detection/detection_runner.py |

|                   |                                          |
|-------------------|------------------------------------------|
| __getattr__(name) |                                          |
| ▶                 | core/src/crackz/ai/detection/__init__.py |
| ▶                 | core/src/crackz/ai/data/dataset.py       |

|                 |                                    |
|-----------------|------------------------------------|
| apply(im)       |                                    |
| ▶               | core/src/crackz/ai/data/dataset.py |
| Dataset[Sample] |                                    |
| ▶               | core/src/crackz/ai/data/dataset.py |

```
load_image_with_crackz_image(image_path)
```

|            |      |  |          |
|------------|------|--|----------|
|            |      |  |          |
| image_path | Path |  | required |

|       |  |
|-------|--|
|       |  |
| Image |  |

|                                   |  |
|-----------------------------------|--|
|                                   |  |
| <a href="#">ImageLoadingError</a> |  |

► core/src/crackz/ai/data/dataset.py

```
CrackzDetectionDataset(
 image_dir, transform=None, normalization_config=None
)
```

|                      |                            |  |          |
|----------------------|----------------------------|--|----------|
|                      |                            |  |          |
| image_dir            | Path   str                 |  | required |
| transform            | ImageTransform   None      |  | None     |
| normalization_config | NormalizationConfig   None |  | None     |

|                               |  |
|-------------------------------|--|
|                               |  |
| <a href="#">CrackzDataset</a> |  |

► core/src/crackz/ai/data/dataset.py

```
create_training_datasets(
 train_path,
 train_mask_path,
 val_path,
 val_mask_path,
 test_path,
 test_mask_path,
)
```

|                 |      |  |          |
|-----------------|------|--|----------|
|                 |      |  |          |
| train_path      | Path |  | required |
| train_mask_path | Path |  | required |
| val_path        | Path |  | required |
| val_mask_path   | Path |  | required |
| test_path       | Path |  | required |
| test_mask_path  | Path |  | required |

|  |  |
|--|--|
|  |  |
|--|--|

`tuple[CrackzDataset, CrackzDataset, CrackzDataset]`



► `core/src/crackz/ai/data/dataset.py`

```
create_dataset(
 project,
 transforms,
 mask_transforms,
 image_dir=None,
 image_masks_dir=None,
)
```

|                 |                         |  |          |
|-----------------|-------------------------|--|----------|
|                 |                         |  |          |
| project         | <a href="#">Project</a> |  | required |
| transforms      | ImageTransform   None   |  | required |
| mask_transforms | MaskTransform   None    |  | required |
| image_dir       | Path   None             |  | None     |
| image_masks_dir | Path   None             |  | None     |

|  |  |
|--|--|
|  |  |
|--|--|

[CrackzDataset](#)

► `core/src/crackz/ai/data/dataset.py`

```
collate(batch)
```

► core/src/crackz/ai/data/dataloader.py

```
make_crackz_loader(
 dataset, batch_size=32, num_workers=8, *, shuffle=True
)
```

► core/src/crackz/ai/data/dataloader.py

```
create_training_data loaders(
 train_ds,
 val_ds,
 test_ds,
 batch_size,
 num_workers,
 *,
 shuffle_train=True,
)
```

| train_ds      | <a href="#">CrackzDataset</a> |      | required |
|---------------|-------------------------------|------|----------|
| val_ds        | <a href="#">CrackzDataset</a> |      | required |
| test_ds       | <a href="#">CrackzDataset</a> |      | required |
| batch_size    | int                           |      | required |
| num_workers   | int                           |      | required |
| shuffle_train | bool                          | True |          |

```
tuple[DataLoader, DataLoader, DataLoader]
```

► core/src/crackz/ai/data/dataloader.py

CrackzModel

crackz.ai.model.segmentation.tasks

CrackzModel

TaskStrategy

task\_mode

Module

- aux\_weight
- aux\_weight

focal\_weight ~crackz.ai.model.segmentation.model.CrackzConfig aux\_weight

► core/src/crackz/ai/model/segmentation/loss.py

```
forward(y_pred, y_true)
```

dice\_weight \* Dice + aux\_weight \* (focal | cross\_entropy)

► core/src/crackz/ai/model/segmentation/loss.py

**dataclass**

CrackzModel

lr float

backbone str

in\_channels int

num\_classes int

threshold float

dice\_weight float

focal\_weight float

task\_mode TaskMode

category\_keys tuple[str, ...]

► core/src/crackz/ai/model/segmentation/model.py

**staticmethod**

```
from_project(project, lr)
```

CrackzConfig      Project

►      core/src/crackz/ai/model/segmentation/model.py

LightningModule

►      core/src/crackz/ai/model/segmentation/model.py

training\_step(batch, \_batch\_idx)

►      core/src/crackz/ai/model/segmentation/model.py

validation\_step(batch, \_batch\_idx)

►      core/src/crackz/ai/model/segmentation/model.py

test\_step(batch, \_batch\_idx)

►      core/src/crackz/ai/model/segmentation/model.py

on\_train\_epoch\_end()

►      core/src/crackz/ai/model/segmentation/model.py

on\_validation\_epoch\_end()

►      core/src/crackz/ai/model/segmentation/model.py

on\_test\_epoch\_end()

►      core/src/crackz/ai/model/segmentation/model.py

configure\_optimizers()

►      core/src/crackz/ai/model/segmentation/model.py

get\_loss\_function(config)

```
loss_name == "combo"
```

```
ComboLoss
```

► `core/src/crackz/ai/model/segmentation/model.py`

## Callback

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(
 trainer, pl_module, outputs, batch, batch_idx
)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_validation_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_test_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

`Callback`

► `core/src/crackz/ai/training/callbacks.py`

```
__call__(progress, message=None)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_start(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_batch_end(
 trainer, pl_module, outputs, batch, batch_idx
)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_train_epoch_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_fit_end(trainer, pl_module)
```

► `core/src/crackz/ai/training/callbacks.py`

```
on_test_end(trainer, pl_module)
```

► core/src/crackz/ai/training/callbacks.py

Callback

► core/src/crackz/ai/training/callbacks.py

```
on_validation_epoch_end(trainer, pl_module)
```

► core/src/crackz/ai/training/callbacks.py

```
make_reporter(ui_callback)
```

► core/src/crackz/ai/training/callbacks.py

```
create_training_callbacks(
 project,
 val_ds,
 tb_logger,
 progress_reporter,
 runtime_tracker,
 cancellation_token,
)
```

| project            | <a href="#">Project</a>                  |  | required |
|--------------------|------------------------------------------|--|----------|
| val_ds             | <a href="#">CrackzDataset</a>            |  | required |
| tb_logger          | TensorBoardLogger   None                 |  | required |
| progress_reporter  | ProgressReporter   None                  |  | required |
| runtime_tracker    | RuntimeTracker                           |  | required |
| cancellation_token | <a href="#">CancellationToken</a>   None |  | required |

`list[Any]`

► `core/src/crackz/ai/training/callbacks.py`

`get_available_checkpoints(checkpoints_dir)`

`checkpoints_dir` `Path` *required*

`list[Path]`

► `core/src/crackz/ai/model/checkpoints.py`

`has_trained_model(project)`

`project` [Project](#) *required*

`bool`

► `core/src/crackz/ai/model/checkpoints.py`

`select_checkpoint(checkpoints_dir)`

`checkpoints_dir` `Path` *required*

Path | None

► core/src/crackz/ai/model/checkpoints.py

```
peek_checkpoint_backbone(checkpoint_path)
```

► core/src/crackz/ai/model/checkpoints.py

```
get_checkpoint_compatibility(project)
```

► core/src/crackz/ai/model/checkpoints.py

```
load_model_from_checkpoint(
 project,
 lr,
 checkpoint_name=None,
 *,
 on_incompatible="original",
)
```

project [Project](#) *required*

lr float *required*

checkpoint\_name str | None None

on\_incompatible IncompatibleMode "raise" "untrained" "original" 'original'  
CheckpointIncompatibleError

[CrackzModel](#)

[CheckpointIncompatibleError](#) *on\_incompatible* "raise"

► core/src/crackz/ai/model/checkpoints.py

create\_checkpoint\_callback(project)

project [Project](#) *required*

ModelCheckpoint

► core/src/crackz/ai/model/checkpoints.py

prune\_checkpoints(project, keep)

project [Project](#) *required*

keep int *required*

list[Path]

► core/src/crackz/ai/model/checkpoints.py

plot\_image(img\_path, out\_file, pred\_mask)

► core/src/crackz/ai/visualization/visualization.py

```
save_detection_figure(fig, out_file)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
save_metrics_json(results, output_dir)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
save_training_metrics_json(results, output_dir)
```

► `core/src/crackz/ai/visualization/visualization.py`

```
show_image(ax, index, title, img, alpha=None, cmap='gray')
```

► `core/src/crackz/ai/visualization/visualization.py`

```
to_numpy_img(tensor)
```

► `core/src/crackz/ai/visualization/visualization.py`

Enum

► `core/src/crackz/ai/device/device.py`

**staticmethod**

```
available_devices()
```

► `core/src/crackz/ai/device/device.py`

**staticmethod**

```
default()
```

► core/src/crackz/ai/device/device.py

staticmethod

test\_device(device\_name)

device\_name str required

bool

str •

dict[str, str | None] •

tuple[bool, str, dict[str, str | None]] •

► core/src/crackz/ai/device/device.py

get\_accelerator()

► core/src/crackz/ai/device/utils.py

resolve\_device\_and\_accelerator(project)

project [Project](#) required

str

► core/src/crackz/ai/device/utils.py

[AIPipelineError](#)

[ModelNotFoundError](#)

[DataLoadingError](#)

CrackzError

► `core/src/crackz/ai/exceptions.py`

CrackzError

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

[AIPipelineError](#)

► `core/src/crackz/ai/exceptions.py`

`detection_results.json`

▼

▼

`crackz.ai.eval`

[BaseModel](#)

► `core/src/crackz/ai/detection/results.py`

```
load_detection_results(project_or_path, *, strict=False)
```

► `core/src/crackz/ai/detection/results.py`

```
result_has_finding(result)
```

► `core/src/crackz/ai/detection/results.py`

`result_present_categories(result)`

► `core/src/crackz/ai/detection/results.py`

`result_primary_category(result)`

► `core/src/crackz/ai/detection/results.py`

`result_area_pct(result)`

► `core/src/crackz/ai/detection/results.py`

`result_category_label(category)`

► `core/src/crackz/ai/detection/results.py`

`iter_iou(results)`

► `core/src/crackz/ai/detection/results.py`

`DetectionResult`

`aggregate_iou(results)`

► `core/src/crackz/ai/detection/eval.py`

basic\_summary(results)

► core/src/crackz/ai/detection/eval.py

get\_available\_checkpoints(checkpoints\_dir)

checkpoints\_dir Path required

list[Path]

► core/src/crackz/ai/model/checkpoints.py

has\_trained\_model(project)

project [Project](#) required

bool

► core/src/crackz/ai/model/checkpoints.py

select\_checkpoint(checkpoints\_dir)

checkpoints\_dir

Path

required

Path | None

►

core/src/crackz/ai/model/checkpoints.py

peek\_checkpoint\_backbone(checkpoint\_path)

►

core/src/crackz/ai/model/checkpoints.py

get\_checkpoint\_compatibility(project)

►

core/src/crackz/ai/model/checkpoints.py

load\_model\_from\_checkpoint(  
 project,  
 lr,  
 checkpoint\_name=None,  
 \*,  
 on\_incompatible="original",  
)

project

[Project](#)

required

lr

float

required

checkpoint\_name

str | None

None

on\_incompatible

IncompatibleMode

"raise"

"untrained"

"original"

CheckpointIncompatibleError

'original'

[CrackzModel](#)

[CheckpointIncompatibleError](#)

*on\_incompatible* "raise"

► core/src/crackz/ai/model/checkpoints.py

create\_checkpoint\_callback(project)

project [Project](#) *required*

ModelCheckpoint

► core/src/crackz/ai/model/checkpoints.py

prune\_checkpoints(project, keep)

project [Project](#) *required*

keep int *required*

list[Path]

► core/src/crackz/ai/model/checkpoints.py

► `core/src/crackz/ai/pipeline/cancellation.py`

`__init__()`

► `core/src/crackz/ai/pipeline/cancellation.py`

`cancel()`

► `core/src/crackz/ai/pipeline/cancellation.py`

`is_cancelled()`

`bool`

`bool`

► `core/src/crackz/ai/pipeline/cancellation.py`

`check_cancelled()`

`CancellationError`

`CancellationError`

► `core/src/crackz/ai/pipeline/cancellation.py`

---

`Handler`

► `core/src/crackz/log/crackz_logger.py`

emit(record)

►

core/src/crackz/log/crackz\_logger.py

use\_rich\_console()

bool

►

core/src/crackz/log/crackz\_logger.py

setup\_logging(level=None, log\_dir=None, \*, use\_rich=None)

|          |             |      |
|----------|-------------|------|
| level    | str   None  | None |
| log_dir  | Path   None | None |
| use_rich | bool   None | None |

Logger

►

core/src/crackz/log/crackz\_logger.py

get\_logger(name, filename=None, log\_dir='logs')

logging

InterceptHandler

filename

►

core/src/crackz/log/crackz\_logger.py

TyperGroup

► core/src/crackz/cli/crackz\_cli.py

```
main(
 args=None,
 prog_name=None,
 complete_var=None,
 standalone_mode=True,
 windows_expand_args=True,
 **extra,
)
```

► core/src/crackz/cli/crackz\_cli.py

list\_commands(ctx)

► core/src/crackz/cli/crackz\_cli.py

get\_command(ctx, cmd\_name)

► core/src/crackz/cli/crackz\_cli.py

format\_commands(ctx, formatter)

► core/src/crackz/cli/crackz\_cli.py

```
root_callback(
 ctx,
 version=False,
 debug=False,
 quiet=False,
 log_dir=None,
 telemetry=False,
)
```

► core/src/crackz/cli/crackz\_cli.py

main()

► core/src/crackz/cli/crackz\_cli.py

```

init_project(
 name=NAME_OPTION,
 image_path=IMAGE_PATH_OPTION,
 image_mask_path=IMAGE_MASK_PATH_OPTION,
 artifacts_path=ARTIFACTS_PATH_OPTION,
 description=DESCRIPTION_OPTION,
 category_template=CATEGORY_TEMPLATE_OPTION,
 task_mode=TASK_MODE_OPTION,
 log_dir=LOG_DIR_OPTION,
 *,
 force=FORCE_OPTION,
)

```

► `core/src/crackz/cli/project_cli.py`

```

export_project_config(
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
 out=EXPORT_OUTPUT_OPTION,
)

```

► `core/src/crackz/cli/project_cli.py`

```

migrate_project_config(
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
 *,
 dry_run=DRY_RUN_OPTION,
 no_backup=NO_BACKUP_OPTION,
 migrate_files=typer.Option(
 default=False,
 help="Also migrate artifacts directory structure (checkpoints to models, etc.)",
),
)

```

► `core/src/crackz/cli/project_cli.py`

```

clean_project(
 project_path=PROJECT_PATH_OPTION,
 *,
 dry_run=DRY_RUN_OPTION,
 logs_only=LOGS_ONLY_OPTION,
 keep_best_model=KEEP_BEST_MODEL_OPTION,
)

```

► core/src/crackz/cli/project\_cli.py

```
import_images(
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
 source_path=SOURCE_PATH_OPTION,
 masks_source_path=MASKS_SOURCE_PATH_OPTION,
 image_type=TYPE_OPTION,
 image_size=SIZE_OPTION,
 quality=QUALITY_OPTION,
 target_size=TARGET_SIZE_OPTION,
 *,
 overwrite=OVERWRITE_OPTION,
 split=SPLIT_OPTION,
 train_ratio=TRAIN_RATIO_OPTION,
 val_ratio=VAL_RATIO_OPTION,
 test_ratio=TEST_RATIO_OPTION,
 random_seed=RANDOM_SEED_OPTION,
 mask_label_map=MASK_LABEL_MAP_OPTION,
)
```

| cfg               | Path  <br>None             |             | CFG_OPTION               |
|-------------------|----------------------------|-------------|--------------------------|
| project_path      | Path  <br>None             |             | PROJECT_PATH_OPTION      |
| source_path       | Path                       |             | SOURCE_PATH_OPTION       |
| masks_source_path | Path  <br>None             | source_path | MASKS_SOURCE_PATH_OPTION |
| image_type        | <a href="#">ImagesType</a> |             | TYPE_OPTION              |
| image_size        | SizeType                   |             | SIZE_OPTION              |
| quality           | int                        |             | QUALITY_OPTION           |
| target_size       | int   None                 |             | TARGET_SIZE_OPTION       |
| overwrite         | bool                       |             | OVERWRITE_OPTION         |
| split             | bool                       |             | SPLIT_OPTION             |
| train_ratio       | float                      |             | TRAIN_RATIO_OPTION       |

|                |            |  |                       |
|----------------|------------|--|-----------------------|
|                |            |  |                       |
| val_ratio      | float      |  | VAL_RATIO_OPTION      |
| test_ratio     | float      |  | TEST_RATIO_OPTION     |
| random_seed    | int   None |  | RANDOM_SEED_OPTION    |
| mask_label_map | str   None |  | MASK_LABEL_MAP_OPTION |

|  |  |
|--|--|
|  |  |
|--|--|

BadParameter

Exit

|  |  |
|--|--|
|  |  |
|--|--|

None

► core/src/crackz/cli/image\_import\_cli.py

```
run_detection(
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
 model=CHECKPOINT_NAME_OPTION,
)
```

► core/src/crackz/cli/detection\_cli.py

```
detection_train(
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
 epochs=EPOCHS_OPTION,
 batch_size=BATCH_SIZE_OPTION,
 resume=RESUME_FROM_OPTION,
)
```

|              |                |              |                     |
|--------------|----------------|--------------|---------------------|
|              |                |              |                     |
| cfg          | Path  <br>None | project_path | CFG_OPTION          |
| project_path | Path  <br>None |              | PROJECT_PATH_OPTION |

|            |               |  |                    |
|------------|---------------|--|--------------------|
|            |               |  |                    |
| epochs     | int  <br>None |  | EPOCHS_OPTION      |
| batch_size | int  <br>None |  | BATCH_SIZE_OPTION  |
| resume     | str  <br>None |  | RESUME_FROM_OPTION |

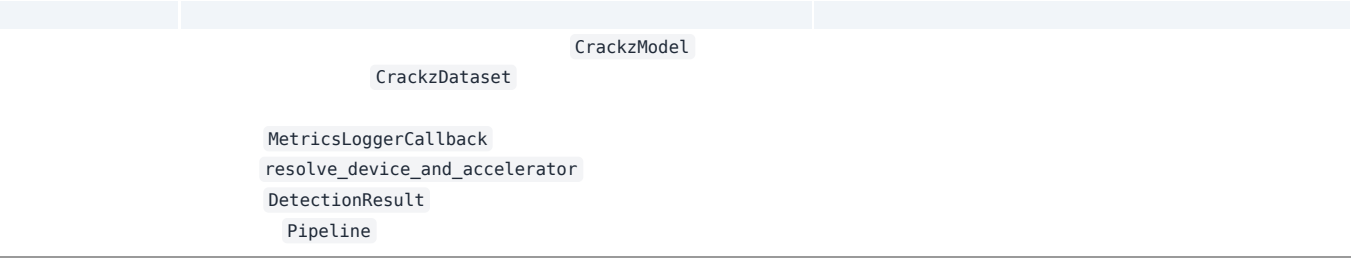
BadParameter      cfg      project\_path

Exit

►      core/src/crackz/cli/detection\_cli.py

```
prune_checkpoints_cmd(
 keep=KEEP_CHECKPOINTS_OPTION,
 cfg=CFG_OPTION,
 project_path=PROJECT_PATH_OPTION,
)
```

►      core/src/crackz/cli/detection\_cli.py



1.  
2.  
3.  
4.  
5.  
6.  
7.  
8.  
9.  
10.  
11.  
12.  
13.  
14.  
15.  
16.  
17.  
18.
- DetectionResult
- evaluation.py
- ProjectConfig

- 19.
- 20.
- 21.
- 22. Pipeline

---

```
Project -> load_model_from_checkpoint -> CrackzModel.eval() -> DataLoader -> forward -> sigmoid -> threshold -> mask -> metrics(json)
```

---

requirements.txt

---

---

# Async Callbacks

---

# Async Callback Architecture

---

- 
- 
- 
- 

```
from crackz.ai.core.async_callbacks import CallbackDispatcher

dispatcher = CallbackDispatcher(maxsize=100)

Create sync reporter for PyTorch Lightning (training thread)
progress_reporter = dispatcher.create_sync_progress_reporter()

Use in training (synchronous context)
train(project, progress_reporter=progress_reporter)

Consume updates asynchronously (main thread)
async for progress, message in dispatcher.iter_progress():
 update_progress_bar(progress, message)
```

```
from crackz.ai.core.async_callbacks import (
 create_async_progress_consumer,
 create_async_metrics_consumer,
)

Define callback
def update_ui(progress: float, message: str | None = None) -> None:
 progress_bar.set_value(progress)

Create async consumer
consumer = create_async_progress_consumer(dispatcher, update_ui)

Run in async context
await consumer()
```

```
from crackz_gui.qt.async_bridge import (
 start_gui_callback_consumers,
 stop_gui_callback_consumers,
)

Start consumers (manages event loop)
loop, tasks = start_gui_callback_consumers(
 dispatcher,
 progress_callback=update_progress,
 metrics_callback=update_metrics,
)

... run training ...
```

```
Cleanup
stop_gui_callback_consumers(dispatcher, loop)
```

TaskSignals

async\_bridge.py

```
from crackz_gui.qt.async_bridge import TaskSignals, run_in_background

def _do_work(signals: TaskSignals) -> None:
 signals.progress.emit(0.5) # Safe from any thread
 signals.status.emit("Processing...")
 result = expensive_operation()
 signals.data.emit(result)
 signals.finished.emit()

signals = run_in_background(_do_work)
signals.progress.connect(self._on_progress)
signals.finished.connect(self._on_complete)
signals.error.connect(self._on_error)
```

threading.Lock

after()

crackz\_gui.qt.async\_bridge

```
def progress_callback(progress: float, message: str | None = None) -> None:
 print(f"Progress: {progress:.2%}")

train(project, progress_reporter=progress_callback)
```

```
dispatcher = CallbackDispatcher(maxsize=100)

Create sync reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

Start training in background
training_thread = threading.Thread(
 target=train,
 args=(project,),
 kwargs={
 "progress_reporter": progress_reporter,
 "metrics_callback": metrics_reporter,
 }
)
training_thread.start()

Consume updates asynchronously
async def consume_updates():
 async for progress, message in dispatcher.iter_progress():
 print(f"Progress: {progress:.2%} - {message}")

asyncio.run(consume_updates())
```

```
from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import (
 start_gui_callback_consumers,
 stop_gui_callback_consumers,
)

dispatcher = CallbackDispatcher(maxsize=100)

Start async consumers with Qt-thread-safe GUI updates
loop, tasks, emitters = start_gui_callback_consumers(
 dispatcher,
 progress_callback=self.update_progress_bar,
```

```

 metrics_callback=self.update_metrics_display,
)
 self.qt_emitters = emitters # Keep signal emitters alive

Create reporters
progress_reporter = dispatcher.create_sync_progress_reporter()
metrics_reporter = dispatcher.create_sync_metrics_reporter()

try:
 # Run training in background thread
 training_thread = threading.Thread(
 target=train,
 args=(project,),
 kwargs={
 "progress_reporter": progress_reporter,
 "metrics_callback": metrics_reporter,
 }
)
 training_thread.start()
 training_thread.join()
finally:
 stop_gui_callback_consumers(dispatcher, loop)

```

```

Training Loop:
 Process batch → Update metrics → **Wait for UI** → Next batch
 ↑ 1-5ms overhead

```

```

Training Loop:
 Process batch → Put in queue → Next batch (no wait!)
 ↓ 0.1ms
 Async Consumer: Takes from queue → Updates UI

```

```
dispatcher = CallbackDispatcher(maxsize=100)

... use reporters ...

Check progress queue stats
stats = dispatcher.progress_stats
print(f"Total items put: {stats.total_put}")
print(f"Total items retrieved: {stats.total_get}")
print(f"Total items dropped: {stats.total_dropped}")
print(f"Current queue size: {stats.current_size}")
print(f"Max queue size: {stats.max_size}")

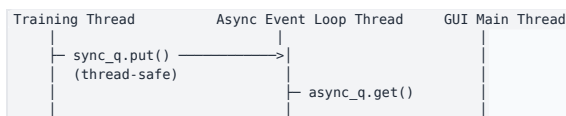
Check metrics queue stats
metrics_stats = dispatcher.metrics_stats
```

```
total_dropped
```

```

 sync_q
 async_q
 threading.Event

```





```

Before (still works)
train(project, progress_reporter=my_callback)

After (optional upgrade for better performance)
dispatcher = CallbackDispatcher()
train(project, progress_reporter=dispatcher.create_sync_progress_reporter())

```

```

from crackz.ai.core.async_callbacks import CallbackDispatcher
from crackz_gui.qt.async_bridge import start_gui_callback_consumers

dispatcher = CallbackDispatcher(maxsize=100)
loop, tasks = start_gui_callback_consumers(
 dispatcher,
 progress_callback=update_progress,
 metrics_callback=update_metrics,
)

Use dispatcher's sync reporters in training
train(
 project,
 progress_reporter=dispatcher.create_sync_progress_reporter(),
 metrics_callback=dispatcher.create_sync_metrics_reporter(),
)

stop_gui_callback_consumers(dispatcher, loop)

```

### crackz.ai.core.async\_callbacks

- CallbackDispatcher
- QueueStats
- create\_async\_progress\_consumer()
- create\_async\_metrics\_consumer()

### crackz\_gui.qt.async\_bridge

- AsyncEventLoop
- create\_threadsafe\_gui\_callback()
- create\_threadsafe\_metrics\_callback()
- start\_gui\_callback\_consumers()
- stop\_gui\_callback\_consumers()

help()

```

from crackz.ai.core.async_callbacks import CallbackDispatcher
help(CallbackDispatcher)

```

[docs/examples/async\\_callbacks\\_example.py](https://crackz.ai/docs/examples/async_callbacks_example.py)

- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

---

# MCP Integration

---

# MCP (Model Context Protocol) Integration

- 
- 
- 
- 

- 1.
- 2.
- 3.
- 4.

`.github/copilot-instructions.md`

| Development Agents                                                                                                                     | GitHub Copilot                                                           | End-User GUI                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <div><div>4 MCP Servers</div><ul style="list-style-type: none"><li>filesystem</li><li>github</li><li>python</li><li>uv</li></ul></div> | <div><div>× No MCP</div><div>IDE context<br/>Workspace files</div></div> | <div><div>× No MCP</div><div>Simple API<br/>Project info only</div></div> |
| Tools: Claude Desktop, Cline                                                                                                           | Tools: VS Code, IDE                                                      | Tools: Crackz GUI Chat Panel                                              |

- 
- 
- 
-

```

TOOL CALL: filesystem.readDirectory("src/")
→ returns 50 files in context

TOOL CALL: filesystem.readFile("src/file1.py")
→ returns full file in context

TOOL CALL: filesystem.readFile("src/file2.py")
→ returns full file in context

```

```

Discover and filter in code
files = await filesystem.readDirectory("src/")
python_files = [f for f in files if f.endswith(".py")]

Load only what's needed
for filename in python_files[:5]: # Top 5 only
 content = await filesystem.readFile(f"src/{filename}")
 # Process in code, log summary only
 print(f"{filename}: {len(content)} lines")

```

- 1.
- 2.
- 3.
- 4.

<type>/<scope>-<short-slug>

```

feat/gui-live-preview # New GUI feature
fix/pdf-rendering-crash # Bug fix in PDF generation
perf/dataloader-cache # Performance improvement
docs/api-reference-update # Documentation update
refactor/config-cleanup # Code refactoring

```

<type>(<scope>): <description>

```

feat(gui): add real-time detection preview
fix(cli): correct path handling in Windows
perf(dataloader): implement batch caching
docs(api): update training configuration reference

```

<type>(<scope>): <description>

```

feat(gui): add export panel for detection results
fix(pdf): handle missing hero image gracefully
perf(training): optimize data loading pipeline

```

```

Fix GUI bugs and add new feature
Update documentation
Add export functionality

```

```

feat: feat(<scope>):
perf: perf(<scope>):

```

```

fix: fix(<scope>):

```

```

test: docs:
chore:

```

```

refactor:
ci:

```

```

build:

```

```
typer.secho()
crackz.cli.options

logger
crackz.log

print()
```

```
uv run task tests
```

```
uv run task mypy
```

```
crackz mcp init
```

```
.mcp/config.json
```

```
crackz mcp show
```

```
crackz mcp validate
```

```
.mcp/config.json
```

```
{
 "mcpServers": {
 "filesystem": {
 "type": "stdio",
 "command": "npx",
 "args": ["-y", "@modelcontextprotocol/server-filesystem", "."],
 "description": "Filesystem operations with progressive disclosure"
 },
 "github": {
 "type": "stdio",
 "command": "npx",
 "args": ["-y", "@modelcontextprotocol/server-github"],
 "env": {
 "GITHUB_PERSONAL_ACCESS_TOKEN": "${GITHUB_TOKEN}"
 },
 "description": "GitHub operations (issues, PRs, releases)"
 },
 "python": {
 "type": "stdio",
 "command": "npx",
 "args": ["-y", "@modelcontextprotocol/server-python"],
 "description": "Python code execution environment"
 },
 "uv": {
 "type": "stdio",
 "command": "npx",
 "args": ["-y", "@modelcontextprotocol/server-uv"],
 "description": "UV package manager operations"
 }
 }
}
```

```
{
 "metadata": {
 "project": "Crackz",
 "language": "python",
 "version": "3.13",
 "package_manager": "uv",
 "framework": "pytorch-lightning",
 "cli": "typer",
 "gui": "pyside6",
 "testing": "pytest",
 "ci": "github-actions",
 "release": "semantic-release",
 "code_execution": true,
 "progressive_disclosure": true
 }
}
```

```
{
 "patterns": {
 "code_execution": {
 "enabled": true,
 "description": "Use code execution to interact with MCP servers efficiently",
 "examples": [
 "Filter large datasets in code before returning results",
 "Loop over operations without alternating tool calls",
 "Compose multiple MCP operations in a single execution"
]
 },
 "progressive_disclosure": {
 "enabled": true,
 "description": "Load tool definitions on-demand",
 "workflow": [
 "List directories to discover modules",
 "Read specific files as needed",
 "Load documentation when relevant"
]
 }
 }
}
```

```
--output -o
```

```
.mcp/config.json
```

```
--force -f
```

```
Create with defaults
crackz mcp init

Create in custom location
crackz mcp init --output custom/mcp.json

Overwrite existing config
crackz mcp init --force
```

```
--config -c
```

```
.mcp/config.json
```

```
Show default config
crackz mcp show

Show custom config
crackz mcp show --config custom/mcp.json
```

`--config -c``.mcp/config.json`

```
Validate default config
crackz mcp validate

Validate custom config
crackz mcp validate --config custom/mcp.json
```

`.mcp/config.json`

```
{
 "mcpServers": {
 "filesystem": { ... },
 "custom-api": {
 "type": "stdio",
 "command": "node",
 "args": ["/custom-mcp-server.js"],
 "description": "Custom MCP server for specialized operations"
 }
 }
}
```

```
{
 "metadata": {
 "project": "Crackz",
 "language": "python",
 "team": "ML Ops",
 "repository": "https://github.com/Anaxiatech-AB/crackz",
 "custom_field": "your_value"
 }
}
```

- 1.
- 2.
- 3.
- 4.
- 5.

```
List Python files in src/
files = await filesystem.readDirectory("src/crackz")
py_files = [f for f in files if f.endswith(".py")]

Read only the first 3 for initial analysis
for filename in py_files[:3]:
 content = await filesystem.readFile(f"core/src/crackz/{filename}")
 lines = len(content.split('\n'))
 print(f"{filename}: {lines} lines")
```

```
Get recent PRs and filter in code
prs = await github.listPullRequests(repo="Anaxiatech-AB/crackz", state="open")
priority_prs = [pr for pr in prs if "feat:" in pr.title or "fix:" in pr.title]

print(f"Found {len(priority_prs)} priority PRs out of {len(prs)} total")
for pr in priority_prs[:5]: # Show top 5 only
 print(f"#{pr.number}: {pr.title}")
```

```
Install multiple packages in one operation
packages = ["pytest", "ruff", "mypy"]
for pkg in packages:
 await uv.install(package=pkg)
 print(f"✓ Installed {pkg}")
```

- 
- 
- 
- 



logger

print()

- 1.
2. CRACKZ\_LOG\_DIR --log-dir
- 3.
- 4.
5. <project>/logs/
- 6.
- 7.
- 8.

## Info level - operational progress

---

## Warning level - non-critical issues

---

## Error level - failures with details

---

# Debug level - detailed troubleshooting info

---

- ---
- ---
- ---
- ---

---

# Tracing & Observability

---

# Tracing with OpenTelemetry

---

```
uv sync --extra tracing
```

```
opentelemetry-instrumentation-anthropic
opentelemetry-exporter-otlp-proto-http
```

```
opentelemetry-sdk
```

```
Ctrl+Shift+P Cmd+Shift+P
```

```
AI Toolkit: Open Tracing
```

```
ai-mlstudio.tracing.open
```

```
export CRACKZ_TRACING_ENABLED=1
crackz-gui # Tracing auto-enabled
```

```
from crackz.tracing import setup_tracing

Initialize with default AI Toolkit endpoint
setup_tracing()

Or custom endpoint
setup_tracing(
 endpoint="http://custom-collector:4318/v1/traces",
 service_name="crackz-production"
)
```

```
•
•
•
•
•
```

```
crackz
└─ anthropic.messages.create
 └─ model: claude-sonnet-4-5-20250929
 └─ input_tokens: 1245
 └─ output_tokens: 382
 └─ duration: 2.3s
```

- 
- 
- 
- 

|                        |   |
|------------------------|---|
| CRACKZ_TRACING_ENABLED | 0 |
|------------------------|---|

|   |   |
|---|---|
| 1 | 0 |
|---|---|

<http://localhost:4318/v1/traces>
<http://localhost:4317>

```
setup_tracing(endpoint="http://your-collector:4318/v1/traces")
```

```
Enable tracing for all CLI operations
export CRACKZ_TRACING_ENABLED=1
crackz detect run --project my_project
```

```
Disable for specific run
export CRACKZ_TRACING_ENABLED=0
crackz detect run --project my_project
```

```
In main.py or startup script
from crackz.tracing import setup_tracing
```

```
Enable before starting GUI
if os.getenv("CRACKZ_TRACING_ENABLED") == "1":
 setup_tracing()
```

```
Run GUI
from crackz_gui.qt.app import main
main()
```

```
from crackz.tracing import setup_tracing, shutdown_tracing
```

```
Initialize tracing
setup_tracing(service_name="crackz-batch-processor")
```

```
Your AI operations (automatically traced)
from crackz_gui.services.ai_chat import AIChatService
chat = AIChatService(...)
chat.send_message("Analyze this crack detection result")
```

```
Clean shutdown (flushes pending spans)
shutdown_tracing()
```

```
uv sync --extra tracing
```

Ctrl+Shift+P → AI Toolkit: Open Tracing

CRACKZ\_TRACING\_ENABLED=1

uv sync --extra tracing

shutdown\_tracing()

```
import atexit
from crackz.tracing import shutdown_tracing
atexit.register(shutdown_tracing)
```

CRACKZ\_TRACING\_ENABLED=1

```
from opentelemetry import trace
tracer = trace.get_tracer(__name__)

with tracer.start_as_current_span("custom-operation") as span:
 span.set_attribute("project.name", "harbor_alpha")
 span.set_attribute("image.count", 1024)
 # Your code here
```

```
from opentelemetry.sdk.trace.sampling import TraceIdRatioBased

Sample 10% of traces
sampler = TraceIdRatioBased(0.1)

Pass to setup_tracing (requires custom implementation)
```

- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

⋮

\_\_\_\_\_

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# Contributing

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# Contributing Guide



- 
- `.github/copilot-instructions.md`
- `.github/workflows/copilot-setup-steps.yml`

`<type>/<scope>--<short-slug>`

```
AI creates feature branch
git checkout -b feat/gui-export-panel

AI makes changes with proper commit messages
git commit -m "feat(gui): add export panel component"
git commit -m "feat(gui): integrate export panel with main view"

When creating PR, AI suggests proper title
[] PR Title: "feat(gui): add export panel for detection results"
This triggers a MINOR version release (e.g., 1.2.0 → 1.3.0)
```



- 
- 
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- 
- 

```
create & activate venv
uv venv
source .venv/bin/activate # Windows: .venv\Scripts\activate

install with dev dependencies
uv sync --dev

run CLI help
uv run crackz --help

run GUI (optional)
uv run crackz-gui
```

`pyproject.toml`      `[tool.taskipy.tasks]`

```
uv run task lint # ruff lint (auto-fix)
uv run task format # ruff format
uv run task mypy # static type check
uv run task tests # unit tests only
uv run task integration # integration tests
uv run task all_tests # full test matrix
uv run task docs # live docs (mkdocs serve)
uv run task build_docs # strict build
```

- `tests/unit` `core/tests/unit` `gui/tests/unit`
- `tests/integration` `@pytest.mark.integration`
- `backend/backend/tests/unit/crackz/cli/test_cli_benchmark.py` `@pytest.mark.benchmark`

- `tests/test_data`
- `core/tests` `gui/tests`

```
uv run task tests
```

```
uv run task all_tests
```

```
uv run task benchmark
```

```
pyproject.toml fail_under = 75
```

```
tests/integration/crackz/gui/
```

```
Run all integration tests (includes GUI smoke tests)
uv run task integration

Run only GUI smoke tests
uv run pytest tests/integration/crackz/gui/test_gui_smoke.py

Run GUI in smoke mode (auto-close after initialization)
set CRACKZ_GUI_SMOKE=1 # Windows CMD
export CRACKZ_GUI_SMOKE=1 # Linux/macOS
uv run crackz-gui
```

```
CRACKZ_GUI_SMOKE=1
```

```
CI GITHUB_ACTIONS
```

```
tests/integration/crackz/gui/
├─ test_gui_smoke.py # Smoke tests for GUI initialization
└─ ... # Future GUI integration tests
```

1.

```
env = os.environ.copy()
env["CRACKZ_GUI_SMOKE"] = "1"
subprocess.run([sys.executable, "gui/src/crackz_gui/crackz_gui.py"], env=env)
```

2.

```
@pytest.mark.skipif(not has_display() or os.environ.get("CI"),
 reason="Requires display")
def test_component(qapp):
 component = MyComponent()
 assert component is not None
```

3.

```
def test_no_unsupported_parameters():
 # Ensure widget initialization works without errors
 browser.toggle_toolbar_buttons(visible=True) # Should not raise
```

### benchmark

```
Run benchmarks - compares to last saved baseline and fails on regression
uv run task benchmark
```

```
Update the baseline after intentional performance-impacting changes
uv run task benchmark_save
```

```
Manual benchmark with comparison
uv run pytest -m benchmark --benchmark-only --benchmark-autosave --benchmark-compare --benchmark-compare-fail=mean:20%
```

`.benchmarks/`

- `uv run task format lint`
- 
- 
- `loguru` `print`
- 

`<type>(<optional-scope>): <description>`

```
feat: feat(<scope>): fix: fix(<scope>):
perf: perf(<scope>):
docs: refactor:
build: test: chore: ci:
style:
```

```
feat(gui): add return-to-run-view functionality
fix(pdf): correct missing hero image
perf(dataloader): optimize batch processing
```

```
Fix GUI image display and add return-to-run-view functionality
Add new feature to handle user authentication
Update documentation for new feature
```

```
feat: fix: :
feat(gui): fix(cli): :
Feat: Fix: :
```

```
Install pre-commit hooks (one-time setup)
pip install pre-commit # or: uv tool install pre-commit
pre-commit install
pre-commit install --hook-type commit-msg
```

```
Test with a valid message
echo "feat(gui): add export panel" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt

Test with an invalid message
echo "Fix bug" > temp_msg.txt
cz check --commit-msg-file temp_msg.txt
```

```
git commit --no-verify -m "your message"
```

---

- - 
  - 
  - 
  -
- 

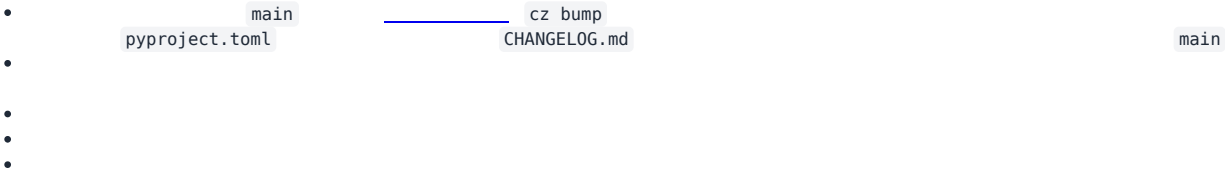
```
uv run task docs # serves at http://127.0.0.1:8000
```

```
uv run task build_docs
```

```
scripts/export_pdf.py
```

```
uv run task pdf
```

---



```
logs/crackz.log
```

---

main

<type>/<scope>-<short-slug>

optimization

feat/gui-live-preview

docs/api-reference-update

fix/pdf-rendering-crash

perf/dataloader-refactor/config-module-cleanup

feat(gui): add live preview panel

fix/pdf-crash

feat/gui-live-preview

fix(pdf): handle null hero image

- main

<type>/<scope>-<short-slug>

feat/gui-live-preview

fix/pdf-hero-image

refactor/logging-bootstrap

perf/dataloader-prefetch

chore/ci-release-step

hotfix/0.17.1-pdf-null-crash

exp/vector-db-poc

chore

feat

refactor

ci

build

exp

fix

docs

test

style

perf

hotfix/\*

main

fix:

<type>(<optional-scope>): <imperative summary>

<body – optional>

<footer – BREAKING CHANGE / issue refs>

feat(gui): add real-time detection preview panel

fix(pdf): correct missing hero when logo == cover image

refactor(logging): unify central + project logger bootstrap

chore(ci): refine release workflow version verification

feat!: remove legacy --batch flag

BREAKING CHANGE: The --batch flag was replaced by --batch-size.

feat

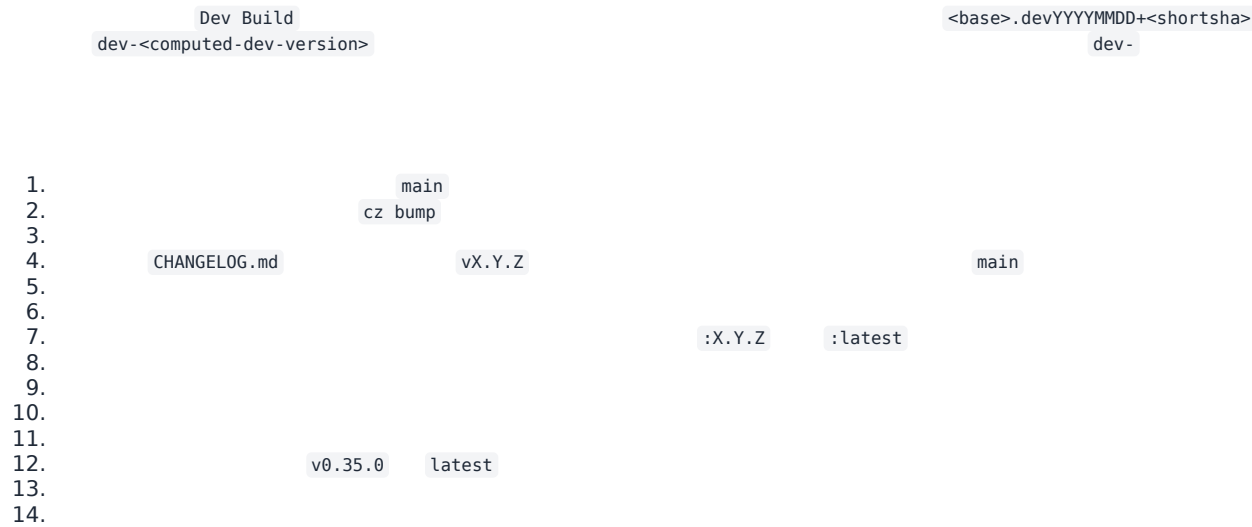
fix

!

BREAKING CHANGE:

- 
- feat(gui): add live preview
- fix(cli): correct path handling
- docs: update installation guide
- Fix GUI bugs
- Add new feature
- 
- 
- 
- 
- 

fix: docs: <type>(<scope>): <description> feat:



1. hotfix/<next-patch>-<short-desc> main
2. fix:
3. cz bump

TrainingConfig

tests/unit/crackz/project/test\_training\_config\_schema.py

.pre-commit-config.yaml

```
pip install pre-commit # or uv tool install pre-commit
pre-commit install
```

test\_project.py

docs/configuration.md

CHANGELOG.md

TrainingConfig Contract & Compatibility

ModelConfig    DetectionConfig

```
feat/cli-batch-export
fix/train-metrics-path
perf/dataloader-cache
refactor/model-config
docs/pdf-section
test/cli-train-e2e
chore/update-ruff
ci/release-tag-fix
build/multiarch-images
style/ruff-cleanup
hotfix/0.17.2-threshold-null
exp/alt-model-head
wip/gui-layout-pass1
```

feat(gui):

BREAKING CHANGE:

- - 
  -
- feat/\*

typer.secho

```
Success messages
typer.secho("✅ Operation completed successfully", fg=typer.colors.GREEN)

Error messages
typer.secho("❌ Operation failed", fg=typer.colors.RED)

Warning messages
typer.secho("⚠️ Warning message", fg=typer.colors.YELLOW)

Info messages
typer.secho("ℹ️ Information", fg=typer.colors.CYAN)
```

logger

```
from crackz.log import logger

Detailed progress, technical info, debugging
logger.info("Detailed operational information")
logger.warning("Warning about internal state")
logger.error("Error details for troubleshooting")
```

NEVER USE

- print()
- typer.echo()            typer.secho()
- console.print()

- 

#### core/src/crackz/cli/options.py

```
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION
```

```
def command_function(
 project_path: Path | None = PROJECT_PATH_OPTION,
 force: bool = FORCE_OPTION,
) -> None:
```

```
Only for options not reused elsewhere
LOCAL_OPTION = typer.Option(
 None,
 "--local-flag",
 help="Module-specific option not reused elsewhere",
)
```

## NEVER

- 
- 
- 

```
typer.Option()
```

```
FORCE_OPTION
```

```
"""Module docstring explaining CLI command purpose."""

from __future__ import annotations

from pathlib import Path
import typer

from crackz.cli.exceptions import cli_failure
from crackz.cli.options import PROJECT_PATH_OPTION, FORCE_OPTION
from crackz.log import logger # Always import logger

app = typer.Typer(no_args_is_help=True)

@app.command(name="command")
def command_function(
 project_path: Path | None = PROJECT_PATH_OPTION,
 force: bool = FORCE_OPTION,
) -> None:
 """Command description."""
 try:
 # Business logic here
 logger.info("Processing started")
 typer.secho("✅ Operation completed", fg=typer.colors.GREEN)
 except Exception as e:
 cli_failure(e, "Operation name")
```

- 
- 
- 
- 
- 

```
loguru
```

```
print()
```

- 
- 
- 
-

```
•
•
•
•
 tests/unit/
 tests/integration/
 @pytest.mark.integration
 @pytest.mark.benchmark
```

```
1.
2.
3.
4.
5.
6.
 options.py
 uv run task lint && uv run task format
 uv run task mypy
 uv run task tests
```

```
1.
2.
3.
4.
5.
 typer.secho()
 logger
 options.py
```

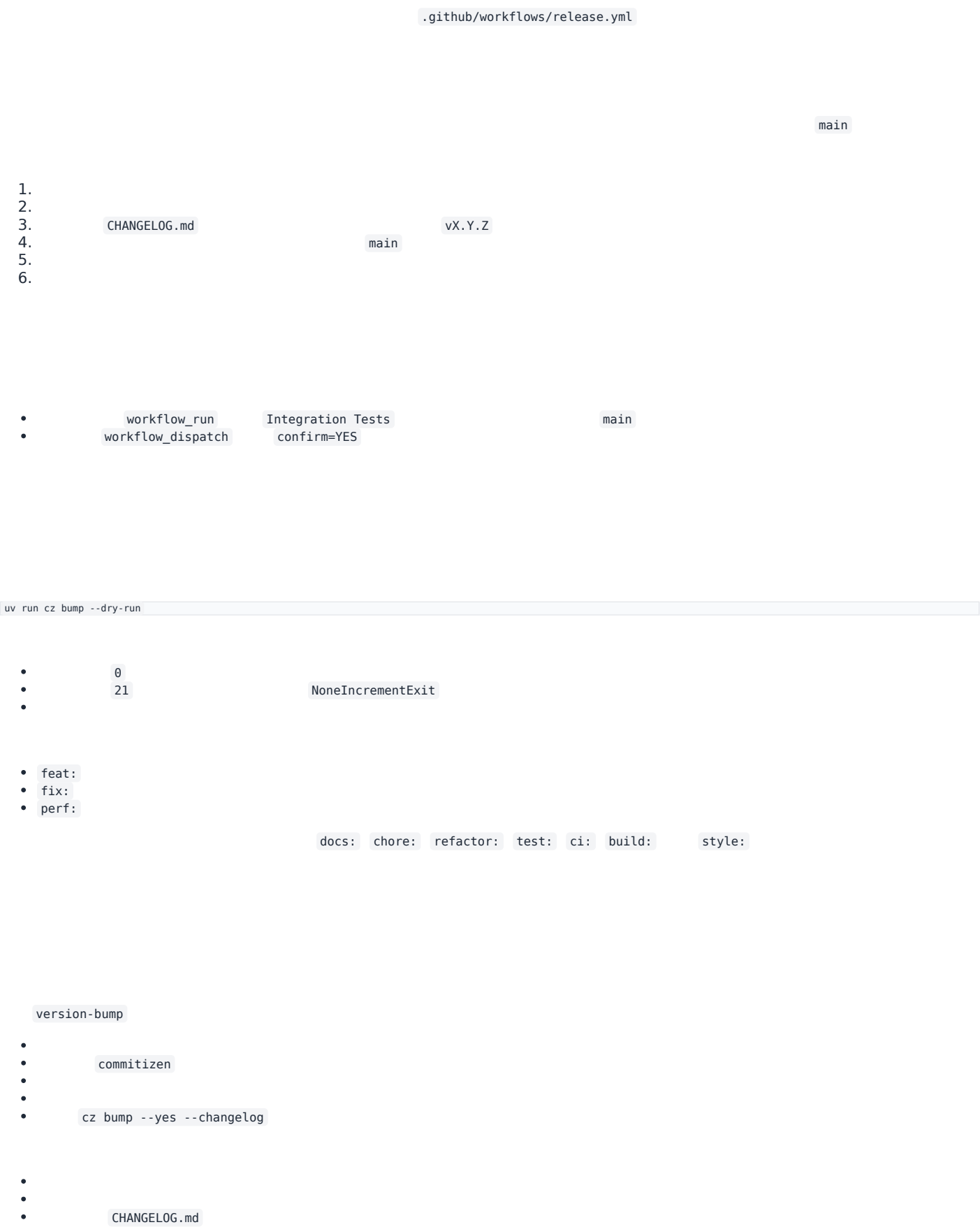
```
1.
2.
3.
4.
5.
6.
7.
 print()
 typer.echo()
 typer.secho()
 options.py
```

---

# Release Flow

---

# Release Flow



- 
- 

uv lock

uv.lock

main

- HEAD main
- 

version-bump

- 
- vx.Y.Z CHANGELOG.md

- build-wheels
- 
- 
- 
- build-sdist
- .tar.gz

crackz

release

- 
- 
- dist/
- site/crackz-docs.pdf .0
- CHANGELOG.md
- SHA256SUMS.txt
- 
- 
- --draft=false

- 
- cu118 cu128
- 
- CHANGELOG.md
- SHA256SUMS.txt
- .0

dist-combined

- `ai-release-notes.yml`
- 
- 
- `pyinstaller-build.yml`
- 
- 
- `docker-build.yml`
- 
- 

- 
- 
- 
- 

- `feat:` `fix:` `perf:` `main`
- 

- 
- 
- 
- `.0`

```
gh run list --limit 10
gh run view <run-id>
gh run watch
```

# PR Review Workflow

---

# Pull Request Review Workflow with Claude Code

- gh
- 
- 

```
List open PRs
gh pr list

View PR details
gh pr view <number>

View PR diff
gh pr diff <number>

Check CI status
gh pr checks <number>

Review PR
gh pr review <number> --approve
gh pr review <number> --comment -b "your comment"
gh pr review <number> --request-changes -b "needs changes"
```

```
List all open PRs
gh pr list

List PRs with specific labels
gh pr list --label "ready-for-review"

List your own PRs
gh pr list --author @me
```

```
View PR summary
gh pr view 254

View PR with comments
gh pr view 254 --comments

Open PR in browser
gh pr view 254 --web
```

```
fix(docs): resolve build failures and clarify release standards #254
Open • theresiasnow wants to merge 4 commits into main from fix/docs-build-and-standards

This PR fixes critical documentation build failures...

View this pull request on GitHub: https://github.com/owner/repo/pull/254
```

```
View full diff
gh pr diff 254

View only changed filenames
gh pr diff 254 --name-only

View diff in browser
gh pr diff 254 --web
```

```
View all checks
gh pr checks 254
```

```
View only required checks
gh pr checks 254 --required

Watch checks until completion
gh pr checks 254 --watch

Open checks in browser
gh pr checks 254 --web
```

```
✓ Build and Test (ubuntu-latest) – pass
✓ Build and Test (windows-latest) – pass
✓ Integration Tests – pass
✓ Lint and Format – pass
```

```
Checkout PR to review code locally
gh pr checkout 254

This creates/switches to the PR branch
Now you can:
- Run tests: uv run task tests
- Run linting: uv run task lint
- Build docs: uv run task build_docs
- Run the code locally
```

```
Review the changes in scripts/validate_commit_msg.py for:
- Code quality and readability
- Error handling
- Edge cases
- Documentation completeness
```

```
Review the changes in docs/contributing.md for:
- Clarity and completeness
- Correct formatting
- Broken links
- Consistency with existing docs
```

```
Check if the changes need additional tests and identify any edge cases not covered.
```

```
gh pr review 254 --approve -b "LGTM! Code quality looks good, tests pass."
```

```
gh pr review 254 --comment -b "Good work! Minor suggestion: consider adding error handling for X."
```

```
gh pr review 254 --request-changes -b "Please address the linting issues in validate_commit_msg.py before merging."
```

```
gh pr review 254 --approve -b "$(cat <<'EOF'
Great PR! Here's my review:

☐ Code quality: Excellent
☐ Tests: All passing
☐ Documentation: Clear and complete
☐ Linting: All checks pass

Minor suggestions:
- Consider adding a note about pre-commit hook in README
- The error messages are very helpful

Approved for merge.
EOF
)"
```

```
After review, switch back to main
git checkout main
```

```
List all PRs and store numbers
gh pr list

Review each PR in sequence
for pr in 254 255 256; do
 echo "Reviewing PR #$pr"
 gh pr view $pr
 gh pr diff $pr --name-only
 gh pr checks $pr
 # Ask Claude Code to review changes
done
```

```
Get PR data as JSON
gh pr view 254 --json title,body,author,createdAt,additions,deletions,files

Get check status as JSON
gh pr checks 254 --json name,state,completedAt
```

```
Continuously monitor checks until completion
gh pr checks 254 --watch --interval 30

This will update every 30 seconds until all checks complete
```

## Always Check CI Before Reviewing

```
Wait for checks to complete before reviewing
gh pr checks 254 --watch
```

## Review Locally for Complex Changes

```
For complex PRs, checkout and test locally
gh pr checkout 254
uv run task lint
uv run task tests
uv run task build_docs
```

## Use Claude Code for Deep Review

## Provide Constructive Feedback

```
Instead of just "needs work"
gh pr review 254 --request-changes -b "$(cat <<'EOF'
Thanks for the PR! A few items to address:

1. Linting: Please run 'uv run task lint' to fix PLR2004 error
2. Tests: Add test case for empty commit message
3. Docs: Update contributing.md with hook installation steps

Otherwise looks great!
EOF
)"
```

## Verify Release Triggers

```
Check PR title format
gh pr view 254 --json title

Should match: <type>(<scope>): <description>
Examples:
```

```
[] feat(gui): add export panel
[] fix(docs): resolve build failures
x Fix bugs
```

```
1. View the PR
gh pr view <number>

2. Check what files changed
gh pr diff <number> --name-only

3. View the actual changes
gh pr diff <number>

4. Checkout and test the fix
gh pr checkout <number>
Test the fix...

5. Run all pre-commit checks (lint, format, mypy, tests)
uv run task pre_commit

6. Approve if all looks good
gh pr review <number> --approve -b "Bug fix verified, all pre-commit checks pass."
```

```
1. View the PR
gh pr view <number>

2. Checkout the branch
gh pr checkout <number>

3. Build docs to check for errors
uv run task build_docs

4. Review locally with live server
uv run task docs
Opens at http://127.0.0.1:8000

5. Ask Claude Code to review
"Review the documentation changes for clarity, completeness, and broken links"

6. Approve or request changes
gh pr review <number> --approve
```

```
1. View PR and check size
gh pr view <number> --json additions,deletions,files

2. Check CI status
gh pr checks <number>

3. Checkout locally
gh pr checkout <number>

4. Run full test suite
uv run task all_tests

5. Test the feature manually
... manual testing ...

6. Ask Claude Code for code review
"Review this feature for:
- Code quality and maintainability
- Test coverage
- Documentation completeness
- Potential edge cases"

7. Provide detailed review
gh pr review <number> --approve -b "Feature works well, tests comprehensive."
```

- 1.
- 2.
- 3.
- 4.

```
📦 Release workflow completed successfully
📦 New release published: v0.52.1
📦 Artifacts built and uploaded
📦 CHANGELOG updated
📦 Git tag created
```

```
ⓘ Release workflow completed - No release created
📦 Reason: No conventional commits found that trigger releases
📦 This is expected behavior for documentation and maintenance commits
```

```
gh auth login
```

```
Make sure you're in the right repository
gh repo view

Or specify repository explicitly
gh pr view 254 -R Anaxiatech-AB/crackz
```

```
Stash local changes first
git stash

Then checkout PR
gh pr checkout <number>

Restore changes later
git stash pop
```

- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

---

## Distribution

---

# Distribution & Commercial Delivery

```
.github/workflows/docker-build.yml
```

```
.github/workflows/docker-build.yml
```

```
uv build --wheel
python scripts/build_cuda_wheel.py
uv build --sdist
uv run task build_exe
docker build -f Dockerfile -t crackz-cli:latest .
docker build -f Dockerfile.slim -t crackz-cli:slim .
docker build -f Dockerfile.gpu -t crackz-cli:gpu .
```

```
Build with CUDA 11.8 (default, most compatible for Windows)
python scripts/build_cuda_wheel.py

Build with CUDA 12.1
python scripts/build_cuda_wheel.py --cuda-version cu121

Build with CUDA 12.8 (default for Linux)
python scripts/build_cuda_wheel.py --cuda-version cu128

Build without cleaning artifacts first
python scripts/build_cuda_wheel.py --no-clean
```

```
Method 1: Install with specific CUDA index
pip install dist/crackz-*.whl --extra-index-url https://download.pytorch.org/whl/cu118

Method 2: Install with CUDA extra (uses pyproject.toml config)
pip install dist/crackz-*.whl[cu128] # For CUDA 12.8 on Linux

Method 3: Install CPU-only version
pip install dist/crackz-*.whl[cpu]
```

```
cu118 cu121

cu128

index-url pyproject.toml --extra-
python -c "import torch; print(f'CUDA available: {torch.cuda.is_available()}')"
```

```
Build both CLI and GUI executables (CPU-only)
uv run task build_exe

Build only CLI
uv run task build_exe_cli

Build only GUI
uv run task build_exe_gui

Clean and rebuild
uv run task build_exe_clean
```

```
Build with CUDA 11.8 (default, recommended for Windows)
python scripts/build_cuda_exe.py
```

```
Build with CUDA 12.8 (for newer GPUs/Linux)
python scripts/build_cuda_exe.py --cuda-version cu128

Build only GUI with CUDA
python scripts/build_cuda_exe.py --gui-only

Skip CUDA check if already installed
python scripts/build_cuda_exe.py --skip-cuda-check
```

dist/crackz

dist/crackz-gui

```
Use the build script directly for more control
python scripts/build_exe.py --help
python scripts/build_exe.py --clean # Clean then build both
python scripts/build_exe.py --no-build # Only clean, don't build

CUDA build script
python scripts/build_cuda_exe.py --help
python scripts/build_cuda_exe.py --clean --cuda-version cu128
```

- - 
  -
- upload\_to\_pypi = false

dist/

v0.14.0

- 1.
- 2.
- 3.
- 4.

```
Install specific version from GitHub release
pip install https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

Or download then install
wget https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz-0.30.2-py3-none-any.whl
```

```
Set GitHub token
export GITHUB_TOKEN="ghp_XXXXXXXXXX" # gitleaks:allow
```

```
Download using gh CLI
gh release download v0.30.2 --pattern '*.whl' --repo OWNER/REPO
pip install crackz-*.whl

Or use curl with authentication
curl -L \
 -H "Authorization: token $GITHUB_TOKEN" \
 -o crackz.whl \
 https://github.com/OWNER/REPO/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
pip install crackz.whl
```

PRIVATE\_INDEX\_USER

\_\_token\_\_

PRIVATE\_INDEX\_PASSWORD

PRIVATE\_INDEX\_URL

```
pip install --index-url https://pypi.company.com/simple/ crackz
```

.github/workflows/release.yml

```
name: release
on:
 push:
 tags:
 - 'v*.*.*'
jobs:
 build-release:
 runs-on: ubuntu-latest
 permissions:
 contents: write # allow releasing
 steps:
 - uses: actions/checkout@v4
 - uses: actions/setup-python@v5
 with:
 python-version: '3.13'
 - name: Install uv
 run: pip install uv
 - name: Sync deps (locked)
 run: uv sync --frozen
 - name: Build wheel + sdist
 run: uv build --wheel --sdist
 - name: Build documentation PDF
 run: |
 uv run mkdocs build --clean --strict
 uv run playwright install chromium
 uv run python scripts/export_pdf.py --outline \
 --cover-image 'assets/crackz-logo.png' \
 --logo 'assets/crackz-logo.png' \
 --out site/crackz-docs.pdf
 cp site/crackz-docs.pdf dist/
 - name: Upload to private index (twine)
 if: env.PRIVATE_INDEX_URL != ''
 env:
 TWINE_USERNAME: ${ secrets.PRIVATE_INDEX_USER }
 TWINE_PASSWORD: ${ secrets.PRIVATE_INDEX_PASSWORD }
 run: |
 pip install twine
 twine upload --repository-url ${ secrets.PRIVATE_INDEX_URL } dist/*.whl dist/*.tar.gz
 - name: Attach artifacts to Release
 uses: softprops/action-gh-release@v2
 with:
 files: dist/*
```

publish-release.yml

crackz-gui

v0.35.0

latest

https://pypi.company.com/simple/

PRIVATE\_INDEX\_USER

PRIVATE\_INDEX\_URL

\_\_token\_\_

PRIVATE\_INDEX\_PASSWORD

`.github/workflows/docker-build.yml``.github/workflows/docker-build.yml`

```

docker login ghcr.io -u USER -p TOKEN
Build variants
docker build -f Dockerfile -t ghcr.io/anaxiatech-ab/crackz:latest .
docker build -f Dockerfile.slim -t ghcr.io/anaxiatech-ab/crackz:slim .
docker build -f Dockerfile.gpu -t ghcr.io/anaxiatech-ab/crackz:gpu .
Push
for tag in latest slim gpu; do docker push ghcr.io/anaxiatech-ab/crackz:$tag; done

```

```

docker pull ghcr.io/anaxiatech-ab/crackz:slim

```

```

pip install --index-url https://pypi.company.com/simple/ crackz
pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
pip install torch torchvision --index-url <TORCH_URL> && pip install <crackz-wheel-url>
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
pip install --require-hashes -r requirements.txt

```

1.

```

Download Crackz wheel from GitHub Release first
wget https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl

Create wheelhouse directory and download all dependencies
mkdir wheelhouse
pip download crackz-VERSION-py3-none-any.whl -d wheelhouse

```

2.

```

(cd wheelhouse && shasum -a 256 * > SHA256SUMS.txt)

```

3.

4.

```

pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl

```

- 
- `pip install pip-audit cyclonedx-bom`
- `.asc`

`~/.config/pip/pip.conf`

```
[global]
extra-index-url = https://TOKEN@pkg.company.com/simple

%APPDATA%/pip/pip.ini
```

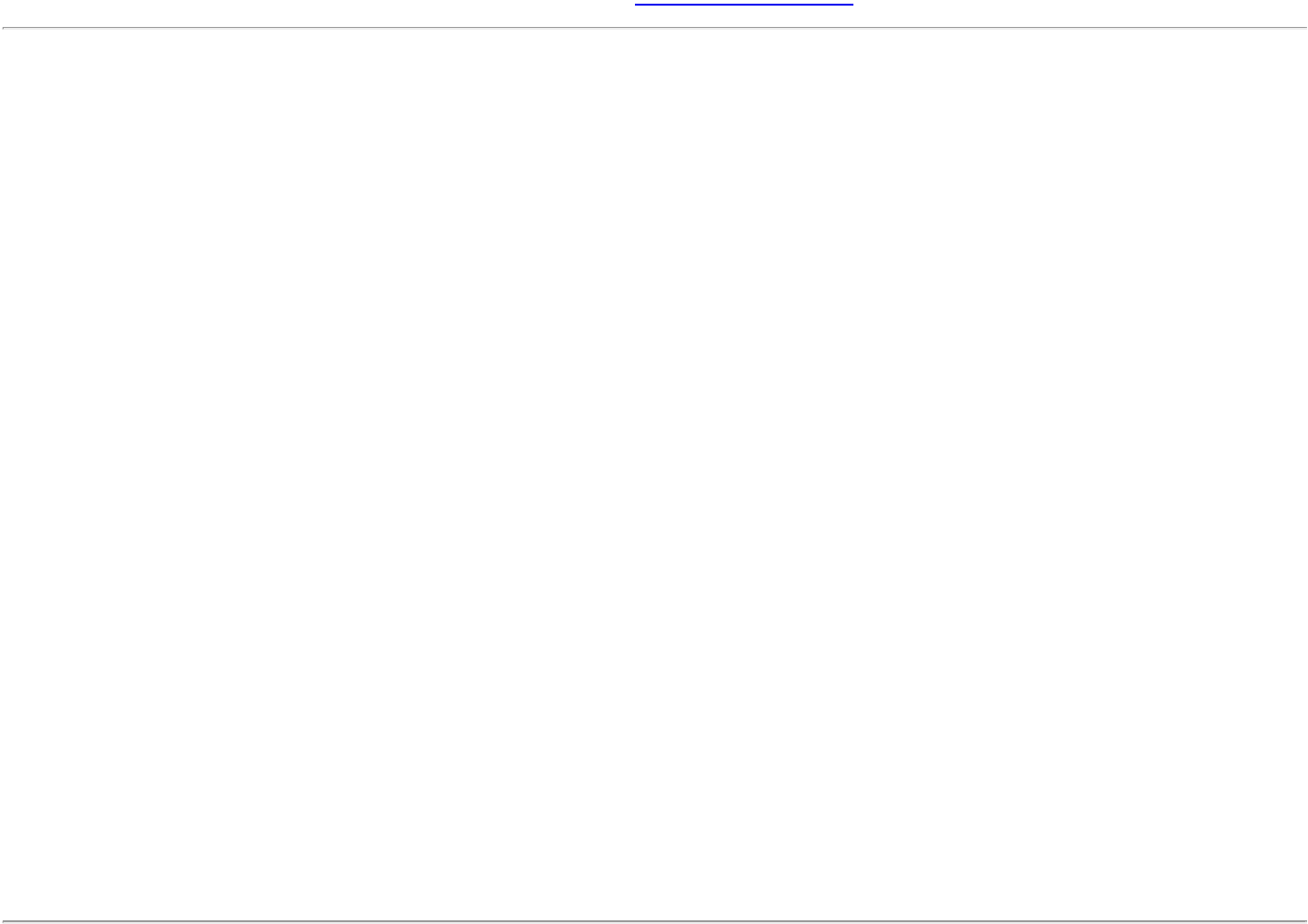
- 
- `escrow-*`

```
Build wheel
uv build --wheel
Build + attach release (manual example)
git tag v0.14.0 && git push origin v0.14.0
Install from private index
pip install --index-url https://pypi.company.com/simple/ crackz
Install from GitHub Release
pip install https://github.com/OWNER/REPO/releases/download/vVERSION/crackz-VERSION-py3-none-any.whl
Offline install
pip install --no-index --find-links wheelhouse crackz-VERSION-py3-none-any.whl
```

# Customer Guide

# Customer Guide: Crackz for Infrastructure Management

---



## What is a Mask?

| Original Image      | Mask Image                          | Result                                  |
|---------------------|-------------------------------------|-----------------------------------------|
| [photo of concrete] | → [white lines on black background] | = Trained model learns to detect cracks |

## When Do You Need Masks?

- 
- 
- 
- 
- 

## Recommended Tools for Creating Masks

1. CVAT (Computer Vision Annotation Tool) - Best for Teams

```
Self-hosted option (Docker)
docker run -d -p 8080:8080 cvat/server
Or use cloud-hosted version at app.cvat.ai
```

2. Labelme - Best for Individual Users

```
pip install labelme
labelme
```

labelme\_json\_to\_dataset <json\_file>      label.png

3. Photoshop / GIMP - Best for Fine Control

masks/val/      masks/test/      masks/train/

4. Python Scripting - Best for Automation

```
Example: Convert polygon annotations to masks
from PIL import Image, ImageDraw
import json

def polygon_to_mask(image_path, polygon_coors, output_path):
 # Load original image to get dimensions
 img = Image.open(image_path)
 width, height = img.size

 # Create black background
 mask = Image.new('L', (width, height), 0)
 draw = ImageDraw.Draw(mask)

 # Draw white polygon for crack
 draw.polygon(polygon_coors, fill=255)

 # Save mask
 mask.save(output_path)

Usage
polygon_coors = [(100, 200), (150, 250), (200, 230), ...]
polygon_to_mask('image_01.jpg', polygon_coors, 'masks/train/image_01.png')
```

Best Practices for Mask Creation

Quality Guidelines

- 
- 
- 
- 
- image\_01.jpg      image\_01.png

File Naming Convention

```
✓ CORRECT:
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001.png

✗ INCORRECT:
```

```
images/train/harbor_crack_001.jpg
masks/train/harbor_crack_001_mask.png ← Extra suffix
masks/train/HarborCrack001.png ← Different name
```

Organizational Tips

- 1.
- 2.
- 3.
- 4.
- 5.

Train/Val/Test Split

```
masks/
train/ # 70% of labeled data (200-350 images)
val/ # 15% of labeled data (30-50 images)
test/ # 15% of labeled data (30-50 images)
```

images/

```
images/
train/ # Same filenames as masks/train/
val/ # Same filenames as masks/val/
test/ # Same filenames as masks/test/
```

Using Crackz with Your Masks

images/train/

```
Automatic mask detection (if masks are in source_folder/masks/)
crackz import-images \
--project my-project \
--src ./labeled-data \
--type train \
--size 512 512

Explicit mask source
crackz import-images \
--project my-project \
--src ./images \
--masks_src ./masks \
--type train \
--size 512 512
```

```
crackz import-images \
--project my-project \
--src ./labeled-data \
--masks_src ./labeled-data/masks \
--split \
--train-ratio 0.7 \
--val-ratio 0.15 \
--test-ratio 0.15 \
--size 512 512 \
--random-seed 42
```

Troubleshooting Mask Issues

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

masks/train/

masks/

Advanced: Semi-Automated Mask Generation

Time Estimates

Professional Labeling Services

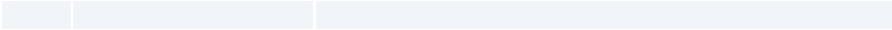
```
inspection-campaign-2025/
 images/ # Organized by train/val/test/raw splits
 masks/ # Ground truth for training (if available)
 artifacts/ # Detection results, metrics, checkpoints
 logs/ # Complete audit trail
 project.yaml # Configuration snapshot
```

- 
- 
- 
- 



- 
- 
- 
- 





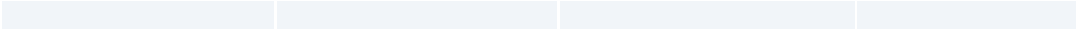
- 
- 
- 
- 

- 
- 
- 
- 

- 
- 
- 
- 



---



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# Customer Access

- 1.
- 2.
- 3.
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- 1.
- 2. templates/welcome-email.md
- 3. templates/access-revocation-email.md
- 4. templates/renewal-reminder-email.md
- 5.
- 6.
- 7.
- 8.

- 1.
- 2.
- 3.

# Customer Access Management

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- 
- 1.
  - 2.
  3. `pip install crackz`
- 

`pip install crackz`

- 
1. 

```
Via GitHub web UI:
Settings > Teams > New Team
Team name: "Customer-Acme-Corp"
Privacy: Secret
Repository access: crackz (Read)
```

- 2.

- 3.
- 4.
- 5.

Subject: Crackz Access - Welcome to [Customer Name]

Hi [Contact Name],

Thank you for your Crackz license purchase!

To access Crackz packages and Docker images, please:

1. Create a GitHub account (if you don't have one): <https://github.com/signup>
2. Reply with your GitHub username
3. We'll add you to the private repository

Once added, you can access:

- Releases: <https://github.com/Anaxiatech-AB/crackz/releases>
- Docker images: [https://github.com/anaxiatech?tab=packages&repo\\_name=crackz](https://github.com/anaxiatech?tab=packages&repo_name=crackz)
- Documentation: <https://github.com/Anaxiatech-AB/crackz/tree/main/docs>

License: [LICENSE-KEY or CONTRACT-REF]

Now the `release.yml` workflow will automatically publish to Azure Artifacts when configured!

Best regards,  
Theresia

Subject: Crackz Access Confirmed

Hi [Contact Name],

You now have access to the Crackz repository!

Installation instructions:  
<https://github.com/Anaxiatech-AB/crackz/blob/main/docs/installation.md>

Getting started:  
<https://github.com/Anaxiatech-AB/crackz/blob/main/docs/getting-started.md>

Docker registry:  
ghcr.io/anaxiatech-ab/crackz:latest

Need help? Email [theresia.sofia.snow@gmail.com](mailto:theresia.sofia.snow@gmail.com)

Best regards,  
Theresia

```
Find latest version at: https://github.com/Anaxiatech-AB/crackz/releases
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
Download from GitHub Releases page
wget https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl

Install locally
pip install crackz-0.30.2-py3-none-any.whl
```

```
Install GitHub CLI: https://cli.github.com/
gh auth login

Download latest release
gh release download --repo anaxiatech/crackz --pattern '*.whl'

Install
pip install crackz-*.whl
```

```
Generate GitHub Personal Access Token

1. Go to: https://github.com/settings/tokens/new
```

2. Note: "Crackz Docker Access"
3. Expiration: 1 year (or custom)
4. Scopes: Select ONLY:
  - ☐ read:packages
5. Click "Generate token"
6. Copy the token (starts with ghp\_)

```
Customer runs this once
export CR_PAT=ghp_XXXXXXXXXXXXXXXXXXXX # Their PAT # gitleaks:allow
echo $CR_PAT | docker login ghcr.io -u GITHUB_USERNAME --password-stdin
```

```
Now they can pull
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker pull ghcr.io/anaxiatech-ab/crackz:slim
docker pull ghcr.io/anaxiatech-ab/crackz:gpu
```

```
Use as normal
docker run --rm ghcr.io/anaxiatech-ab/crackz:latest --version
```

- 1.
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- 4.

```
Via Azure DevOps UI:
Artifacts > Create Feed
Name: crackz-packages
Visibility: Private (organization)
Upstream sources: Enable PyPI (optional)
```

- 5.
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  - o
- 8.

```
https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
```

```
Settings > Secrets and variables > Actions > New repository secret
```

```
PRIVATE_INDEX_URL: https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/upload
PRIVATE_INDEX_USER: azure
PRIVATE_INDEX_PASSWORD: <Azure-PAT-with-Packaging-Write-scope> # gitleaks:allow
```

```
publish-release.yml
```

```
Option 1: Install with explicit index
pip install crackz \
 --index-url https://pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/

Option 2: Configure pip globally
Linux/macOS: ~/.config/pip/pip.conf
Windows: %APPDATA%\pip\pip.ini
```

```
[global]
index-url = https://YOUR-PAT@pkgs.dev.azure.com/YOUR-ORG/_packaging/crackz-packages/pypi/simple/
```

```
pip install crackz
crackz --version
```

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```
Install devpi
pip install devpi-server devpi-web

Initialize server
devpi-init

Start server
devpi-server --start --host=0.0.0.0 --port=3141

Create user for yourself
devpi use http://localhost:3141
devpi user -c admin password=SECRET # gitleaks:allow

Create index
devpi index -c crackz bases=root/pypi
devpi use crackz
```

```
Install devpi client
pip install devpi-client

Login
devpi use http://YOUR-SERVER:3141
devpi login admin --password=SECRET # gitleaks:allow
devpi use admin/crackz

Upload wheel
devpi upload dist/crackz-*.whl
```

```
Customer configures pip
pip install --index-url http://YOUR-SERVER:3141/admin/crackz/+simple/ crackz
```

```
docker run -p 3141:3141 mucg/devpi
```

- 2.
- 3.
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- 7.
- 8.
- 9.

```
Subject: Crackz License Expired - Access Revoked

Hi [Contact],

Your Crackz license expired on [DATE].
Repository and package access has been revoked.

To renew, please contact theresia.snow@gmail.com

Thank you for using Crackz.
```

- 1.
- 2.
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- 1.
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- 

```
Ask customer to verify:
1. Go to: https://github.com/Anaxiatech-AB/crackz
2. If you see "404 Not Found", you don't have access yet
3. Reply with your GitHub username for access
```

```
read:packages
```

```
Ask customer to:
1. Generate new PAT: https://github.com/settings/tokens/new
2. Select ONLY: read:packages
3. Run: echo PAT | docker login ghcr.io -u USERNAME --password-stdin
4. Try again: docker pull ghcr.io/anaxiatech-ab/crackz:latest
```

```
Recommend:
pip install https://github.com/Anaxiatech-AB/crackz/releases/download/v0.30.2/crackz-0.30.2-py3-none-any.whl
```

```
You prepare:
pip download crackz -d wheelhouse
tar -czf crackz-offline-bundle.tar.gz wheelhouse/

Customer installs:
tar -xzf crackz-offline-bundle.tar.gz
pip install --no-index --find-links wheelhouse crackz
```

1.

```
Customer pulls and retags
docker pull ghcr.io/anaxiatech-ab/crackz:latest
docker tag ghcr.io/anaxiatech-ab/crackz:latest registry.company.com/crackz:latest
docker push registry.company.com/crackz:latest
```

2.

```
Get release download counts
gh api repos/anaxiatech/crackz/releases \
--jq '.[] | {name: .name, downloads: [.assets[]].download_count} | add}'
```

```
{"name": "v0.30.2", "downloads": 47}
{"name": "v0.30.1", "downloads": 23}
```

```
Requires GitHub GraphQL API (complex)
Easier to use web UI for now
```

```
Crackz Usage Report - October 2025
```

```
Active Customers: 5
```

| Customer  | License | Status  | Last Access |
|-----------|---------|---------|-------------|
| Acme Corp | LIC-001 | Active  | 2025-10-08  |
| Beta Inc  | LIC-002 | Active  | 2025-10-05  |
| Gamma LLC | LIC-003 | Expired | 2025-09-30  |

```
Download Statistics
```

```
- Total releases: 12
- Total downloads (wheels): 156
- Docker pulls (latest): 89
- Docker pulls (gpu): 34
```

```
Top Versions
```

```
1. v0.30.2: 47 downloads
2. v0.30.1: 23 downloads
```

3. v0.30.0: 18 downloads

## Action Items

- [ ] Contact Gamma LLC for renewal
- [ ] Send v0.31.0 release announcement

- 1.
2. PRIVATE\_INDEX\_\*
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- 4.
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- 9.
- 10.
- 11.
- 12.

- 1.
- 2.
3. read:packages
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.

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```
#!/bin/bash
scripts/customer-access/onboard-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
GITHUB_USERNAME="$2"
LICENSE_KEY="$3"

echo "👤 Onboarding customer: $CUSTOMER_NAME"
echo "GitHub username: $GITHUB_USERNAME"
echo "License: $LICENSE_KEY"

Create GitHub team (requires gh CLI with admin access)
TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"
gh api orgs/YOUR-ORG/teams -f name="Customer-$CUSTOMER_NAME" -f privacy=secret

Add member to team
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/memberships/$GITHUB_USERNAME -X PUT

Add team to repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X PUT -f permission=pull

echo "👤 Customer onboarded successfully!"
echo ""
echo "Next steps:"
echo "1. Send welcome email to customer"
echo "2. Add to tracking spreadsheet"
echo "3. Schedule renewal reminder for 1 year from now"
```

```
#!/bin/bash
scripts/customer-access/revoke-customer.sh

set -euo pipefail

CUSTOMER_NAME="$1"
REASON="${2:-License expired}"

echo "👤 Revoking access for: $CUSTOMER_NAME"
echo "Reason: $REASON"

TEAM_SLUG="customer-$(echo $CUSTOMER_NAME | tr '[:upper:]' '[:lower:]' | tr ' ' '-')"

Remove team from repository
gh api orgs/YOUR-ORG/teams/$TEAM_SLUG/repos/YOUR-ORG/crackz -X DELETE

Optionally delete team entirely
read -p "Delete team entirely? (y/n) " -n 1 -r
echo
if [[$REPLY =~ ^[Yy]$]]; then
 gh api orgs/YOUR-ORG/teams/$TEAM_SLUG -X DELETE
 echo "👤 Team deleted"
else
 echo "👤 Team kept (access removed from repository only)"
fi

echo "👤 Access revoked successfully!"
echo ""
echo "Next steps:"
echo "1. Send termination notice to customer"
echo "2. Update tracking spreadsheet"
echo "3. Archive customer records"
```

```
pip install crackz
```

```
PRIVATE_INDEX_*
```

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## License

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# License

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